

Stretched-exponential barriers for typical SK initial states

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Abstract

At every fixed inverse temperature $\beta > 1$, we prove a stretched-exponential obstruction to mixing for the zero-field Sherrington–Kirkpatrick model from typical equilibrium configurations held fixed as initial states. The Gibbs mass of states whose total-variation distance from equilibrium at time $\exp(n^{1/10000})$ exceeds $1/4$ tends to one in probability over the disorder. This holds for rate-one-per-site heat-bath dynamics and after the same stated number of discrete update attempts.

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1 Introduction

Local spin updates can preserve information about a realized equilibrium configuration for a long time, even though the unconditioned chain is stationary. We quantify this persistence throughout the low-temperature phase of the zero-field Sherrington–Kirkpatrick model. The central issue is the mass of slowly mixing initial states: a worst-state lower bound alone gives no information about whether a Gibbs sample encounters the obstruction.

For an integer $n \geq 2$, let $\Sigma_n = \{-1, 1\}^n$, and let $(J_{ij})_{1 \leq i < j \leq n}$ be independent standard Gaussian variables. At inverse temperature $\beta > 0$, the zero-field Sherrington–Kirkpatrick model has Hamiltonian and Gibbs measure

$$H^J(x) = H_n^J(x) = \frac{1}{\sqrt{n}} \sum_{i < j} J_{ij} x_i x_j, \quad \pi^J(x) = \pi_{n,\beta}^J(x) = \frac{e^{\beta H^J(x)}}{Z_{n,\beta}^J}, \quad x \in \Sigma_n, \quad (1.1)$$

where $Z^J = Z_{n,\beta}^J = \sum_{x \in \Sigma_n} e^{\beta H^J(x)}$. For $x, y \in \Sigma_n$, their overlap is

$$R(x, y) = \frac{1}{n} \sum_{i=1}^n x_i y_i.$$

The measure π^J is invariant under the global spin reversal $x \mapsto -x$. We fix $\beta > 1$ before sending n to infinity.

In continuous-time heat-bath dynamics, every site has an independent rate-one clock. At a ring of site i , its spin is resampled according to

$$\mathbb{P}(x_i \leftarrow s \mid x_{[n] \setminus \{i\}}) = \frac{\exp\{\beta s h_i^J(x)\}}{2 \cosh(\beta h_i^J(x))}, \quad h_i^J(x) = \frac{1}{\sqrt{n}} \sum_{j \neq i} J_{ij} x_j, \quad s \in \{-1, 1\},$$

with $J_{ji} = J_{ij}$ and $J_{ii} = 0$. Write P_t^J for its transition kernel. The total update-attempt rate is n , and an attempt may leave the configuration unchanged. We also consider the discrete kernel K^J

that chooses one site uniformly and performs one such conditional resampling. Both dynamics have stationary measure π^J .

For probability measures ν, η on Σ_n , set $\|\nu - \eta\|_{\text{TV}} = \frac{1}{2} \sum_x |\nu(x) - \eta(x)|$. For a fixed realized state x , define

$$d^J(x, t) = \|P_t^J(x, \cdot) - \pi^J\|_{\text{TV}}, \quad T^J(x) = \inf\{t \geq 0 : d^J(x, t) \leq 1/4\}, \quad (1.2)$$

$$d_{\text{disc}}^J(x, k) = \|(K^J)^k(x, \cdot) - \pi^J\|_{\text{TV}}, \quad T_{\text{disc}}^J(x) = \min\{k \geq 0 : d_{\text{disc}}^J(x, k) \leq 1/4\}. \quad (1.3)$$

Here k is an integer, and neither mixing time takes a supremum over the initial state.

Theorem 1.1. *Set $\kappa = 1/10000$. For every fixed $\beta > 1$,*

$$\begin{aligned} \mathbb{P}_{J, x_0 \sim \pi^J} \left\{ d^J(x_0, \exp(n^\kappa)) > \frac{1}{4} \right\} &\longrightarrow 1, \\ \mathbb{P}_{J, x_0 \sim \pi^J} \left\{ d_{\text{disc}}^J(x_0, \lfloor \exp(n^\kappa) \rfloor) > \frac{1}{4} \right\} &\longrightarrow 1. \end{aligned}$$

The order of conditioning in this statement is essential. We first draw x_0 from π^J , then regard it as a fixed argument of the transition kernel. Integrating that kernel over x_0 would instead recover π^J at every time. The joint formulation above is equivalent to convergence of the corresponding Gibbs mass to one in disorder probability. Indeed, if $F_n(J) \in [0, 1]$ is the mass of the complementary set, the joint assertion says $\mathbb{E}F_n \rightarrow 0$. Markov's inequality gives convergence in probability, and boundedness proves the converse.

Total variation contracts under a Markov kernel. In particular, for $s, t \geq 0$,

$$d^J(x, t+s) = \|(P_t^J(x, \cdot) - \pi^J)P_s^J\|_{\text{TV}} \leq d^J(x, t),$$

and the discrete distances are nonincreasing in k as well. Since the state space is finite, $t \mapsto d^J(x, t)$ is continuous. Consequently $d^J(x, t) > 1/4$ implies $T^J(x) > t$; in discrete time, $d_{\text{disc}}^J(x, k) > 1/4$ implies $T_{\text{disc}}^J(x) > k$. It follows that the theorem implies a lower bound at every fixed polynomial time scale. Appendix B will also give an independent proof using a fixed number of crossing levels.

Corollary 1.2. *For every fixed $\beta > 1$ and every fixed $a > 0$,*

$$\mathbb{P}_{J, x_0 \sim \pi^J} \{T^J(x_0) > n^a\} \longrightarrow 1, \quad \mathbb{P}_{J, x_0 \sim \pi^J} \{T_{\text{disc}}^J(x_0) > n^a\} \longrightarrow 1.$$

Equivalently, each corresponding Gibbs mass tends to one in probability over the disorder.

Proof. Fix $a > 0$. Eventually n^a is smaller than both $\exp(n^\kappa)$ and $\lfloor \exp(n^\kappa) \rfloor$. Apply Theorem 1.1 at those larger times and use contraction of total variation to pass to n^a . $\square$

1.1 Background and relation to earlier work

Sherrington and Kirkpatrick introduced the model as a mean-field description of a spin glass [16]. Parisi's variational description of its equilibrium free energy [14] was established rigorously through Guerra's interpolation bound and Talagrand's matching lower bound for the limiting pressure [9, 17]. These equilibrium results provide the scalar variational structure behind our argument. Passing from free energy to relaxation requires additional information about which overlap configurations a local trajectory can visit.

Ben Arous and Jagannath developed criteria that turn overlap free-energy barriers into exponentially small spectral gaps for mean-field spin glasses [4]. Their use of constrained replica pressures

is an important antecedent of the overlap comparisons used here. Their barrier criteria require additional landscape hypotheses; the explicit Ising sufficient conditions in that work exclude the pure quadratic covariance of SK. They therefore do not directly give the present result throughout $\beta > 1$. For SK itself, Sellke proves exponentially slow worst-case Glauber mixing at sufficiently large inverse temperature [15, Theorem 1.1]. That conclusion concerns the slowest initial states. It does not determine the Gibbs mass of the states that remain far from equilibrium after their initial configuration has been fixed.

The role of the external field is also substantial. Bandeira, El Alaoui, and Rödder prove polynomial worst-case mixing for lazy discrete heat-bath dynamics at every fixed inverse temperature when a sufficiently large constant uniform field is present, with high probability over the disorder [2, Corollary 1.2]. Our result concerns zero field. Its distinctive combination is the entire range of fixed $\beta > 1$, Gibbs mass tending to one for realized initial states, and a stretched-exponential time scale. The exponent $1/10000$ is not asserted to be optimal, and the constants in the proof may depend on β ; the argument does not give uniformity as β decreases to one.

El Alaoui, Montanari and Sellke prove a complementary obstruction to sampling at $\beta > 1$ for algorithms stable under disorder perturbations, in normalized 2-Wasserstein distance [8, Theorem 2.6]. Their stability hypothesis and approximation metric differ from the conditional Glauber transition law considered here.

Numerical studies have examined much sharper predictions for low-temperature relaxation. Billoire studies equilibrium autocorrelation times for binary couplings and sequential Metropolis updates [5]; Monthus and Garel compute inverse spectral gaps for small SK systems with Metropolis dynamics [11]. Both report evidence associated with an $n^{1/3}$ scale for logarithmic relaxation times. Their observables and dynamics differ from the state-conditioned total variation distance studied here, and these numerical predictions do not fix the exponent in Theorem 1.1.

The static comparisons build on Guerra’s Gaussian interpolation framework [9] and Talagrand’s extensions to systems with prescribed replica overlaps [18, Propositions 3.4 and 4.4]. Panchenko developed a general vector-spin formulation and proved the asymptotic sharpness of these constrained-overlap bounds [13]. Talagrand’s two- and three-copy bounds already retain an explicit cost when the prescribed overlaps differ from the covariance endpoint. In Section 2, we derive the finite-hierarchy matrix form needed here for the quadratic SK covariance, allowing arbitrary finite replica dimension and nonunit endpoint diagonals. The same section obtains a quantitative scalar free-energy comparison by choosing an interpolation path through a differential equation driven by overlap expectations. This choice has a methodological parallel in the adaptive interpolation method of Barbier and Macris [3]; their theorems are not inputs to our finite-size SK estimate. Gaussian concentration and the Brascamp–Lieb variance inequality [6] supply the probabilistic estimates used in these comparisons.

1.2 Proof strategy

Set $L = n^\kappa$ and $\rho = L^{-1}$. Our task is to exclude an overlap sign passage during the first e^L units of time. To connect that task to mixing, choose a reference v_1 with $R(v_1, x_0) \geq q_*$ for a fixed $q_* > 0$. The half-space $\{z : R(v_1, z) \leq 0\}$ has Gibbs mass at least $1/2$ by spin reversal. A transition law within $1/4$ of equilibrium must therefore enter it with probability at least $1/4$. The reference will be chosen from independent equilibrium samples, using the initial state but no future updates.

The first part of the paper establishes the static estimates that make this path argument possible. Section 2 compares the expected normalized log partition function $n^{-1}\mathbb{E} \log Z^J$ with the minimum of the scalar Parisi functional, with error $O_\beta(n^{-1/24})$, and proves the matrix comparison for constrained replica overlaps. Section 3 develops the scalar evolution and the variational identities of a minimizing

measure. Section 4 uses these identities to compare three replicas: when one pair has a substantial positive overlap, their small signed overlaps with the third replica are forced to remain close, apart from a set of very small averaged Gibbs probability. Section 5 proves an equilibrium energy bound and a quantitative estimate for small Gaussian rotations of the disorder. All of these specialized inputs are proved within the paper.

There are two further static steps. First, the energy and rotation bounds imply a coverage statement in Section 6: simultaneously for every subset S , outside a set of targets of absolute Gibbs mass less than $e^{-D_1 L}$, each $x \in S$ has Gibbs mass at least $e^{-D_3 L}$ of $z \in S$ with $|R(x, z)| \geq q_*$. Here D_3 is fixed after D_1 . Sampling $\exp(O(L))$ independent Gibbs configurations can therefore supply a reference inside S . Second, Section 7 treats triples whose two smaller overlaps lie near a positive level r , while the larger overlap exceeds r by only a shrinking amount. It supplies locking at the scales required by the backwards searches. The estimate uses a scalar stationarity identity on a fixed interval; if that identity fails, a pair-overlap pressure gap supplies a forbidden interval instead. This alternative avoids assuming a complete description of the minimizing measure's support.

Section 8 applies these estimates to a stationary discrete path (x_k) , and obtains the continuous-time path by an independent rate- n Poisson clock. At a crossing of a small positive overlap level, coverage supplies a new reference with large overlap with the current configuration. Searching backwards for the new reference's crossing, the locking estimates keep the overlaps with previously retained references almost unchanged. Iterating creates many simultaneous overlap constraints at a single path index.

The reference samples are arranged in blocks of a fixed length B . Within a block, each new reference is chosen in the set of configurations having small absolute overlap with all the earlier references in that block. At the start of the next block this restriction is reset. A conflict with older retained references can then delete only a suffix of length at most a fixed $H_0 < B$. Since $H_0 < B$, deletion cannot reach the first reference of the preceding block or any older retained reference. In particular, v_1 survives; Figure 2 shows this invariant. This construction permits p levels of order L with only $\exp(O(L))$ total reference samples, while retaining a fixed positive fraction of the levels.

The final probability estimate uses the order in which randomness is revealed. At level i , the reference is already determined when the independent threshold r_i is drawn uniformly from an interval of length ρ . Its overlap at any fixed path index is unlikely to fall in the much narrower permitted window. Removing constraints by conditional expectation in decreasing order of level bounds their joint probability, even though later reference choices depend on earlier thresholds. We sum over deterministic path indices and deterministic subsets of retained levels. This avoids conditioning on the adaptively chosen retained list and completes both clock versions.

Two auxiliary conclusions have a separate role. Appendix A proves that every disorder-selected set of Gibbs mass at least n^{-D_1} has polynomially large conditional mass of pairs with absolute overlap at least a fixed $q_*(\beta)$; this threshold is chosen before D_1 , and the resulting exponent D_2 is fixed before n grows. Its Gaussian-gain lemma and cell corollary are proved in full. Appendix B combines that result with the fixed-scale locking estimate to obtain a second proof of the polynomial-time obstruction. There the number of levels is fixed and the reference banks have polynomial sizes. These appendices reuse the energy, rotation and static locking proofs already established in the body, while retaining their own complete coverage and crossing arguments.

Notation. We suppress J , n , and the fixed β from H_n^J , $\pi_{n,\beta}^J$, and $Z_{n,\beta}^J$ when harmless. All constants may depend on β and on other parameters fixed before n grows. Gibbs replicas and reference samples are independent conditional on the disorder unless stated otherwise. For matrices, $U : V = \sum_{a,b} U_{ab} V_{ab}$ and $|U|^2 = U : U$ denote the Frobenius inner product and squared norm.

2 Gaussian hierarchies and quantitative free energy comparison

The overlap estimates in the later sections require two pressure comparisons: an upper bound with prescribed replica overlaps, and a lower bound for the unconstrained pressure with a controlled finite-size error. We derive both from finite Gaussian hierarchies. The upper bound allows the hierarchy's covariance endpoint to differ from the prescribed overlap matrix; the cost of that difference remains explicit. For the lower bound, we choose the hierarchy along an overlap-dependent differential equation and estimate its error before making that choice.

The constrained comparison belongs to Guerra's Gaussian interpolation framework [9], including Talagrand's bounds for coupled replicas [18, Propositions 3.4 and 4.4] and Panchenko's vector-spin formulation [13]. Talagrand's bounds already retain a cost when prescribed overlaps differ from the covariance endpoint. We derive the finite-hierarchy matrix form needed here directly, allowing nonunit endpoint diagonals as well. The quantitative lower comparison is proved separately below. We fix $\beta > 0$ throughout. A constant C_β may vary between occurrences, but never depends on n .

We first remove a constant term from the covariance calculation. Add independent standard Gaussian variables $J_{11}, \dots, J_{nn}$ to the disorder and, only in free-energy calculations, use the Hamiltonian

$$\hat{H}_n(\sigma) = H_n(\sigma) + \frac{1}{\sqrt{2n}} \sum_{i=1}^n J_{ii}.$$

The extra term has mean zero and is the same for every σ . The Gibbs measure and the expected log partition function are therefore unchanged. The covariance now takes the exact form

$$\mathbb{E} \hat{H}_n(\sigma) \hat{H}_n(\tau) = \frac{n}{2} R(\sigma, \tau)^2. \quad (2.1)$$

2.1 Two Gaussian tools and a hierarchy identity

We will need concentration for Gaussian disorder and a variance estimate after conditioning on a spin configuration. The following standard Gaussian and Brascamp–Lieb inequalities supply these two estimates.

Lemma 2.1 (Gaussian tools). *Let G be a standard Gaussian vector in a finite-dimensional Euclidean space. If f is L -Lipschitz, then for $a > 0$ and $\theta \in \mathbb{R}$,*

$$\mathbb{P}[|f(G) - \mathbb{E}f(G)| \geq a] \leq 2 \exp\left(-\frac{a^2}{2L^2}\right), \quad (2.2)$$

$$\log \mathbb{E} \exp\{\theta(f(G) - \mathbb{E}f(G))\} \leq \frac{\theta^2 L^2}{2}. \quad (2.3)$$

If a probability density on $\mathbb{R}^m$ is proportional to e^{-V} , where V is twice differentiable and $\nabla^2 V \succeq I$, then

$$\text{Var}(f) \leq \mathbb{E} \|\nabla f\|^2. \quad (2.4)$$

In particular, the variance of a linear functional $a \cdot G$ under this density is at most $\|a\|^2$.

The variance bound is the strongly log-concave case of the Brascamp–Lieb inequality [6]. For clarity, we prove the concentration estimate and the specialization of the variance bound to linear functions, which is the only specialization used below.

Proof of the needed Gaussian tools. For the concentration estimate, write H_s for convolution with a centered Gaussian of covariance sI , and let B_t be standard Brownian motion. The heat equation and Itô's formula represent a smooth L -Lipschitz function as

$$f(B_1) - \mathbb{E}f(B_1) = \int_0^1 \nabla H_{1-t}f(B_t) \cdot dB_t, \quad |\nabla H_s f| \leq L.$$

The integrand is bounded, so its exponential martingale has expectation one. Its quadratic variation is at most L^2 ; this proves Equation (2.3) for every real θ . Apply exponential Markov's inequality and optimize separately for the two tails to obtain Equation (2.2). Smooth convolution and localization, followed by Gaussian integrability, extend the argument to Lipschitz functions of at most linear growth. Every application here lies in that class.

For the linear variance estimate, we may work with the potentials that occur in the proof. They are smooth and satisfy $I \preceq \nabla^2 V \preceq CI$, where the finite constant C may depend on the fixed finite system. To see the upper bound, note that the nonquadratic terms are finite logarithmic partition functions or hierarchy prefix values: their Hessians are bounded sums of covariances. A linear tilt leaves the Hessian unchanged. Consider the diffusion

$$dY_t = \sqrt{2} dB_t - \nabla V(Y_t) dt$$

and denote its semigroup by Q_t . The drift is globally Lipschitz, which gives a unique solution. Under synchronous coupling, strong convexity contracts distances by e^{-t} . Thus an L -Lipschitz function becomes $e^{-t}L$ -Lipschitz under Q_t . Integration by parts shows that $\nu(dy) \propto e^{-V(y)} dy$ is invariant and that

$$\frac{d}{dt} \text{Var}_\nu(Q_t f) = -2 \int |\nabla Q_t f|^2 d\nu.$$

The Gaussian tails supplied by strong convexity justify the same identity for $f(y) = a \cdot y$ by a cutoff argument. For independent Y, Y' with law ν , contraction also gives

$$\text{Var}_\nu(Q_t f) \leq \frac{1}{2} e^{-2t} |a|^2 \mathbb{E}|Y - Y'|^2 \longrightarrow 0.$$

We can therefore integrate the dissipation identity to infinity. The gradient bound $|\nabla Q_t f| \leq e^{-t}|a|$ yields $\text{Var}_\nu(a \cdot Y) \leq 2|a|^2 \int_0^\infty e^{-2t} dt = |a|^2$. This proves the linear estimate needed for the conditional disorder laws. The finite upper Hessian bound justifies the diffusion argument; at each application we will check its lower Hessian bound explicitly. $\square$

We next define the hierarchy and its probability law. Besides any quenched root randomness, take independent Gaussian fields $g_0, \dots, g_k$. The field g_0 belongs to the quenched root. Choose masses

$$0 = x_{-1} \leq x_0 \leq \dots \leq x_{k-1} \leq x_k = 1, \quad w_l = x_l - x_{l-1},$$

and a terminal log partition function F_k on a finite configuration space. Integrate the fields backwards by setting, for $1 \leq j \leq k$,

$$F_{j-1} = \frac{1}{x_{j-1}} \log \mathbb{E}_j e^{x_{j-1} F_j}. \quad (2.5)$$

Here $\mathbb{E}_j$ integrates the field g_j . At a zero mass the right-hand side is interpreted as $\mathbb{E}_j F_j$. After these recursive integrations, the hierarchy pressure is $\mathbb{E} F_0$, with expectation also over all quenched randomness.

The recursion also specifies a probability law on fields and terminal configurations. Conditional on the prefix through level $j-1$, multiply the original Gaussian law of g_j by the density

$$W_j = \exp\{x_{j-1}(F_j - F_{j-1})\}. \quad (2.6)$$

Equation (2.5) normalizes this density. After sampling level k , draw a configuration from the terminal Gibbs law. For a pair of paths indexed by a level l , sample their root and fields through l once, and then sample the two continuations independently conditional on this prefix. We denote expectation under that pair law by $\mathbb{E}_l^{(2)}$. At $l = k$ the only independent choices are the terminal thermal draws. More generally, a finite tree is sampled by drawing each shared prefix once and making its descendants conditionally independent. In particular, every branch has the single-path marginal law. This convention fixes the shared randomness in all subsequent integration-by-parts calculations.

Lemma 2.2 (Hierarchy differentiation). *Suppose an independent standard Gaussian vector g at level j enters the terminal exponent as $\sqrt{v} g \cdot e(S)$, where e is a fixed vector-valued function of the terminal configuration. With all other parameters fixed,*

$$\frac{\partial}{\partial v} \mathbb{E} F_0 = \frac{1}{2} \left(\mathbb{E} \|e(S)\|^2 - \sum_{l=j}^k w_l \mathbb{E}_l^{(2)} e(S^1) \cdot e(S^2) \right). \quad (2.7)$$

A quenched Gaussian is treated as level 0. The identity holds for $v > 0$ and extends to interpolation endpoints by integration and continuity.

Proof. To differentiate the pressure, first add a deterministic vector in the same direction as $\sqrt{v} g$. Differentiating the terminal log partition function and then the hierarchy recursion shows that, for $l \geq j$, the gradient of F_l is the conditional mean m_l^e of $e(S)$ given the prefix through l . A second differentiation gives

$$\nabla^2 F_l = \mathbb{E}[\nabla^2 F_{l+1} \mid l] + x_l \text{Cov}(m_{l+1}^e \mid l). \quad (2.8)$$

All conditional expectations in this calculation use the path law. At the terminal level, the Hessian is the thermal covariance of $e(S)$.

For $j \geq 1$, differentiate the j th logarithmic expectation with respect to the variance and integrate by parts in its Gaussian field. The result is one half of the tilted expectation of

$$\text{tr } \nabla^2 F_j + x_{j-1} \|m_j^e\|^2.$$

At level $j = 0$, ordinary Gaussian differentiation omits the second term, exactly as prescribed by $x_{-1} = 0$. Earlier levels propagate the derivative by conditional expectation. Iteration of Equation (2.8) now expresses twice the pressure derivative as

$$\mathbb{E} \|e(S)\|^2 - \mathbb{E} \|m_k^e\|^2 + \sum_{l=j}^{k-1} x_l (\mathbb{E} \|m_{l+1}^e\|^2 - \mathbb{E} \|m_l^e\|^2) + x_{j-1} \mathbb{E} \|m_j^e\|^2.$$

The sum telescopes to $\mathbb{E} \|e(S)\|^2 - \sum_{l=j}^k w_l \mathbb{E} \|m_l^e\|^2$. Two descendants are independent conditional on their common prefix, so $\mathbb{E} \|m_l^e\|^2 = \mathbb{E}_l^{(2)} e(S^1) \cdot e(S^2)$. This is the claimed identity.

Gaussian integrability justifies each differentiation: the configuration space and the number of levels are finite, the coupled gradients are bounded, and the recursive values grow at most linearly in the Gaussian variables. A zero mass can be handled directly by the expectation recursion, or obtained as a limit. Integration and continuity give the stated endpoint interpretation. $\square$

2.2 A matrix interpolation bound

We now apply the differentiation identity to a prescribed overlap constraint. For d configurations $\sigma^1, \dots, \sigma^d$, write $D_{ab} = R(\sigma^a, \sigma^b)$ for their Gram matrix. If D is feasible, meaning that at least one

such tuple realizes it, define

$$F_{d,n}(D) = \frac{1}{n} \mathbb{E} \log \sum_{\substack{\sigma^1, \dots, \sigma^d \in \{-1, 1\}^n \\ R(\sigma^a, \sigma^b) = D_{ab} \quad \forall a, b}} \exp \left\{ \beta \sum_{a=1}^d H_n(\sigma^a) \right\}. \quad (2.9)$$

The sum is unnormalized. We use the Frobenius notation $U : V = \sum_{a,b} U_{ab} V_{ab}$ and $|U|^2 = U : U$. To compare this pressure with a single-site recursion, choose a symmetric matrix path

$$Q_{-1} = 0, \quad Q_j - Q_{j-1} \succeq 0 \quad (0 \leq j \leq k), \quad Q_k = Q_{\text{end}}.$$

Within these conditions, the endpoint is free: it may differ from D and may have nonunit diagonal. Increments of zero covariance are allowed. If the first nonzero increment should carry positive mass, insert a quenched level $Q_0 = 0$; the first genuine increment is then integrated at mass x_0 . At a single site, take independent d -dimensional Gaussian fields of covariance $\beta^2(Q_j - Q_{j-1})$. For a symmetric multiplier λ , define $\Psi(Q, x, \lambda)$ as the expected hierarchy value with terminal function

$$\log \sum_{s \in \{-1, 1\}^d} \exp \left\{ \left(\sum_{j=0}^k g_j \right) \cdot s + s^\top \lambda s \right\}.$$

For the deterministic correction to the pressure, set

$$\int x \, d|Q|^2 = \sum_{j=0}^k x_{j-1} (|Q_j|^2 - |Q_{j-1}|^2). \quad (2.10)$$

The initial quenched increment therefore carries mass zero in this integral. The next bound keeps the cost of the free endpoint explicit.

Proposition 2.3 (Matrix interpolation). *For every feasible D and every path and multiplier as above,*

$$F_{d,n}(D) \leq \Psi(Q, x, \lambda) - \lambda : D + \frac{\beta^2}{4} \left(|D - Q_{\text{end}}|^2 - \int x \, d|Q|^2 \right). \quad (2.11)$$

For fixed D and λ , the bound passes to any sequence of finite paths for which Q_{end} , $\Psi(Q, x, \lambda)$, and the displayed path integral converge.

Proof. Place independent copies of the single-site fields at all n sites. Interpolate from those fields to the common Hamiltonian disorder, retaining the overlap constraint throughout. The terminal exponent is

$$\sqrt{t} \beta \sum_{a=1}^d \hat{H}_n(\sigma^a) + \sqrt{1-t} \sum_{i=1}^n \sum_{j=0}^k g_{j,i} \cdot s_i + \sum_{i=1}^n s_i^\top \lambda s_i, \quad s_i = (\sigma_i^1, \dots, \sigma_i^d).$$

Denote the expected recursive value, divided by n , by $\varphi(t)$. For two terminal draws, define the cross-overlap matrix by $R_{ab} = R(\sigma^a, \tau^b)$; this matrix need not be symmetric. Equation (2.1) gives Hamiltonian covariance $n\beta^2|R|^2/2$, whereas the j th site-field increment has covariance $n\beta^2(Q_j - Q_{j-1}) : R$. The self-overlap is D . Sum Lemma 2.2 over these Gaussian increments to obtain

$$\begin{aligned} \varphi'(t) &= \frac{\beta^2}{4} \left[|D|^2 - 2Q_k : D - \sum_{l=0}^k w_l \mathbb{E}_l^{(2)}(|R|^2 - 2Q_l : R) \right] \\ &\leq \frac{\beta^2}{4} \left(|D|^2 - 2Q_k : D + \sum_{l=0}^k w_l |Q_l|^2 \right). \end{aligned}$$

The inequality discards the nonnegative squares $|R - Q_l|^2$ and therefore does not require symmetry of R . Summation by parts gives

$$|Q_k|^2 - \sum_{l=0}^k w_l |Q_l|^2 = \int x \, d|Q|^2.$$

It remains to identify the endpoints. At $t = 1$ the fields vanish and $\varphi(1) = F_{d,n}(D) + \lambda : D$. At $t = 0$, remove the overlap constraint. This increases the terminal sum and hence every recursive value, because each integration is monotone. The resulting recursion factors over sites, giving $\varphi(0) \leq \Psi(Q, x, \lambda)$. Integration of the derivative bound proves the proposition. Singular covariance increments are permitted by continuity, or directly by representing a field through a square root of its covariance. $\square$

2.3 The scalar functional

The scalar version of the hierarchy gives the variational quantity against which we compare the unconstrained pressure. Let μ be a probability measure on $[0, 1]$ and put $\alpha(s) = \mu([0, s])$. Define $\Phi_\alpha(s, z)$ by the terminal condition $\Phi_\alpha(1, z) = \log(2 \cosh z)$ and the backward equation

$$-\partial_s \Phi_\alpha = \frac{\beta^2}{2} \left(\partial_{zz} \Phi_\alpha + \alpha(s) (\partial_z \Phi_\alpha)^2 \right). \quad (2.12)$$

For a step function α , the equation means the Gaussian recursion already defined: over an interval of length $b - a$ with constant mass x , its terminal function f is mapped to

$$z \mapsto \frac{1}{x} \log \mathbb{E} \exp \{ x f(z + \beta \sqrt{b - a} G) \},$$

with ordinary expectation at $x = 0$. For general α , take L^1 limits of step functions; Lemma 3.1 proves that this definition exists, is independent of the approximation, and has the stability properties we use. Define

$$\mathcal{P}_\beta(\mu) = \Phi_\alpha(0, 0) - \frac{\beta^2}{2} \int_0^1 s \alpha(s) \, ds, \quad P_\beta = \inf_\mu \mathcal{P}_\beta(\mu). \quad (2.13)$$

The infimum is attained by compactness and continuity, as proved in Proposition 3.2. These scalar approximation and compactness arguments are independent of the pressure comparison; the lower comparison below will use only finitely supported trial measures. Apply Proposition 2.3 with $d = 1$, $D = Q_{\text{end}} = 1$, and $\lambda = 0$. First for step measures and then by approximation, this gives the upper comparison

$$p_n(\beta) := \frac{1}{n} \mathbb{E} \log Z_{n,\beta} \leq P_\beta. \quad (2.14)$$

Theorem 2.4 (Quantitative comparison). *For every fixed $\beta > 0$, there is $C_\beta < \infty$ such that for all sufficiently large n ,*

$$P_\beta - C_\beta n^{-1/24} \leq p_n(\beta) \leq P_\beta. \quad (2.15)$$

The remaining task is the lower comparison. We will choose an interpolation path from expectations of overlaps, using a hierarchical backward system. Such expectation-dependent choices are also a feature of the adaptive interpolation method of Barbier and Macris [3]. Here we first prove an explicit overlap error estimate for the hierarchy, without assuming a prior description of the Gibbs measure, and then use it along the chosen path. Mourrat's enriched-free-energy and Hamilton–Jacobi viewpoint [12, Section 3, equations (3.24)–(3.27)] provides related motivation for organizing errors by hierarchical overlaps. This is a methodological connection only: the direct covariance and concentration argument and the quantitative bound here do not invoke a Hamilton–Jacobi theorem.

2.4 Concentration along a hierarchy

To obtain an error estimate that can be summed over the levels, we use equally spaced masses for the remainder of the section:

$$w_j = \frac{1}{k+1}, \quad x_j = \frac{j+1}{k+1} \quad (0 \leq j \leq k).$$

Fix $0 \leq t \leq 1$ and levels $0 = q_{-1} < q_0 < \dots < q_k$. Set $\Delta_j = \beta^2(q_j - q_{j-1})$ and use the terminal exponent

$$U(\sigma) = \sqrt{t} \beta \hat{H}_n(\sigma) + \sum_{j=0}^k \sqrt{\Delta_j} z_j \cdot \sigma, \quad (2.16)$$

where $z_0, \dots, z_k$ are independent standard Gaussian vectors in $\mathbb{R}^n$. Let $\mathcal{A}_j$ contain the prefix through level j , including the quenched Hamiltonian disorder. Under the path law, put

$$m_j = \mathbb{E}[\sigma \mid \mathcal{A}_j], \quad M_j = \frac{\|m_j\|^2}{n}, \quad r_j = \mathbb{E}M_j. \quad (2.17)$$

The conditional means m_j form a martingale. Consequently $0 \leq r_0 \leq \dots \leq r_k \leq 1$. They can also be obtained by differentiating F_j with respect to an additional deterministic field coupled linearly to the spins. We begin by controlling the Gaussian field energy under the same path law.

Lemma 2.5 (Field-energy concentration). *For every $0 \leq j \leq k$, under the single-path law,*

$$\text{Var} \left(\frac{z_j \cdot \sigma}{n} \right) \leq \frac{1}{n}. \quad (2.18)$$

Proof. Condition first on the terminal spin σ . The product of the tilts in Equation (2.6) and the terminal Gibbs density telescopes. Thus the conditional joint density of all standard Gaussian disorder variables and fields is proportional to their original Gaussian density multiplied by

$$\exp \left\{ U(\sigma) - \sum_{l=0}^k w_l F_l \right\}. \quad (2.19)$$

Each F_l is jointly convex in its prefix Gaussian variables. At the terminal level this follows from log-sum-exp with linearly entering variables. Backwards integration preserves convexity: expectation does so at mass zero, and Hölder's inequality does so for a positive-mass logarithmic expectation. Extend F_l to all variables by making it constant in its descendants; it is still convex. The weights w_l are nonnegative, and $U(\sigma)$ is linear for fixed σ . The negative logarithm of the conditional density therefore has Hessian at least I . The linear specialization of Lemma 2.1 bounds the conditional variance of $z_j \cdot \sigma/n$ by $1/n$.

To remove the conditioning, we must also check the conditional mean. It is independent of σ , as follows from the gauge transformation with signs $\varepsilon_i = \sigma_i$:

$$J_{ab} \mapsto \varepsilon_a \varepsilon_b J_{ab} \quad (a < b), \quad z_{l,i} \mapsto \varepsilon_i z_{l,i} \quad (0 \leq l \leq k).$$

Leave the diagonal variables fixed. This transformation preserves the Gaussian law and carries the conditional law at σ to the conditional law at the all-plus spin. It also preserves $z_j \cdot \sigma$. Thus the conditional means agree, and the law of total variance proves Equation (2.18). $\square$

Lemma 2.6 (Average overlap concentration). *Let $\Delta_{\min} = \min_j \Delta_j$. If $k \geq 1$ and $k/\sqrt{n\Delta_{\min}} \leq 1$, then*

$$\sum_{j=0}^k w_j \mathbb{E}_j^{(2)}(R - r_j)^2 \leq C \left(\frac{k}{\sqrt{n\Delta_{\min}}} + k^{-1/2} \right), \quad (2.20)$$

where C is an absolute constant.

Proof. The law of total variance splits the overlap error into two terms: the fluctuation of its conditional mean M_j , and the variance left after the prefix is fixed. We estimate these terms in that order.

Fluctuation of the conditional mean. Fix $j < k$. Sample three paths a, b, c with common prefix through $j + 1$ and conditionally independent continuations. If $B = B(\sigma^a, \sigma^b, \sigma^c)$ satisfies $|B| \leq 1$, integration by parts in the two adjacent Gaussian fields gives

$$\mathbb{E}B \left(\frac{z_j}{\sqrt{\Delta_j}} - \frac{z_{j+1}}{\sqrt{\Delta_{j+1}}} \right) \cdot \frac{\sigma^a}{n} = -w_j \mathbb{E}B \frac{m_j \cdot \sigma^a}{n}. \quad (2.21)$$

Here is the density calculation, including the common prefix. Relative to the product Gaussian law, multiply W_l along each common edge $1 \leq l \leq j + 1$, along each of the three distinct descendant branches, and then by the three terminal Gibbs probabilities. The logarithm of this density is

$$\begin{aligned} & \sum_{l=1}^{j+1} x_{l-1} (F_l - F_{l-1}) \\ & + \sum_{v \in \{a, b, c\}} \left[\sum_{l=j+2}^k x_{l-1} (F_l^v - F_{l-1}^v) + U^v(\sigma^v) - F_k^v \right]. \end{aligned}$$

Apply the derivative difference $\Delta_j^{-1/2} \partial_{z_{j,i}} - \Delta_{j+1}^{-1/2} \partial_{z_{j+1,i}}$ coordinate by coordinate. At level $j + 1$ and below, every term depends on these fields only through $\sqrt{\Delta_j} z_j + \sqrt{\Delta_{j+1}} z_{j+1}$, so its two derivatives cancel. Terms above j depend on neither field. Only F_j remains, with coefficient $x_{j-1} - x_j = -w_j$; the same calculation applies at $j = 0$ because $x_{-1} = 0$. Its normalized derivative is $m_{j,i}$. Throughout this differentiation the spin labels and B are fixed. Integration by parts and summation over i now prove Equation (2.21).

Let Y be the quantity multiplying B on the left-hand side of Equation (2.21). Every branch has the single-path marginal. Lemma 2.5 and the L^2 triangle inequality therefore imply

$$\text{Var}(Y)^{1/2} \leq \frac{2}{\sqrt{n\Delta_{\min}}}.$$

With $B = 1$, Equation (2.21) reads $\mathbb{E}Y = -w_j r_j$. Subtract this mean, use $|B| \leq 1$, and recall $w_j = 1/(k + 1)$ to obtain

$$\left| \mathbb{E}B \left(\frac{m_j \cdot \sigma^a}{n} - r_j \right) \right| \leq \frac{2(k + 1)}{\sqrt{n\Delta_{\min}}}. \quad (2.22)$$

Choose $B = R(\sigma^b, \sigma^c)$. Conditional on $\mathcal{A}_{j+1}$ the three descendants are independent, so the left-hand side becomes

$$\left| \mathbb{E} \left(\frac{m_j \cdot m_{j+1}}{n} - r_j \right) M_{j+1} \right|.$$

We now replace m_{j+1} by m_j in both factors. The change in the expectation is at most $3\sqrt{\gamma_j}$, where

$$\gamma_j = \frac{1}{n} \mathbb{E} \|m_{j+1} - m_j\|^2.$$

This follows from the bounds $\|m_l\| \leq \sqrt{n}$ and $|M_j - r_j| \leq 1$. After replacement the expectation is $\mathbb{E}(M_j - r_j)M_j = \text{Var}(M_j)$. Martingale orthogonality gives $\sum_{j < k} \gamma_j \leq 1$; averaging the errors and using Cauchy–Schwarz proves

$$\sum_{j < k} w_j \text{Var}(M_j) \leq C \left(\frac{k}{\sqrt{n\Delta_{\min}}} + k^{-1/2} \right). \quad (2.23)$$

Conditional variance. The conditional mean fluctuation is now controlled. To bound the variance of the overlap of two descendants after their common prefix is fixed, define

$$C_j = \text{Cov}(\sigma \mid \mathcal{A}_j), \quad D_j = \nabla^2 F_j, \quad V_j = \text{Cov}(m_{j+1} \mid \mathcal{A}_j),$$

where all conditional quantities are given $\mathcal{A}_j$, and D_j is the Hessian in an additive deterministic spin field. The law of total covariance and the differentiated recursion give

$$C_j = \mathbb{E}[C_{j+1} \mid \mathcal{A}_j] + V_j, \quad D_j = \mathbb{E}[D_{j+1} \mid \mathcal{A}_j] + x_j V_j.$$

The terminal identity $D_k = C_k$, together with the nondecreasing masses, allows backward induction and gives

$$D_j \succeq x_j C_j. \quad (2.24)$$

For the next field, abbreviate $\Delta = \Delta_{j+1}$ and set $y = \sqrt{\Delta} z_{j+1}$. Conditional on $\mathcal{A}_j$, its density is proportional to $\exp\{-\|y\|^2/(2\Delta) + x_j F_{j+1}\}$. Integration by parts, using $\nabla_y F_{j+1} = m_{j+1}$, yields

$$\text{Cov}(m_{j+1}, y \mid \mathcal{A}_j) = \Delta D_j, \quad (2.25)$$

$$\text{Cov}(y \mid \mathcal{A}_j) = \Delta I + \Delta^2 x_j D_j. \quad (2.26)$$

For the first identity, use $\mathbb{E}[y \mid \mathcal{A}_j] = \Delta x_j m_j$ and subtract the product of means from the integration-by-parts expression for $\mathbb{E}[m_{j+1} y^\top \mid \mathcal{A}_j]$. A further application to $yy^\top$ proves the second identity. The joint covariance matrix of (m_{j+1}, y) is positive semidefinite, so its Schur complement gives

$$V_j \succeq \Delta D_j (I + \Delta x_j D_j)^{-1} D_j. \quad (2.27)$$

Order the eigenvalues of C_j, D_j, V_j decreasingly and denote them by c_i, d_i, v_i , respectively. Equation (2.24) implies $d_i \geq x_j c_i$. The function $d \mapsto \Delta d^2/(1 + \Delta x_j d)$ is increasing on $d \geq 0$. Apply the min–max principle to Equation (2.27) to get

$$v_i \geq \frac{\Delta d_i^2}{1 + \Delta x_j d_i} \geq \frac{\Delta x_j^2 c_i^2}{1 + \Delta x_j^2 c_i}.$$

At or above the threshold $c_i \geq (\Delta x_j^2)^{-1}$, the final expression is at least $c_i/2$. Since also $c_i \leq \text{tr } C_j \leq n$, these eigenvalues contribute at most $2n \text{tr } V_j$ to $\sum_i c_i^2$. Those below the threshold contribute at most $n/(\Delta x_j^2)$. Together the two bounds give

$$\frac{\text{tr } C_j^2}{n^2} \leq \frac{2 \text{tr } V_j}{n} + \frac{1}{n \Delta_{j+1} x_j^2}. \quad (2.28)$$

Martingale orthogonality again supplies the summable quantity: $\sum_{j < k} \mathbb{E} \text{tr } V_j \leq n$. Sum over j , using $x_j = (j+1)/(k+1)$, to obtain

$$B_0 := \sum_{j < k} w_j \mathbb{E} \frac{\text{tr } C_j^2}{n^2} \leq \frac{C}{k} + \frac{Ck}{n \Delta_{\min}}. \quad (2.29)$$

For two independent descendants of the level- j prefix, the conditional overlap has mean M_j and variance

$$\frac{1}{n^2} \left(\text{tr } C_j^2 + 2m_j^\top C_j m_j \right).$$

The contribution of the second term is at most $2\sqrt{\text{tr } C_j^2/n^2}$: use $\|m_j\|^2 \leq n$ and $\|C_j\|_{\text{op}} \leq \sqrt{\text{tr } C_j^2}$. Thus the averaged conditional variance is at most $B_0 + 2\sqrt{B_0}$. Under $k/\sqrt{n\Delta_{\min}} \leq 1$, Equation (2.29) gives $B_0 \leq C/k$. Combine this estimate with the conditional-mean bound (2.23) through the law of total variance. Finally, the level k costs at most $4/(k+1)$ because $|R - r_k| \leq 2$. These bounds prove Equation (2.20). $\square$

2.5 Adaptive interpolation and the finite-size error

Proof of Theorem 2.4. The upper comparison was proved in Equation (2.14). To obtain the lower comparison, we choose the field levels so that the pressure derivative separates into a quadratic expression in the overlap means and the overlap error controlled by Lemma 2.6. After constructing that path, we will integrate the derivative and identify its single-site endpoint with a scalar trial measure.

Choosing the adaptive path. Consider provisionally a differentiable path $q(t)$ with positive gaps, and set $f(t) = n^{-1} \mathbb{E} F_0(t, q(t))$. For a field gap in overlap units, $a_j = q_j - q_{j-1}$, the hierarchy identity gives

$$\frac{\partial}{\partial a_j} \frac{1}{n} \mathbb{E} F_0 = \frac{\beta^2}{2} \left(1 - \sum_{l=j}^k w_l r_l \right). \quad (2.30)$$

The Hamiltonian contribution to the derivative is $\frac{\beta^2}{4} (1 - \sum_l w_l \mathbb{E}_l^{(2)} R^2)$. Summing the field contributions and using $\sum_{j=0}^l a'_j = q'_l$ therefore gives

$$f'(t) = \frac{\beta^2}{4} \left(1 - \sum_{l=0}^k w_l \mathbb{E}_l^{(2)} R^2 \right) + \frac{\beta^2}{2} \left(q'_k - \sum_{l=0}^k w_l r_l q'_l \right).$$

This calculation suggests the system

$$q'_j(t) = -r_j(t, q(t)). \quad (2.31)$$

Indeed, since $\mathbb{E}_j^{(2)} R = r_j$, every positive-gap solution satisfies

$$f'(t) = \frac{\beta^2}{4} \left[1 - 2r_k + \sum_{j=0}^k w_j r_j^2 - \sum_{j=0}^k w_j \mathbb{E}_j^{(2)} (R - r_j)^2 \right]. \quad (2.32)$$

The final term is exactly the error already estimated; the remaining terms can be bounded after integration using the differential equation. We now construct a solution with gaps large enough to apply that estimate. Choose

$$k = \lfloor n^{1/12} \rfloor, \quad \eta = n^{-1/6}, \quad q_j(1) = (j+1)\eta \quad (0 \leq j \leq k),$$

and solve Equation (2.31) backwards from $t = 1$.

Existence and bounds for the adaptive path. We verify that this system has a solution throughout $[0, 1]$ and keeps every gap positive. On the open region of positive gaps, the finite Gaussian

recursions make r_j continuous in t and locally Lipschitz in q . The following domination justifies those assertions. Fix a compact subset of this region; in this local argument constants may depend on n, k, β and that subset. Uniformly in the prefix and in $t \in [0, 1]$, we have $\|\nabla_{z_h} F_h\| = \sqrt{\Delta_h} \|m_h\| \leq L$. Subtract the value of F_h at $z_h = 0$ from both numerator and denominator in the normalized tilt. This gives

$$W_h(z_h) \leq C \exp(L\|z_h\|).$$

The conditional first moments of subsequent fields are consequently uniformly bounded. The derivative $\partial_{q_i} U$ is bounded by a constant times $\|z_i\| + \|z_{i+1}\|$, with the second term omitted at $i = k$. Differentiation through the recursions therefore yields

$$|\partial_{q_i} F_l| \leq C \left(1 + \sum_{h \leq l} \|z_h\| \right).$$

To apply this to r_j , represent it as the expectation of R on a two-path tree. Since terminal Gibbs probabilities are at most one, the preceding estimates dominate its density and the q_i derivative of that density by $C(1 + S)e^{LS}$, where S is the sum of the norms of all fields in this finite tree. The bound is Gaussian integrable, uniformly in t and in the chosen compact set of q values. It justifies the differentiated recursions and gives the required local Lipschitz property. At $t = 0$, continuity follows from the same bound and continuity in $\sqrt{t}$; no derivative of $\sqrt{t}$ is needed.

Along the local solution, $r_j \geq r_{j-1}$ implies that each gap $q_j - q_{j-1}$ is nondecreasing as time runs backwards, with $r_{-1} = 0$. Each gap therefore stays at least η . The bound $0 \leq r_j \leq 1$ also prevents escape to infinity and gives

$$q_k(t) \leq (k+1)\eta + 1 - t, \quad \Delta_j(t) \geq \beta^2 \eta. \quad (2.33)$$

These estimates continue the local solution across the full interval. They also verify the field-gap hypothesis of Lemma 2.6. Thus, for all sufficiently large n , uniformly along the path,

$$\sum_{j=0}^k w_j \mathbb{E}_j^{(2)} (R - r_j)^2 \leq C_\beta n^{-1/24}. \quad (2.34)$$

Indeed, the two terms in that lemma are controlled by $k/\sqrt{n\beta^2\eta} = O_\beta(n^{-1/3})$ and $k^{-1/2} = O(n^{-1/24})$.

Integration of the pressure derivative. The adaptive path has now been constructed and its overlap error is bounded, so we can integrate Equation (2.32). Write $\delta = (k+1)\eta = O(n^{-1/12})$ and $Q_j = q_j(0)$. The derivative in Equation (2.30) lies between 0 and $\beta^2/2$. Consequently, removing the fields present at $t = 1$ costs at most $\beta^2\delta/2$, and

$$f(1) \leq p_n(\beta) + \frac{\beta^2}{2} \delta.$$

The differential equation and Jensen's inequality also give

$$\int_0^1 r_j(t) dt = Q_j - q_j(1), \quad \int_0^1 r_j(t)^2 dt \geq (Q_j - q_j(1))^2.$$

The endpoint bounds are $0 \leq Q_j \leq 1 + \delta$ and $0 \leq q_j(1) \leq \delta$. Integrate Equation (2.32), apply these bounds and the overlap error in Equation (2.34), and conclude that

$$p_n(\beta) \geq f(0) + \frac{\beta^2}{4} \left(1 - 2Q_k + \sum_{j=0}^k w_j Q_j^2 \right) - C_\beta n^{-1/24}. \quad (2.35)$$

Identification of a scalar trial measure. At $t = 0$ there is no Hamiltonian coupling, so the recursion factors over sites. Thus $f(0)$ is the scalar single-site value at levels $Q_0, \dots, Q_k$. The only possible obstruction to using these levels as the support of a trial measure on $[0, 1]$ is that some may exceed one.

Replace each such level by $\bar{Q}_j = \min(Q_j, 1)$. Each gap can only decrease, and their total decrease is $Q_k - \bar{Q}_k \leq \delta$. Equation (2.30) bounds the change in the scalar value by $\beta^2 \delta / 2$. The quadratic expression in Equation (2.35) changes by at most $C_\beta \delta$ as well. We may therefore take $Q_k \leq 1$ within the stated error, interpreting zero gaps by continuity.

Now define $\mu = \sum_{j=0}^k w_j \delta_{Q_j}$. Its distribution function is zero up to Q_0 , equals x_{j-1} on each gap (Q_{j-1}, Q_j) , and equals one between Q_k and 1. Over this final interval, the recursion maps $\log(2 \cosh z)$ to $\log(2 \cosh z) + \beta^2(1 - Q_k)/2$. Hence

$$\Phi_\alpha(0, 0) = f(0) + \frac{\beta^2}{2}(1 - Q_k).$$

The penalty is identified by summation by parts:

$$\frac{\beta^2}{2} \int_0^1 s \alpha(s) \, ds = \frac{\beta^2}{4} \left(1 - \sum_{j=0}^k w_j Q_j^2 \right).$$

The expression preceding the error in Equation (2.35) is therefore exactly $\mathcal{P}_\beta(\mu)$. It is at least P_β , which completes the lower comparison and the proof. $\square$

3 Scalar regularity and properties of a minimizer

We now establish the scalar identities used in the overlap comparisons and the equilibrium energy estimate. The argument has three parts. First, we justify the scalar equation and its stochastic evolution for general distribution functions, including those with atoms. Next, we derive first variations and use them to locate the support of an arbitrary minimizing measure. Finally, the resulting stationarity identity gives a strict energy bound when $\beta > 1$. No further description of the minimizer's support is assumed.

Regularity and step approximation for the Parisi evolution, together with zero-field support properties of its minimizers, were developed by Auffinger and Chen [1, Propositions 1–2 and Theorems 1, 5]. The full first-variation characterization appears in Chen [7, Theorem 2 and Proposition 1] and, independently, in Jagannath and Tobasco [10, Proposition 1.1, Corollary 3.10 and Remark 1.3]. We prove the required statements here with the normalization and atom conventions used by the later pressure comparisons.

Throughout this section, $\alpha(s) = \mu([0, s])$ is the right-continuous distribution function of a probability measure on $[0, 1]$. Recall the scalar terminal-value equation

$$-\partial_s \Phi_\alpha = \frac{\beta^2}{2} (\partial_{zz} \Phi_\alpha + \alpha(s) (\partial_z \Phi_\alpha)^2), \quad \Phi_\alpha(1, z) = \log(2 \cosh z). \quad (3.1)$$

For a step distribution function this is the Gaussian recursion from the preceding section. For general distribution functions, the scalar values are defined by approximation, which the next lemma justifies. The time derivative in the equation is understood almost everywhere; no differentiability at a jump of α is required. We denote the first two spatial derivatives by $u = \partial_z \Phi_\alpha$ and $\chi = \partial_{zz} \Phi_\alpha$.

3.1 Regularity and first variations

Lemma 3.1 (Regularity, stability, and scalar evolution). *The step approximation defining Φ_α is independent of the approximating sequence. The functions Φ_α, u, χ are continuous on $[0, 1] \times \mathbb{R}$, with*

$$|u| \leq 1, \quad 0 < \chi \leq 1, \quad |\chi(s, z) - \chi(s, z')| \leq C_\beta |z - z'|. \quad (3.2)$$

The function $\Phi_\alpha(s, \cdot)$ is even, so $u(s, 0) = 0$. For two distribution functions $\alpha, \tilde{\alpha}$,

$$\sup_{s, z} (|\Phi_\alpha - \Phi_{\tilde{\alpha}}| + |u_\alpha - u_{\tilde{\alpha}}| + |\chi_\alpha - \chi_{\tilde{\alpha}}|) \leq C_\beta \int_0^1 |\alpha(s) - \tilde{\alpha}(s)| ds. \quad (3.3)$$

These functions are also continuous in $\beta > 0$, uniformly in (s, z) for their differences, locally in β .

The stochastic differential equation

$$dX_s = \beta dB_s + \beta^2 \alpha(s) u(s, X_s) ds, \quad X_0 = 0, \quad (3.4)$$

has a unique strong solution. For a step distribution function, its transitions across the constant-mass intervals are the normalized tilted Gaussian kernels of the scalar recursion. Its bounded martingale $U_s = u(s, X_s)$ satisfies

$$U_t - U_s = \beta \int_s^t \chi(v, X_v) dB_v. \quad (3.5)$$

Moreover,

$$\mathbb{E}\chi(t, X_t) - \mathbb{E}\chi(s, X_s) = -\beta^2 \int_s^t \alpha(v) \mathbb{E}\chi(v, X_v)^2 dv. \quad (3.6)$$

For $0 \leq s < t \leq 1$, the conditional form is

$$\mathbb{E}[\chi(t, X_t) \mid X_s] = \chi(s, X_s) - \beta^2 \int_s^t \alpha(v) \mathbb{E}[\chi(v, X_v)^2 \mid X_s] dv. \quad (3.7)$$

Consequently $\Gamma(s) := \mathbb{E}u(s, X_s)^2$ is continuously differentiable, with one-sided derivatives at the endpoints, and

$$\Gamma(0) = 0, \quad \Gamma'(s) = \beta^2 \mathbb{E}\chi(s, X_s)^2. \quad (3.8)$$

The well-posedness statement and the evolution identities (3.5)–(3.6) also hold for a diffusion started at any deterministic time and spatial value.

Proof. We prove estimates uniform over step distribution functions, use them to pass to general α , and then establish continuity in the temperature parameter.

Derivative bounds along a finite step path. Suppose first that α is a step function. On an interval of constant mass m , its recursion is

$$T_m f(z) = m^{-1} \log \mathbb{E} e^{mf(z + \beta\sqrt{\ell} Z)}$$

where ℓ is the interval length and $T_0 f = \mathbb{E}f(z + \beta\sqrt{\ell} Z)$. To keep track of the successive tilts, fix (s, z) and list the remaining breakpoints as $s = t_0 < \dots < t_r = 1$. Let m_j be the mass on $[t_j, t_{j+1})$ and set $f_j = \Phi_\alpha(t_j, \cdot)$. Starting with $Z_0 = z$, sample Z_{j+1} from

$$K_j(v, dy) = \exp\{m_j[f_{j+1}(y) - f_j(v)]\} N(v, \beta^2(t_{j+1} - t_j))(dy),$$

where the exponential equals one at $m_j = 0$. Each kernel is normalized by the scalar recursion on its own interval. Put $\mathcal{F}_j = \sigma(Z_0, \dots, Z_j)$ and $M_j = f'_j(Z_j)$. Differentiating the recursion once and twice gives

$$\begin{aligned}\mathbb{E}[M_{j+1} \mid \mathcal{F}_j] &= M_j, \\ f''_j(Z_j) &= \mathbb{E}[f''_{j+1}(Z_{j+1}) \mid \mathcal{F}_j] + m_j \mathbb{E}[(M_{j+1} - M_j)^2 \mid \mathcal{F}_j].\end{aligned}$$

Thus M is a martingale for the concatenated kernels, even though the tilts on successive intervals can differ. Its terminal value is $M_r = \tanh Z_r$. Iterating the second identity yields

$$\chi(s, z) = \mathbb{E} \operatorname{sech}^2(Z_r) + \sum_{j=0}^{r-1} m_j \mathbb{E}(M_{j+1} - M_j)^2.$$

The increments are orthogonal, so their unweighted squared sum has expectation $\operatorname{Var}(\tanh Z_r)$. Since $0 \leq m_j \leq 1$, we obtain

$$0 < \mathbb{E}[1 - \tanh^2(Z_r)] \leq \chi(s, z) \leq \mathbb{E}[1 - \tanh^2(Z_r)] + \operatorname{Var}(\tanh Z_r) = 1 - u(s, z)^2 \leq 1.$$

Every recursion preserves evenness. Its gradient is a conditional mean of the terminal gradient, which gives $|u| \leq 1$.

Before differentiating the equation, we record why all required spatial derivatives exist and are bounded for a fixed finite path. The k th derivative of one logarithmic Gaussian integral is a polynomial in tilted moments of $f', \dots, f^{(k)}$, with coefficients bounded for $0 \leq m \leq 1$. At $m = 0$, differentiation passes directly through Gaussian expectation. All positive-order derivatives of $\log(2 \cosh z)$ are bounded, so finite induction gives bounded derivatives at the breakpoints and within each interval. The scalar values have at most linear growth, ensuring Gaussian integrability at every step. These preliminary bounds may depend on the path; we next obtain the uniform bound needed for approximation.

Within a constant-mass interval the recursion is smooth and solves Equation (3.1). Its first two spatial derivatives satisfy

$$\begin{aligned}-u_s &= \frac{\beta^2}{2} u_{zz} + \beta^2 \alpha u u_z, \\ -\chi_s &= \frac{\beta^2}{2} \chi_{zz} + \beta^2 \alpha u \chi_z + \beta^2 \alpha \chi^2.\end{aligned}$$

Set $w = \chi_z$. One more differentiation gives a linear equation:

$$-w_s = \frac{\beta^2}{2} w_{zz} + \beta^2 \alpha u w_z + 3\beta^2 \alpha \chi w.$$

The bounds already proved for u and χ control this equation's drift and potential uniformly over all step paths. Its terminal value has absolute value at most 2. Concatenate the diffusion representations over the finitely many intervals to obtain

$$\sup_{s, z} |w(s, z)| \leq 2e^{3\beta^2}.$$

These differentiations may all be performed first inside the finite Gaussian recursions. Their bounded derivatives justify passage under the integrals, so the uniform estimate does not presuppose any regularity of the limiting path.

The scalar diffusion and its evolution identities. We next identify the tilted kernels with the diffusion in Equation (3.4). On a constant-mass interval $[a, b]$, start Brownian motion at z by setting $Y_s = z + \beta(B_s - B_a)$. Under this law, Itô's formula identifies

$$\exp\{m(\Phi(b, Y_b) - \Phi(a, z))\}$$

as the exponential martingale with stochastic logarithm $m\beta \int_a^b u(v, Y_v) dB_v$. Its bounded integrand justifies the change of measure, which adds drift $\beta^2 mu$. Under that drift, a further application of Itô's formula gives

$$du(s, X_s) = \beta\chi(s, X_s) dB_s, \quad d\chi(s, X_s) = \beta\chi_z(s, X_s) dB_s - \beta^2\alpha(s)\chi(s, X_s)^2 ds.$$

This establishes both evolution identities for step paths.

Stability and passage to general distribution functions. The remaining task is to make these constructions independent of the step approximation. Subtract the equations for two step paths, keeping the same diffusion coefficient. The difference of the scalar values has drift $\frac{\beta^2}{2}\alpha(u + \tilde{u})$ and source $\frac{\beta^2}{2}(\alpha - \tilde{\alpha})\tilde{u}^2$. The equation for $u - \tilde{u}$ has bounded drift and potential, with source bounded by $C_\beta|\alpha - \tilde{\alpha}|$. For $\chi - \tilde{\chi}$, the drift and potential remain bounded, and the source is bounded by

$$C_\beta(|\alpha - \tilde{\alpha}|(s) + \sup_z |u(s, z) - \tilde{u}(s, z)|).$$

To obtain the last bound, expand $\alpha u\chi_z - \tilde{\alpha}\tilde{u}\tilde{\chi}_z$ and use the uniform bound on $\tilde{\chi}_z$. In the remaining reaction term, factor the difference of squares. Each difference has zero terminal value, so the diffusion representation of these linear equations proves Equation (3.3), first for Φ , then for u , and finally for χ .

Approximate an arbitrary distribution function in L^1 by step distribution functions. The stability estimates make Φ, u, χ converge uniformly, with uniform convergence of differences for the unbounded scalar value Φ . The fundamental theorem of calculus in z identifies the limits of u and χ with the first two spatial derivatives of the limiting Φ . Continuity, symmetry, and all non-strict bounds follow by passage to the limit. Passing to the limit in the time-integrated step equation also gives Equation (3.1) almost everywhere in time.

The limiting drift in Equation (3.4) is measurable in time, bounded, and uniformly Lipschitz in space. Hence the equation has a strong solution and pathwise uniqueness. Couple the solutions for two step approximations with the same Brownian motion; Gronwall's inequality gives convergence uniformly in time. We may therefore also pass the terminal lower bound for χ to the limit:

$$\chi(s, z) \geq \mathbb{E}_{s,z} \text{sech}^2(X_1) > 0.$$

Since X_1 is finite almost surely, this lower bound is strictly positive. Itô isometry passes the martingale identity to the limit, and bounded convergence does the same for the expectation identity. Nothing in these arguments requires the starting point to be $(0, 0)$; they apply at every deterministic starting time and spatial value. Applying the expectation identity to a diffusion started at (s, z) and then using the Markov property gives Equation (3.7). Finally, Equation (3.5) and the continuity of χ and X give Equation (3.8).

Continuity in the temperature parameter. Write $b = \beta^2$ and repeat the subtraction argument with different values of b . In the equation for Φ , the additional source is $(\tilde{b} - b)(\tilde{\chi} + \alpha\tilde{u}^2)/2$, bounded by $C|\tilde{b} - b|$. The bound on $\tilde{\chi}_z$ gives the same source bound for u . Thus the differences of Φ and u

are $O(|\tilde{b} - b|)$ locally uniformly for $b > 0$. To obtain continuity of χ , use its uniform spatial Lipschitz bound and approximate it by a difference quotient of u . For any $h > 0$,

$$\sup_{s,z} |\chi - \tilde{\chi}| \leq \frac{2}{h} \sup_{s,z} |u - \tilde{u}| + Ch.$$

Choose h small first, then take $\tilde{b}$ sufficiently close to b . This proves the required uniform continuity of χ . The step approximation transfers these parameter estimates to all distribution functions. $\square$

We next differentiate the scalar functional in two directions: a change of its distribution function and a change of temperature with the distribution function fixed. We prove both formulas by differences, so they require no differentiability of a minimizing measure. For another distribution function $\tilde{\alpha}$, set $\alpha_\theta = \alpha + \theta(\tilde{\alpha} - \alpha)$. The first variation is

$$\left. \frac{d}{d\theta} \mathcal{P}_\beta(\alpha_\theta) \right|_{\theta=0+} = \frac{\beta^2}{2} \int_0^1 (\tilde{\alpha} - \alpha)(s)(\Gamma(s) - s) ds. \quad (3.9)$$

We use $\mathcal{P}_\beta(\alpha)$ to denote the functional of the measure having distribution function α . With α held fixed, the temperature derivative of the scalar value is

$$\frac{\partial}{\partial(\beta^2)} \Phi_\alpha(0,0) = \frac{1}{2} \int_0^1 \mathbb{E}[\chi(s, X_s) + \alpha(s)u(s, X_s)^2] ds. \quad (3.10)$$

For each formula, start with step paths, where the equations are classical between finitely many breakpoints. Subtraction gives an integral representation, and that representation passes to general paths by the stability lemma.

Variation of the distribution function. Subtract the equations for Φ_{α_θ} and Φ_α . The resulting drift is $\beta^2 \alpha(u_{\alpha_\theta} + u_\alpha)/2$, and the source is $\theta \beta^2 (\tilde{\alpha} - \alpha) u_{\alpha_\theta}^2/2$. Introduce Y^θ by $Y_0^\theta = 0$ and

$$dY_s^\theta = \beta dB_s + \frac{\beta^2}{2} \alpha(s)(u_{\alpha_\theta} + u_\alpha)(s, Y_s^\theta) ds.$$

The diffusion representation of the difference equation is the exact identity

$$\frac{\Phi_{\alpha_\theta}(0,0) - \Phi_\alpha(0,0)}{\theta} = \frac{\beta^2}{2} \mathbb{E} \int_0^1 (\tilde{\alpha} - \alpha)(s) u_{\alpha_\theta}(s, Y_s^\theta)^2 ds. \quad (3.11)$$

For general $\alpha, \tilde{\alpha}$, choose step approximations converging to each in L^1 , and take their convex combinations at a fixed $\theta > 0$. Equation (3.3) shows that the mixed drifts converge in integrated spatial supremum norm. They also have a common spatial Lipschitz bound. A coupling with the same Brownian motion and Gronwall's inequality therefore gives uniform convergence of the corresponding Y^θ processes. The sources converge in integrated expectation by $|u| \leq 1$, the spatial Lipschitz bound for u , its uniform convergence, and the L^1 convergence of $\tilde{\alpha} - \alpha$. Hence Equation (3.11) holds for arbitrary distribution functions.

We may now send $\theta \downarrow 0$. The same estimates give $Y^\theta \rightarrow X$ and uniform convergence of u_{α_θ} to u_α . The right-hand side converges to $\frac{\beta^2}{2} \int (\tilde{\alpha} - \alpha) \Gamma$. Subtract the derivative of the linear penalty to obtain Equation (3.9).

Variation of the temperature. Set $b = \beta^2$ and use a superscript b for scalar functions with parameter $\sqrt{b}$. For $\tilde{b} \neq b$, let $Y_0^{b,\tilde{b}} = 0$ and define its evolution by

$$dY_s^{b,\tilde{b}} = \sqrt{b} dB_s + \frac{b}{2} \alpha(s)(u^b + u^{\tilde{b}})(s, Y_s^{b,\tilde{b}}) ds.$$

Subtract the two step-path equations with diffusion coefficient $b/2$ to obtain

$$\frac{\Phi^{\tilde{b}}(0,0) - \Phi^b(0,0)}{\tilde{b} - b} = \frac{1}{2} \mathbb{E} \int_0^1 [\chi^{\tilde{b}} + \alpha(u^{\tilde{b}})^2](s, Y_s^{b,\tilde{b}}) ds. \quad (3.12)$$

For fixed $b, \tilde{b}$, pass to general α by the same step approximation and Brownian coupling used above. The additional term involving χ is controlled by its uniform spatial Lipschitz bound. Next let $\tilde{b} \rightarrow b$. Parameter continuity in Lemma 3.1 and the coupling give $Y^{b,\tilde{b}} \rightarrow X$. The integrands remain bounded and converge in integrated expectation. The limit in Equation (3.12) therefore proves Equation (3.10) for approach from either side.

3.2 Minimizers and the energy gap

The mass variation converts minimization into a condition on the support. We will use that condition both to identify the expected scalar susceptibility at every time and to show that the minimizing measure is nontrivial when $\beta > 1$.

Proposition 3.2 (Support and stationarity identities). *The scalar functional has a minimizer μ . For every such minimizer, with its associated scalar process, the following assertions hold:*

- (i) $0 \in \text{supp } \mu$, and $\text{supp } \mu \subset [0, 1 - \varepsilon_0]$ for some $\varepsilon_0 > 0$.
- (ii) At every $q \in \text{supp } \mu \cap (0, 1)$, $\Gamma(q) = q$ and $\Gamma'(q) \leq 1$. At $q = 0$, $\Gamma(0) = 0$ and $\Gamma'(0) \leq 1$.
- (iii) Define

$$d_*(s) = 1 - \int_{(s,1]} t \mu(dt), \quad A(s) = \int_s^1 \alpha(t) dt = d_*(s) - s\alpha(s). \quad (3.13)$$

Then, at every $s \in [0, 1]$,

$$\mathbb{E}[\chi(s, X_s) + \alpha(s)u(s, X_s)^2] = d_*(s). \quad (3.14)$$

Equivalently, since $d_*(s) = A(s) + s\alpha(s)$,

$$\mathbb{E}\chi(s, X_s) = A(s) + \alpha(s)(s - \Gamma(s)) \quad (0 \leq s \leq 1). \quad (3.15)$$

In particular

$$\chi(0,0) = A(0), \quad \beta A(0) \leq 1. \quad (3.16)$$

Furthermore $\alpha(s) > 0$ for every $s \in (0, 1]$, $A(h) < A(0)$ for every $h \in (0, 1]$, and $d_*(0+) = A(0)$.

Proof. **Existence and variational support condition.** Weak convergence of probability measures on $[0, 1]$ implies pointwise convergence of their distribution functions at continuity points of the limit. Bounded convergence then gives convergence in L^1 . By Lemma 3.1, the functional $\mathcal{P}_\beta$ is continuous for this topology. The space of probability measures on $[0, 1]$ is compact, so a minimizer exists.

Fix any minimizer and define

$$G(q) = \int_q^1 (\Gamma(s) - s) ds.$$

Fubini's theorem and Equation (3.9) give $\int G d\tilde{\mu} \geq \int G d\mu$ for every probability measure $\tilde{\mu}$. Testing with point masses shows that μ is supported on the set of global minima of G . The function G is twice continuously differentiable and satisfies

$$G'(q) = q - \Gamma(q), \quad G''(q) = 1 - \Gamma'(q).$$

The first- and second-derivative tests now give the claimed identities and inequalities at interior support points. At the right endpoint, $\Gamma(1) = \mathbb{E} \tanh^2(X_1) < 1$, and hence $G'(1) > 0$. Thus 1 does not minimize G . Its minimum set is compact, so $\text{supp } \mu$ stays a positive distance below 1.

The left endpoint of the support. Let $q_0 = \min \text{supp } \mu$. We claim that $q_0 = 0$. Suppose instead that $q_0 \in (0, 1)$. Then $\alpha = 0$ on $[0, q_0)$, so $X_s = \beta B_s$ there and the equation for χ is the backward heat equation. Consequently $\chi(s, X_s)$ is a bounded martingale on $0 \leq s \leq q_0$. One can see this either from the heat semigroup or by choosing step approximations identically zero below q_0 . Its second moment is nondecreasing, as is $\Gamma'(s)$. But $\Gamma(q_0) = q_0$, $\Gamma(0) = 0$, and $\Gamma'(q_0) \leq 1$. Thus the average of Γ' on this interval is 1, whereas its maximum is at most 1. Continuity forces $\Gamma' \equiv 1$ on $[0, q_0]$.

To contradict this constancy, observe that $\chi(q_0, \cdot)$ is positive everywhere and cannot be constant: otherwise $u(q_0, \cdot)$ would have a positive constant derivative, contrary to $|u| \leq 1$. The function $\chi(q_0, \cdot)$ is continuous, and X_{q_0} is a nondegenerate Gaussian with full support. Therefore $\text{Var}(\chi(q_0, X_{q_0})) > 0$, while $\chi(0, X_0)$ is deterministic. The martingale's second moment strictly increases between these endpoints, which contradicts the constancy of Γ' . Hence $q_0 = 0$. At this endpoint, G has a minimum and symmetry gives $G'(0) = 0$. The one-sided second-derivative test then yields $\Gamma'(0) \leq 1$.

Stationarity with atoms included. We have located the support and established $\Gamma(t) = t$ on it. It remains to turn these support identities into an identity at every time. Use Equation (3.6), the terminal relation $\chi(1, z) = 1 - u(1, z)^2$, and Equation (3.8) to write

$$\mathbb{E}\chi(s, X_s) = 1 - \Gamma(1) + \int_s^1 \alpha(t) \Gamma'(t) dt.$$

Stieltjes integration by parts with the right-continuous α gives the following formula, with every atom retained:

$$\int_s^1 \alpha(t) \Gamma'(t) dt = \Gamma(1) - \alpha(s) \Gamma(s) - \int_{(s, 1]} \Gamma(t) \mu(dt).$$

In particular, the integral uses $(s, 1]$, while the endpoint term uses $\alpha(s)$. Substituting $\Gamma(t) = t$ on $\text{supp } \mu$, including at zero, proves Equation (3.14). Fubini's theorem gives the expression for A in Equation (3.13). At $s = 0$ the stationarity identity becomes $\chi(0, 0) = A(0)$. Finally, $\Gamma'(0) = \beta^2 \chi(0, 0)^2 \leq 1$ proves Equation (3.16).

Because $0 \in \text{supp } \mu$, each interval $[0, s]$ with $s > 0$ has positive μ -mass. Thus $A(0) - A(h) = \int_0^h \alpha(s) ds > 0$ for every $h \in (0, 1]$. The identity $d_*(s) = A(s) + s\alpha(s)$, together with continuity of A at zero, also gives $d_*(0+) = A(0)$. $\square$

For $\beta > 1$, the susceptibility bound excludes the measure δ_0 . This makes the temperature derivative at a fixed minimizer strictly smaller than $\beta/2$. A finite-volume secant estimate transfers that strict inequality to the equilibrium energy.

Lemma 3.3 (Strict bound on the equilibrium energy). *For each fixed $\beta > 1$, there are $e_0 = e_0(\beta) < \beta/2$ and $N = N(\beta)$ such that*

$$p'_n(\beta) = \frac{1}{n} \mathbb{E} \pi_{n, \beta}^J(H_n^J) \leq e_0 \quad (n \geq N). \quad (3.17)$$

Proof. Choose a minimizing measure μ at the specified β and hold that measure fixed during the temperature variation. Combine Equations (3.10) and (3.14), and differentiate the penalty, to obtain

$$\left. \frac{d}{d\gamma} \mathcal{P}_\gamma(\mu) \right|_{\gamma=\beta} = \beta \int_0^1 (d_*(s) - s\alpha(s)) ds = \beta \int_0^1 A(s) ds = \frac{\beta}{2} \left(1 - \int_0^1 t^2 \mu(dt) \right). \quad (3.18)$$

The last equality follows from two applications of Fubini's theorem. If $\mu = \delta_0$, then $A(0) = 1$, contradicting $\beta A(0) \leq 1$ for $\beta > 1$. The second moment of μ is therefore positive. Denote the derivative in Equation (3.18) by L ; we have proved $L < \beta/2$.

Choose e_0 with $L < e_0 < \beta/2$. Then fix $\varepsilon > 0$ sufficiently small that

$$\frac{\mathcal{P}_{\beta+\varepsilon}(\mu) - \mathcal{P}_\beta(\mu)}{\varepsilon} < e_0.$$

For every n , the finite-volume pressure is differentiable and convex in β . Bound its derivative by the forward secant, use the scalar upper trial bound at $\beta + \varepsilon$, and apply Theorem 2.4 at β . This gives

$$p'_n(\beta) \leq \frac{p_n(\beta + \varepsilon) - p_n(\beta)}{\varepsilon} \leq \frac{\mathcal{P}_{\beta+\varepsilon}(\mu) - P_\beta}{\varepsilon} + \frac{C_\beta}{\varepsilon} n^{-1/24}.$$

Since $\mathcal{P}_\beta(\mu) = P_\beta$, the fixed secant slope is strictly below e_0 . Its strict margin absorbs the error once n is sufficiently large. Finally, differentiate the finite partition sum and then take Gaussian expectation to obtain the energy identity in Equation (3.17). The integrable bound $\max_{\sigma \in \{-1,1\}^n} |H_n^J(\sigma)|$ justifies this passage. The argument has varied only the temperature in the functional of one fixed measure; it requires no differentiability of P_β or of a choice of minimizer as a function of β . $\square$

4 Locking small overlaps across a large overlap

When two configurations have a positive overlap bounded away from zero, we will show that their small overlaps with a third configuration are close, including their signs. This is the fixed-scale estimate used by the crossing construction. Spin reversal will then give the absolute-overlap version needed in Appendix B.

Throughout the section, fix $\beta > 1$ and a minimizer μ of the scalar functional. We use the notation of Section 3 and write $\pi = \pi_{n,\beta}^J$. The scalar inputs are

$$\begin{aligned} 0 \leq \chi \leq 1, \quad |u| \leq 1, \quad |\chi(s, z) - \chi(s, z')| &\leq C_\beta |z - z'|, \\ \mathbb{E} \chi(s, X_s) &= A(s) + \alpha(s)(s - \Gamma(s)), \quad \mathbb{E} [\chi(s, X_s) + \alpha(s)u(s, X_s)^2] = d_*(s), \\ \beta A(0) &\leq 1, \quad A(h) < A(0) \quad (0 < h \leq 1), \quad d_*(s) \longrightarrow A(0) \quad (s \downarrow 0). \end{aligned} \quad (4.1)$$

We also use the scalar martingale evolution from Lemma 3.1 and the bounds $\alpha(s) > 0$ for $s > 0$ and $0 \leq d_*(s) \leq 1$ from Proposition 3.2.

For a nonempty Gram constraint D , write $Z_{d,n}(D)$ for the unnormalized partition sum in (2.9). Thus $F_{d,n}(D) = n^{-1} \mathbb{E} \log Z_{d,n}(D)$, and division by $Z_{n,\beta}^d$ gives the probability of D under d independent Gibbs draws.

Lemma 4.1 (Signed fixed-scale locking). *Let $e_* = 1/100$. For every fixed $h \in (0, 1)$, there is $b > 0$ such that, for all sufficiently large n ,*

$$\mathbb{E} \pi^{\otimes 3} \left\{ \begin{array}{l} R(x, y) \geq h, \quad |R(v, x)|, |R(v, y)| \leq b, \\ |R(v, x) - R(v, y)| > 2n^{-e_*} \end{array} \right\} \leq e^{-n^{9/10}}. \quad (4.2)$$

The same conclusion holds after decreasing b .

We will prove a pressure deficit for each forbidden overlap matrix and then sum probability bounds. The conversion below will also apply to the shrinking gaps in Section 7.

Lemma 4.2 (From a pressure gap to a probability bound). *Fix a replica number d . If a feasible overlap matrix D satisfies*

$$F_{d,n}(D) \leq dP_\beta - \eta, \quad \eta \geq 2dC_\beta n^{-1/24},$$

where C_β is the constant in Theorem 2.4, then, for all sufficiently large n ,

$$\mathbb{E} \frac{Z_{d,n}(D)}{Z_{n,\beta}^d} \leq \exp(-n\eta/4) + \exp(-c_{\beta,d}n\eta^2). \quad (4.3)$$

The constants are uniform over D and over n -dependent gaps satisfying the displayed lower bound.

Proof. For fixed d , the derivative of $\log Z_{d,n}(D)$ with respect to any J_{ij} has absolute value at most $\beta d/\sqrt{n}$. Both this function and $d \log Z_{n,\beta}$ therefore have Gaussian Lipschitz constant at most $\beta d\sqrt{(n-1)/2}$. Their difference

$$T_D := \log Z_{d,n}(D) - d \log Z_{n,\beta}$$

has Lipschitz constant at most $C_{\beta,d}\sqrt{n}$ and satisfies $T_D \leq 0$, since e^{T_D} is a Gibbs probability. Theorem 2.4 and the pressure-gap hypothesis give $\mathbb{E}T_D \leq -n\eta/2$. Gaussian concentration bounds $\mathbb{P}\{T_D > -n\eta/4\}$ by $\exp(-c_{\beta,d}n\eta^2)$. On its complement $e^{T_D} \leq e^{-n\eta/4}$, and on the event itself $e^{T_D} \leq 1$. Summing proves the claim. This calculation uses the original off-diagonal disorder, so no auxiliary diagonal Gaussian appears in the probability bound. $\square$

Only $d = 2, 3$ will be needed. There are at most $(n+1)^{d(d-1)/2}$ feasible overlap matrices, so a union over all constraints costs only a polynomial factor.

4.1 A pair constraint

The three-replica comparison will use a large overlap q for which $\Gamma(q)$ is close to q . We first show that the remaining pair overlaps already have an exponentially small Gibbs probability.

Lemma 4.3 (Pair pressure bound). *For every nonempty pair constraint with overlap $q \in [0, 1]$,*

$$F_{2,n}\left(\begin{smallmatrix} 1 & q \\ q & 1 \end{smallmatrix}\right) \leq 2P_\beta - \frac{1}{2}(\Gamma(q) - q)^2. \quad (4.4)$$

Consequently, for every fixed $\tau > 0$, the averaged Gibbs probability of pairs satisfying $|\Gamma(|R(x, y)|) - |R(x, y)|| > \tau$ is at most $C \exp(-cn)$ for all large n .

Proof. Use Proposition 2.3 with two coordinates sharing their covariance increments up to scalar time q and receiving independent increments thereafter. Give the common increments masses $\alpha(s)/2$ and the independent increments masses $\alpha(s)$. The endpoint is the prescribed Gram matrix.

At zero terminal multiplier this path has recursive value $2\Phi(0, 0)$. Indeed, integrating the independent portion first leaves $\Phi(q, z_x) + \Phi(q, z_y)$. On the earlier common portion $z_x = z_y$; integration of twice a scalar value at mass $\alpha/2$ is twice the scalar integration at mass α . The penalty has the same factor two: before the split the squared norm is $4s^2$ with mass $\alpha/2$, and after the split its varying part is $2s^2$ with mass α .

Now add $LS_x s_y$ to the terminal exponent and denote the recursive value by $\Psi(L)$. In the matrix convention this sets $\lambda_{xy} = \lambda_{yx} = L/2$, so $\lambda : D = Lq$. Differentiating at $L = 0$ transports the

terminal observable $s_x s_y$ backwards by conditional expectation under the unperturbed tilted path. Given the common field at q , the two later branches are independent and each has conditional spin mean $u(q, X_q)$. Hence

$$\Psi'(0) = \mathbb{E}u(q, X_q)^2 = \Gamma(q).$$

We also need the bound $\Psi''(L) \leq 1$ for every L . For a terminal observable $O \in [-1, 1]$, the terminal Hessian is its thermal variance. Backward integration at mass $m \in [0, 1]$ averages the existing Hessian and adds m times the conditional variance of the gradient. Iterating this formula and using the total-variance decomposition bounds the result by $\text{Var}(O) \leq 1$ under the tilted path law. An outer quenched expectation preserves the bound. Taylor's inequality therefore gives

$$F_{2,n}(q) \leq 2P_\beta + L(\Gamma(q) - q) + \frac{L^2}{2}.$$

The choice $L = q - \Gamma(q)$ yields (4.4).

These computations are first made with step approximations of α whose segment endpoints include q . Lemma 3.1 gives convergence of the scalar values, derivatives, and tilted transitions, so the inequality passes to the minimizing α .

Finally, reversing one spin configuration preserves its Gibbs weight and changes the sign of its overlap. The same pressure estimate thus applies with the absolute overlap in place of q . There are at most $n + 1$ pair overlaps; summing (4.3) with the fixed deficit $\tau^2/2$ proves the probability assertion. $\square$

4.2 A three-coordinate path with an asymmetric perturbation

Fix $q \in [h, 1]$ and a nonempty constraint

$$D = \begin{pmatrix} 1 & r + \delta & r - \delta \\ r + \delta & 1 & q \\ r - \delta & q & 1 \end{pmatrix}, \quad r \geq 0. \quad (4.5)$$

We will choose upper bounds for r and $|\delta|$ depending only on β, h . Write D_0 for the same matrix with $\delta = 0$. Starting from a hierarchy ending at D_0 , we will add opposite field increments to the x and y coordinates. This realizes the desired first-order change in their overlaps with v . Keeping the increments positive semidefinite also adds a second-order covariance term. The free-endpoint matrix bound charges only its squared size, so that correction has a fourth-order cost.

The reason to perturb an early increment is the strict inequality $\beta^2 A(h)A(0) < 1$, supplied by (4.1). At a small branching time r , the two scalar branches are nearly independent. Their susceptibility product is then bounded above by $A(h)A(0)$, up to small errors. The Gaussian variance part of the perturbation will cancel the corresponding penalty up to the pair deviation $q - \Gamma(q)$; this strict product inequality controls what remains.

The base hierarchy itself is valid for every $0 \leq r \leq q \leq 1$; only the later perturbation will require r to be small. We use the three-replica construction in Talagrand's discussion following [18, Proposition 4.4], with masses divided by the size of each common block. Its three stages, in local scalar time s , are:

- (i) From 0 to r , all three coordinates share each increment, integrated at mass $\alpha(s)/3$.
- (ii) From r to q , the coordinate v follows an independent branch at mass $\alpha(s)$, while x, y share a branch at mass $\alpha(s)/2$.
- (iii) From q to 1, all three coordinates have independent increments at mass $\alpha(s)$.

In the middle stage each branch has its own clock. If these clocks are a for v and b for the common x, y branch, the current covariance is

$$Q(a, b) = \begin{pmatrix} a & r & r \\ r & b & b \\ r & b & b \end{pmatrix}, \quad r \leq a, b \leq q.$$

Thus one branch can advance while the other is held fixed. For a step function α , merge the two finite lists of increments in nondecreasing order of mass, retaining the order within each list. Each list is nondecreasing, so such a merge exists; ties may be ordered in either way consistent with the two local orders. The first-stage masses do not exceed either middle starting mass, and the middle masses do not exceed the starting mass of the last stage. Thus the entire schedule is a valid nondecreasing hierarchy. Insert a dummy initial covariance $Q_0 = 0$ to ensure that, even when $\alpha(0) > 0$, the first genuine field increment has its prescribed positive mass and the quenched initial level has zero covariance.

The endpoint is D_0 , and the recursive value minus the penalty equals $3\mathcal{P}_\beta(\mu)$. To check both assertions about the value, observe that a common block of size ℓ has scalar value $\ell\Phi$. Its mass α/ℓ gives the ordinary scalar tilted transition, and its squared-norm differential $\ell^2 d(s^2)$ contributes ℓ times the scalar penalty after multiplication by that mass. Independent branches have additive values. When one branch is integrated, the value of a frozen branch is independent of that increment and passes unchanged through the logarithm. Applying these facts at each split proves the claimed value and penalty for every allowed interleaving. It also identifies the base tilted law. Denote the v path and the shared x, y path by X^v and X^x , and their common field at the first split by X_r . Write $K_{r,z}$ for the scalar path law started at (r, z) and run to time q . If $\mathcal{X}_0$ denotes the common prefix through r , then

$$\mathcal{L}(X_{[r,q]}^v, X_{[r,q]}^x \mid \mathcal{X}_0) = K_{r,X_r} \otimes K_{r,X_r}. \quad (4.6)$$

Indeed, the frozen branch cancels from each normalized kernel, while the factor three or two in the common-block value cancels the divided mass. The two future branches are therefore independent conditional on X_r , for every allowed merge. We will use this same base hierarchy in Section 7.

For the fixed-scale perturbation, now fix $0 < s_1 < s_2 < h$ and suppose $r < s_1$. Define

$$k(s) = \frac{\mathbf{1}_{[s_1, s_2]}(s)}{s_2 - s_1}, \quad g(s) = \int_0^s k(t) dt, \quad G(s) = \int_0^s k(t)^2 dt, \quad K = G(s_2) = \frac{1}{s_2 - s_1}.$$

Let $L_{vx} = L_{xv} = 1$, $L_{vy} = L_{yv} = -1$, with all other entries of L zero. Let L_2 vanish outside its (x, y) block and give that block the value $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Then

$$|L|^2 = |L_2|^2 = 4, \quad L : L_2 = 0.$$

During each v -increment with local clock in $[s_1, s_2]$, add δk times that increment to x and its negative to y . At the covariance level, this replaces $ds e_v e_v^\top$ by

$$ds (e_v + \delta k(e_x - e_y))(e_v + \delta k(e_x - e_y))^\top.$$

This rank-one form is positive semidefinite. The accumulated change is $\delta g L + \delta^2 G L_2$ until the modification ends, and is then held fixed. The new endpoint is consequently

$$\tilde{D} = D_0 + \delta L + \delta^2 K L_2 = D + \delta^2 K L_2.$$

Apply Proposition 2.3 with zero terminal multiplier and this endpoint. Its mismatch contribution is

$$\frac{\beta^2}{4}|D - \tilde{D}|^2 = \beta^2\delta^4K^2. \quad (4.7)$$

We next isolate the beneficial increase in the penalty. Along the base path, $Q : L = 0$. Also $Q : L_2 = 0$ until the split at q , by which time $g = 1$ and $G = K$. In the final stage, $Q : L_2 = 2(s - q)$. Thus

$$|\tilde{Q}|^2 - |Q|^2 = 4\delta^2g^2 + 2\delta^2G(Q : L_2) + 4\delta^4G^2.$$

The last term has nonnegative increments. The first changes only during the v -increments, whose masses are $\alpha(s)$; the middle term changes only in the independent final stage. The penalty increase is therefore at least

$$\beta^2\delta^2 \left(\int_0^1 \alpha(s) d(g(s)^2) + KA(q) \right). \quad (4.8)$$

4.3 The recursive value under the perturbation

To compare this penalty gain with the recursive value, first integrate out the unchanged independent final stage. Let z be the common base field of x, y at the end of the middle stage, and let Y be the sum of the raw v -increments weighted by k . The boundary value changes by

$$\Delta = \Phi(q, z + \delta Y) + \Phi(q, z - \delta Y) - 2\Phi(q, z). \quad (4.9)$$

Convexity, the bound $\chi \leq 1$, and its spatial Lipschitz bound yield

$$0 \leq \Delta \leq \delta^2Y^2, \quad |\Delta - \delta^2\chi(q, z)Y^2| \leq C_\beta|\delta Y|^3. \quad (4.10)$$

The remaining backward integrations are nonlinear, so a bound on the boundary change must be propagated through their normalized kernels. If the future value changes by U , an integration at positive mass m changes by

$$\frac{1}{m} \log \mathbb{E}_{\text{base}}[e^{mU} \mid \text{current prefix}],$$

where the expectation uses the unperturbed tilted kernel. At zero mass, the change is its conditional expectation of U . For $0 < m \leq 1$, Jensen's inequality and the power-mean inequality put the displayed expression between $\mathbb{E}_{\text{base}}[U \mid \text{prefix}]$ and $\log \mathbb{E}_{\text{base}}[e^U \mid \text{prefix}]$.

Here is the resulting induction explicitly. Let K_i be the normalized base kernel at level i , and U_i the difference of recursive values at that prefix. Taking the ratio of the two defining integrals gives $U_i = m_i^{-1} \log K_i e^{m_i U_{i+1}}$ for $m_i > 0$; this identity uses the base kernel only. Put

$$A_i = \mathbb{E}_{\text{base}}[\Delta \mid \text{prefix } i], \quad B_i = \log \mathbb{E}_{\text{base}}[e^\Delta \mid \text{prefix } i].$$

At the boundary these equal $U_i = \Delta$. If $A_{i+1} \leq U_{i+1} \leq B_{i+1}$, the kernel inequalities and tower property give

$$U_i \geq K_i U_{i+1} \geq A_i, \quad U_i \leq \log K_i e^{U_{i+1}} \leq \log K_i e^{B_{i+1}} = B_i.$$

Jensen's inequality gives the same conclusions at zero mass. Applying this induction through the hierarchy proves

$$\mathbb{E}_{\text{base}}\Delta \leq \tilde{\Psi} - \Psi \leq \log \mathbb{E}_{\text{base}}e^\Delta. \quad (4.11)$$

Jensen's inequality also preserves these bounds under an outer quenched expectation.

Under the base tilted law, take scalar diffusions X^v, X^x that agree through time r and have independent Brownian noises thereafter. The identified path law gives

$$z = X_q^x, \quad Y = \int_{s_1}^{s_2} k(s) dX_s^v, \quad dX_s^v = \beta dB_s^v + \beta^2 \alpha(s) u(s, X_s^v) ds.$$

Thus Y is a centered Gaussian increment of variance $\beta^2 K$ plus a possibly dependent random term of absolute value at most β^2 . In particular, there are $c, C > 0$, depending only on β, s_1, s_2 , such that

$$\mathbb{E}_{\text{base}} e^{cY^2} \leq C. \quad (4.12)$$

This estimate is uniform in $q \geq h, r < s_1$, and the step approximations of α . For small enough $|\delta|$, (4.10) and this exponential moment imply

$$\mathbb{E}(e^\Delta - 1 - \Delta) \leq \frac{1}{2} \mathbb{E}[\Delta^2 e^\Delta] \leq C\delta^4.$$

Combining with (4.11) bounds the recursive change by

$$\tilde{\Psi} - \Psi \leq \delta^2 \mathbb{E}[\chi(q, X_q^x) Y^2] + C|\delta|^3. \quad (4.13)$$

At this point all matrix paths can still be taken finite. Include r, q, s_1, s_2 in their segment endpoints and choose nondecreasing step approximations converging to α in $L^1([0, 1])$. Lemma 3.1 gives convergence of the scalar values, derivatives, and diffusions. Couple the diffusions with the same Brownian noises, keeping the noises independent after r . The uniform exponential moment (4.12) then justifies passage to the limit in the expectation in (4.13). The penalty integrals converge as well: $d(g^2)/ds$ is bounded and supported on $[s_1, s_2]$, and $A(q)$ is an integral of α . We have therefore applied the matrix comparison only to finite hierarchies and passed their upper bounds to the limit. No infinite hierarchy has been differentiated. In particular, convergence at a possible atom q is unnecessary; only L^1 convergence of the masses is used. The minimizing identities will be applied after this limit.

Subtracting (4.8) from (4.13) and including the fourth-order cost (4.7) gives

$$\begin{aligned} F_{3,n}(D) &\leq 3P_\beta + \delta^2 \mathcal{B}(q, r) + C|\delta|^3, \\ \mathcal{B}(q, r) &= \mathbb{E}[\chi(q, X_q^x) Y^2] - \beta^2 K A(q) - \beta^2 \int_0^1 \alpha(s) d(g(s)^2). \end{aligned} \quad (4.14)$$

The task is now to choose the perturbation interval so that this quadratic coefficient is strictly negative.

4.4 Choosing an interval with a strict quadratic gain

The strict inequality $A(h) < A(0)$ will supply the gain. To expose it, define

$$V_t = \chi(t, X_t^v) + \alpha(t) u(t, X_t^v)^2, \quad Y_s = \int_{s_1}^s k(t) dX_t^v \quad (s \in [s_1, s_2]).$$

The martingale identity $du(s, X_s^v) = \beta \chi(s, X_s^v) dB_s^v$ and Itô's formula, conditional on the common starting field X_r , give for almost every s

$$\frac{d}{ds} \mathbb{E}[Y_s u(s, X_s^v) \mid X_r] = \beta^2 k(s) \mathbb{E}[V_s \mid X_r].$$

Since $Y_{s_1} = 0$, apply Itô's formula once more to Y_s^2 and use conditional independence of the v and x branches. This yields

$$\begin{aligned}\mathbb{E}[\chi(q, X_q^x)Y^2] &= \beta^2 K \mathbb{E}\chi(q, X_q) \\ &\quad + 2\beta^4 \int_{s_1}^{s_2} \alpha(s)k(s) \int_{s_1}^s k(t)\mathbb{E}[\chi(q, X_q^x)V_t] dt ds.\end{aligned}\tag{4.15}$$

All factors except Y are bounded, and its required moments follow from (4.12). The conditioning and integrations are therefore justified.

Only the common field at the small time r prevents the product expectation in this formula from factoring. Its effect is bounded by

$$\left| \mathbb{E}[\chi(q, X_q^x)V_t] - \mathbb{E}\chi(q, X_q) d_*(t) \right| \leq C_\beta \sqrt{r}, \quad s_1 \leq t \leq s_2, \quad h \leq q \leq 1. \tag{4.16}$$

To prove this estimate, restart the x branch at an independent copy X'_r of X_r , independent also of the whole v branch, and retain its post- r Brownian noise. The drift $\beta^2 \alpha u$ is uniformly Lipschitz in space, so the two x solutions at time q differ by at most $e^{\beta^2} |X_r - X'_r|$. Use the spatial Lipschitz bound for χ , the bound $0 \leq V_t \leq 2$, and $\mathbb{E}|X_r - X'_r| \leq C_\beta \sqrt{r}$, the latter following from the bounded drift. The replacement has the same marginal law and is independent of V_t ; its expectation therefore factors, with $\mathbb{E}V_t = d_*(t)$ by (4.1).

The two inequalities $\beta A(0) \leq 1$ and $A(h) < A(0)$ imply

$$\beta^2 A(h)A(0) < 1.$$

Choose $s_2 \in (0, h)$ and $\gamma > 0$ small enough that

$$\beta^2 A(h) \sup_{0 < t \leq s_2} d_*(t) \leq 1 - 4\gamma, \tag{4.17}$$

and set $s_1 = s_2/2$. The functions k, g and the constant K are now fixed. Also set

$$W = 2\beta^2 \int_{s_1}^{s_2} \alpha(s)k(s) \int_{s_1}^s k(t) dt ds > 0.$$

The strict positivity follows from $\alpha(s) > 0$ for every $s > 0$.

Let $\bar{\chi}(q) = \mathbb{E}\chi(q, X_q)$. For $q \geq h$ with $|q - \Gamma(q)| \leq \tau$, the all-time scalar identity gives

$$\bar{\chi}(q) = A(q) + \alpha(q)(q - \Gamma(q)) \leq A(h) + \tau.$$

First choose $\tau > 0$ small enough and then $r_0 \in (0, s_1)$ small enough. Equations (4.16) and (4.17) give

$$\beta^2 \mathbb{E}[\chi(q, X_q^x)V_t] - 1 \leq -2\gamma \tag{4.18}$$

uniformly for $q \in [h, 1]$ with $|q - \Gamma(q)| \leq \tau$, $0 \leq r \leq r_0$, and $t \in [s_1, s_2]$. More explicitly, since $0 \leq d_*(t) \leq 1$, the left side is bounded above by $-4\gamma + \beta^2 \tau + C_\beta \sqrt{r}$.

We can now compare the entire quadratic term with this strict margin. Using $d(g^2) = 2gk ds$, substitute (4.15) into (4.14). The result is the exact identity

$$\begin{aligned}\mathcal{B}(q, r) &= \beta^2 K \alpha(q)(q - \Gamma(q)) \\ &\quad + 2\beta^2 \int_{s_1}^{s_2} \alpha(s)k(s) \int_{s_1}^s k(t) \left\{ \beta^2 \mathbb{E}[\chi(q, X_q^x)V_t] - 1 \right\} dt ds.\end{aligned}\tag{4.19}$$

Decrease τ further until $\beta^2 K \tau \leq \gamma W$. The first term in (4.19) is then at most γW , and (4.18) bounds the double integral by $-2\gamma W$. Hence

$$\mathcal{B}(q, r) \leq \gamma W - 2\gamma W = -\gamma W.$$

The cubic remainder in (4.14) is uniform over these q, r . Choose $\delta_0 > 0$ small enough to absorb it. For some $c_0 > 0$ we have obtained

$$F_{3,n}(D) \leq 3P_\beta - c_0 \delta^2 \quad (4.20)$$

whenever the constraint is nonempty, $0 \leq r \leq r_0$, $|\delta| \leq \delta_0$, $q \in [h, 1]$, and $|q - \Gamma(q)| \leq \tau$. In particular, this estimate places no restriction $r \geq |\delta|$ on the two small signed overlaps.

4.5 From the pressure gap to locking

Proof of Lemma 4.1. Choose $b \leq \min(r_0, \delta_0)$. For a triple in (4.2), set $q = R(x, y) \geq h$ and reverse v if necessary to make

$$r = \frac{R(v, x) + R(v, y)}{2} \geq 0, \quad \delta = \frac{R(v, x) - R(v, y)}{2}.$$

This reversal preserves the Gibbs weight, the large overlap q , and $|\delta|$. The event gives $r, |\delta| \leq b$ and $|\delta| > n^{-e_*}$. In particular, we do not need the two small overlaps to have the same sign.

Lemma 4.3 bounds pairs with $|q - \Gamma(q)| > \tau$ by $C \exp(-cn)$ in averaged probability. Adding an independent Gibbs draw does not change that bound. For every remaining feasible triple, (4.20) gives a pressure deficit $\eta = c_0 \delta^2 \geq c_0 n^{-2e_*}$. Since $2e_* = 1/50 < 1/24$, this dominates the finite-size error in Lemma 4.2, uniformly over the constraints. Their averaged probabilities are bounded by

$$\exp(-cn^{49/50}) + \exp(-cn^{24/25}).$$

There are at most $(n+1)^3$ overlap triples and two choices of the reversal. The resulting bound is at most $C(n+1)^3 \exp(-cn^{24/25})$, which is smaller than $\exp(-n^{9/10})$ for all sufficiently large n . $\square$

Proposition 4.4 (Absolute-overlap locking). *For every fixed $h \in (0, 1)$ there is $b > 0$ such that, for all sufficiently large n ,*

$$\mathbb{E} \pi^{\otimes 3} \left\{ \begin{array}{l} |R(x, y)| \geq h, \quad |R(v, x)|, |R(v, y)| \leq b, \\ ||R(v, x)| - |R(v, y)|| > 2n^{-1/100} \end{array} \right\} \leq \exp(-n^{9/10}). \quad (4.21)$$

The same conclusion holds after decreasing b .

Proof. Take b from Lemma 4.1. Reverse y if needed to make $R(x, y) \geq h$. This preserves the Gibbs weight and all absolute overlaps in the event. The inequality

$$||R(v, x)| - |R(v, y)|| \leq |R(v, x) - R(v, y)|$$

then puts the transformed triple in the signed event. The two possible reversals multiply its probability bound by at most two. The stronger bound $C(n+1)^3 \exp(-cn^{24/25})$ obtained in the preceding proof absorbs this factor and gives the claimed exponent. $\square$

5 Energy and disorder rotation

The strict energy gap from Lemma 3.3 will force overlap coverage on both stretched-exponential and polynomial mass scales. This section supplies the two estimates behind that implication. We first bound the upper tail of the energy of a Gibbs sample, then bound the change of the partition function under a Gaussian rotation. We work with the off-diagonal Hamiltonian of Equation (1.1) throughout; the auxiliary diagonal variables used for interpolation are absent. Write $H^J = H_n^J$, $Z^J = Z_{n,\beta}^J$ and $\pi^J = \pi_{n,\beta}^J$. All disorder vectors in this section are indexed by $i < j$.

Lemma 5.1 (An upper energy tail). *There are constants $e_1 \in (0, \beta/2)$ and $c_E > 0$ such that, for all sufficiently large n ,*

$$\mathbb{P}_J(\pi^J\{H^J > ne_1\}) \leq e^{-c_E n}. \quad (5.1)$$

Proof. The proof separates concentration from the location of the mean. Lemma 3.3 gives $e_0 < \beta/2$ with $p'_n(\beta) \leq e_0$ once n is sufficiently large. Choose $e_1 \in (\max\{0, e_0\}, \beta/2)$ and put $c_E = (e_1 - e_0)^2/2$.

Sample the disorder J and subsequently a configuration $V \sim \pi^J$. Fix $v \in \{-1, 1\}^n$. Bayes' formula expresses the conditional disorder density given $V = v$ as proportional to

$$\exp\left\{-\frac{|J|^2}{2} + \beta H^J(v) - \log Z^J\right\}.$$

The log partition function is convex in the disorder. The negative log-density therefore has Hessian $I + \nabla^2 \log Z^J \succeq I$. Adding a linear tilt $sH^J(v)$, for any $s \in \mathbb{R}$, leaves that lower Hessian bound unchanged. Write $H^J(v) = h_v \cdot J$, where

$$|h_v|^2 = \frac{1}{n} \sum_{i < j} v_i^2 v_j^2 = \frac{n-1}{2}.$$

If $\varphi_v(s) = \log \mathbb{E}[e^{sH^J(v)} \mid V = v]$, differentiation under the linearly tilted conditional law and the Brascamp–Lieb bound from Lemma 2.1 give

$$\varphi_v''(s) = \text{Var}_{v,s}(H^J(v)) \leq \frac{n-1}{2}, \quad s \in \mathbb{R}.$$

Strong convexity ensures integrability under every tilt used here. Two integrations of the variance bound, subtracting the conditional mean, give

$$\mathbb{E}\left[e^{s(H^J(v) - \mathbb{E}[H^J(v) \mid V=v])} \mid V = v\right] \leq \exp\left\{\frac{s^2(n-1)}{4}\right\}. \quad (5.2)$$

To identify that conditional mean, use the gauge transformation $J_{ij} \mapsto J_{ij}v_iv_j$, which preserves the Gaussian law, preserves Z^J , and carries $H^J(v)$ to $H^J(+)$, where $+$ = $(1, \dots, 1)$. More generally, these transformations act transitively on the configurations. Thus V has a uniform marginal under the joint law, and $\mathbb{E}[H^J(v) \mid V = v]$ has the same value for every v . Its average over V is

$$\mathbb{E}\pi^J(H^J) = np'_n(\beta).$$

Differentiating the finite-volume pressure gives this identity; Gaussian integrability permits interchange with expectation. The conditional mean is therefore at most ne_0 . Chernoff's inequality in (5.2), followed by averaging, yields

$$\mathbb{E}_J \pi^J\{H^J > ne_1\} \leq \exp\left\{-\frac{n^2(e_1 - e_0)^2}{n-1}\right\} \leq e^{-2c_E n}.$$

Finally, Markov's inequality turns this averaged Gibbs bound into the disorder estimate (5.1). $\square$

Lemma 5.2 (A disorder rotation estimate). *Let J, G be independent standard Gaussian vectors and, for $\theta \in (0, \pi/2)$, set $J_\theta = \cos \theta J + \sin \theta G$. Then for every $\lambda \in \mathbb{R}$,*

$$\mathbb{E} \exp \left\{ \lambda (\log Z^{J_\theta} - \log Z^J) \right\} \leq \exp \left\{ \frac{\lambda^2 \theta^2 \beta^2 (n-1)}{4} \right\}. \quad (5.3)$$

In particular, for every $u > 0$,

$$\mathbb{P} \left(\log Z^{J_\theta} - \log Z^J \geq u \right) \leq \exp \left\{ -\frac{u^2}{\theta^2 \beta^2 (n-1)} \right\}. \quad (5.4)$$

Proof. We interpolate by rotation through independent Gaussian directions. First put $F(J) = \log Z^J$. The coordinate derivative $\partial_{J_{ij}} F = \beta n^{-1/2} \pi^J(\sigma_i \sigma_j)$ gives

$$|\nabla F(J)|^2 \leq \frac{\beta^2 (n-1)}{2} =: L_F^2.$$

For $0 \leq s \leq \theta$, let $J_s = \cos s J + \sin s G$ and $K_s = -\sin s J + \cos s G$. At a fixed s , orthogonal invariance makes J_s and K_s independent standard Gaussian vectors. Differentiation along the rotation gives

$$F(J_\theta) - F(J) = \int_0^\theta \nabla F(J_s) \cdot K_s \, ds.$$

Apply Jensen's inequality to the uniform measure on $[0, \theta]$. Inside the resulting integral, condition on J_s and integrate the independent Gaussian K_s :

$$\begin{aligned} \mathbb{E} e^{\lambda(F(J_\theta) - F(J))} &\leq \frac{1}{\theta} \int_0^\theta \mathbb{E} e^{\lambda \nabla F(J_s) \cdot K_s} \, ds \\ &\leq \frac{1}{\theta} \int_0^\theta e^{\lambda^2 \theta^2 L_F^2 / 2} \, ds = e^{\lambda^2 \theta^2 L_F^2 / 2}. \end{aligned}$$

This is (5.3). Exponential Markov and optimization over $\lambda > 0$ prove the tail bound. The argument holds for every permitted angle, so it also applies to angles depending on n . $\square$

For coverage we parametrize the rotation by the variance of its fresh disorder component. If $0 < t < 1/2$ and $J' = \sqrt{1-t} J + \sqrt{t} G$, take $\theta = \arcsin \sqrt{t}$. Since $\theta^2 \leq 2t$, the preceding lemma gives, for every $z > 0$,

$$\mathbb{P} \{ \log Z^{J'} - \log Z^J \geq z \} \leq \exp \left(-c_\beta \frac{z^2}{nt} \right), \quad c_\beta = \frac{1}{2\beta^2}. \quad (5.5)$$

This estimate is uniform in t , including rotations whose variance decreases with n .

6 Coverage at stretched-exponential scales

The energy and rotation estimates now supply references for the dynamical construction. Given a target x in a subset S of the cube, we seek a reference $z \in S$ with $|R(x, z)| \geq q_*$ among independent Gibbs samples. The required estimate has two parts: a small total Gibbs mass of targets may be excluded, and every remaining target must have a neighborhood of sufficient Gibbs mass. Both masses are measured by the original Gibbs law, without conditioning on S . Thus the neighborhood mass is the success probability of one sample from a reference bank.

We prove the estimate with both mass bounds on the scale $L = n^\kappa$. If a set has too few overlapping pairs, fresh Gaussian disorder produces a gain in its restricted partition sum. The energy upper tail limits the loss from decreasing the original disorder, leaving a net increase that the rotation estimate makes unlikely. To make this argument uniform over all subsets, we select a violating set using the original disorder before sampling the perturbation.

Proposition 6.1 (Stretched-exponential coverage). *There exists $q_* \in (0, 1)$, depending only on the fixed inverse temperature $\beta > 1$, with the following property. For each fixed $D_1 > 0$, there exists a fixed $D_3 = D_3(\beta, D_1) > 0$ and a sequence of events $\mathcal{C}_n(D_1)$, measurable with respect to the disorder, such that $\mathbb{P}_J(\mathcal{C}_n(D_1)) \rightarrow 1$ and, on $\mathcal{C}_n(D_1)$,*

$$\pi^J\left(\left\{x \in S : \pi^J\{z \in S : |R(x, z)| \geq q_*\} < e^{-D_3 L}\right\}\right) < e^{-D_1 L} \quad \text{for every } S \subset \{-1, 1\}^n. \quad (6.1)$$

Proof. We first control conditional pair mass and then pass to individual targets. Let $e_1 \in (0, \beta/2)$ and $c_E > 0$ be given by Lemma 5.1, and define

$$\mathcal{E}_n = \left\{\pi^J\{x : H^J(x) > ne_1\} \leq e^{-c_E n}\right\}.$$

Thus $\mathbb{P}_J(\mathcal{E}_n) \rightarrow 1$. Define

$$g = \frac{\beta^2}{4} - \frac{\beta e_1}{2} > 0, \quad 0 < q_* < 1, \quad \beta^2 q_* < \frac{g}{8}.$$

Fix this threshold once for both the present proposition and Appendix A; the choice precedes D_1 . Given $D_1 > 0$, choose fixed constants

$$M > \frac{8(D_1 + 1)}{g}, \quad D_2 > \frac{\beta^2 M}{2} + 1, \quad t = \frac{ML}{n}. \quad (6.2)$$

For all sufficiently large n , we have $0 < t < 1/2$; moreover, $t \rightarrow 0$ and $nt = ML \rightarrow \infty$.

Pair mass in every sufficiently large set. Our first claim is that, with disorder probability tending to one,

$$(\pi^J(\cdot | S))^{\otimes 2}\{|R(x, y)| \geq q_*\} \geq e^{-D_2 L} \quad \text{for every } S \subset \{-1, 1\}^n \text{ with } \pi^J(S) \geq e^{-D_1 L}. \quad (6.3)$$

Write $\mathcal{V}_n$ for the failure event. Order the subsets of $\{-1, 1\}^n$ deterministically and, for $J \in \mathcal{E}_n \cap \mathcal{V}_n$, select the first violating member $S(J)$. The Gibbs quantities for each fixed subset are measurable in J , so the selection is measurable. Once we condition on J , the chosen set is fixed and cannot depend on the independent disorder used later.

Fix such a disorder, write $\pi = \pi^J$, and remove the upper-energy tail from the selected set:

$$T = S \cap \{x : H^J(x) \leq ne_1\}, \quad \nu = \pi(\cdot | T).$$

Since $L = o(n)$, for all sufficiently large n ,

$$e^{-c_E n} \leq \frac{1}{2}e^{-D_1 L}, \quad \pi(T) \geq \frac{1}{2}\pi(S) \geq \frac{1}{2}e^{-D_1 L}.$$

The retained mass is at least half the original mass. Comparing the conditional pair laws gives

$$\nu^{\otimes 2}\{|R(x, y)| \geq q_*\} \leq \left(\frac{\pi(S)}{\pi(T)}\right)^2 (\pi(\cdot | S))^{\otimes 2}\{|R(x, y)| \geq q_*\} < 4e^{-D_2 L}. \quad (6.4)$$

Gain from independent disorder. Sample an independent Gaussian disorder G and set

$$W = \nu \left(e^{\beta \sqrt{t} H^G} \right), \quad J' = \sqrt{1-t} J + \sqrt{t} G.$$

The subscript G means that the original disorder and its selected set are held fixed. The off-diagonal normalization gives

$$\mathbb{E}_G H^G(x)^2 = \frac{n-1}{2}, \quad \mathbb{E}_G [H^G(x) H^G(y)] = \frac{nR(x,y)^2 - 1}{2}.$$

The Gaussian exponential moment formula consequently yields

$$\begin{aligned} \mathbb{E}_G W &= e^{\beta^2 t(n-1)/4}, \\ \frac{\mathbb{E}_G W^2}{(\mathbb{E}_G W)^2} &= \nu^{\otimes 2} \left(e^{\beta^2 t(nR(x,y)^2 - 1)/2} \right) \\ &\leq e^{\beta^2 ntq_*^2/2} + 4e^{(\beta^2 M/2 - D_2)L} \leq 2e^{\beta^2 ntq_*^2/2} \end{aligned} \quad (6.5)$$

for all sufficiently large n . Split the pair sum at $|R(x,y)| = q_*$. On the large-overlap part use (6.4), then the choice of D_2 in (6.2). Paley–Zygmund consequently yields

$$\mathbb{P}_G \left(W \geq \frac{1}{2} \mathbb{E}_G W \right) \geq \frac{1}{8} e^{-A_0}, \quad A_0 = \frac{\beta^2 ntq_*^2}{2}. \quad (6.6)$$

Concentration raises this lower probability to one half after a controlled loss in the threshold. For $f(G) = \log W$, differentiation gives

$$\left| \frac{\partial f}{\partial G_{ij}} \right| \leq \beta \sqrt{\frac{t}{n}}, \quad \|\nabla f\|_2^2 \leq \frac{\beta^2 t(n-1)}{2} =: \sigma^2.$$

Thus Gaussian concentration applies to f with Lipschitz constant at most σ . Write

$$a = \frac{\beta^2 t(n-1)}{4} - \log 2, \quad \bar{f} = \mathbb{E}_G f.$$

When $a > \bar{f}$, compare the upper concentration tail with (6.6):

$$\frac{1}{8} e^{-A_0} \leq \mathbb{P}_G(f \geq a) \leq \exp \left(-\frac{(a - \bar{f})^2}{2\sigma^2} \right),$$

so $\bar{f} \geq a - \sigma \sqrt{2(A_0 + \log 8)}$. If $a \leq \bar{f}$, the same lower bound holds immediately. In either case the lower concentration tail yields

$$\mathbb{P}_G \left(f \geq a - \sigma \sqrt{2(A_0 + \log 8)} - \sigma \sqrt{2 \log 2} \right) \geq \frac{1}{2}.$$

Since $\sigma \leq \beta \sqrt{nt/2}$, we have

$$\sigma \sqrt{2(A_0 + \log 8)} + \sigma \sqrt{2 \log 2} \leq \frac{\beta^2 ntq_*}{\sqrt{2}} + C_\beta \sqrt{nt} \leq \beta^2 ntq_* + C_\beta \sqrt{nt}.$$

Consequently, with conditional probability at least one half,

$$\log W \geq \frac{\beta^2 t(n-1)}{4} - \beta^2 ntq_* - C_\beta \sqrt{nt} - \log 2. \quad (6.7)$$

The constants and the large- n threshold are independent of the selected disorder and its violating set.

Comparison with the rotation cost. The gain from G must be compared with the change in the original energy. Linearity in the disorder gives

$$\begin{aligned}\frac{Z^{J'}}{Z^J} &= \pi\left(e^{\beta(\sqrt{1-t}-1)H^J + \beta\sqrt{t}H^G}\right) \\ &\geq \pi(T)e^{-\beta ne_1(1-\sqrt{1-t})}W.\end{aligned}$$

On T , the original energy is at most ne_1 . Its negative coefficient $\sqrt{1-t}-1$ gives the last lower bound. Combining this with (6.7), we obtain

$$\begin{aligned}\log \frac{Z^{J'}}{Z^J} &\geq -D_1L - \log 2 - \beta ne_1(1 - \sqrt{1-t}) + \log W \\ &\geq (g - \beta^2 q_*)nt - D_1L - C_\beta \sqrt{nt} - C_\beta(nt^2 + 1).\end{aligned}$$

Here we used $1 - \sqrt{1-t} = t/2 + O(t^2)$, uniformly for $0 < t < 1/2$. Dividing the final lower bound by nt , its limit inferior is at least

$$g - \beta^2 q_* - \frac{D_1}{M} > \frac{3g}{4}.$$

It follows that, for all sufficiently large n and every $J \in \mathcal{E}_n \cap \mathcal{V}_n$,

$$\mathbb{P}_G\left(\log Z^{J'} - \log Z^J \geq \frac{g}{2}nt\right) \geq \frac{1}{2}. \quad (6.8)$$

Average the conditional gain over J and use the unconditional rotation bound (5.5):

$$\begin{aligned}\mathbb{P}_J(\mathcal{E}_n \cap \mathcal{V}_n) &\leq 2\mathbb{P}_{J,G}\left(\log Z^{J'} - \log Z^J \geq \frac{g}{2}nt\right) \\ &\leq 2e^{-c'_\beta nt} = 2e^{-c'_\beta ML} \longrightarrow 0\end{aligned}$$

for a fixed $c'_\beta > 0$. Adding $\mathbb{P}_J(\mathcal{E}_n^c) = o(1)$ proves $\mathbb{P}_J(\mathcal{V}_n) = o(1)$, and hence the simultaneous pair statement in Equation (6.3). Since $\mathcal{V}_n$ is the event that any violating subset exists, this proves the simultaneous claim. No union bound over subsets is needed.

From pairs to individual targets. Choose a fixed $D_3 > D_1 + D_2$, and set $\mathcal{C}_n(D_1) = \mathcal{V}_n^c$. Fix J in this event and any $S \subset \{-1, 1\}^n$, and let

$$B = \{x \in S : \pi^J\{z \in S : |R(x, z)| \geq q_*\} < e^{-D_3L}\}.$$

Suppose that $\pi^J(B) \geq e^{-D_1L}$. Since $B \subset S$,

$$\begin{aligned}(\pi^J(\cdot | B))^{\otimes 2}\{|R(x, y)| \geq q_*\} &= \frac{1}{\pi^J(B)^2} \sum_{x \in B} \pi^J(x) \pi^J\{y \in B : |R(x, y)| \geq q_*\} \\ &< \frac{e^{-D_3L}}{\pi^J(B)} \leq e^{-(D_3-D_1)L} < e^{-D_2L}.\end{aligned}$$

The set B itself satisfies the mass hypothesis of (6.3), so this is a contradiction. Thus $\pi^J(B) < e^{-D_1L}$ for every S , proving the target-wise statement. $\square$

7 Locking on a narrow annulus

Lemma 4.1 applies when one pair overlap exceeds a fixed positive cutoff and the other two are close to zero. The path argument also needs a different regime: the latter overlaps are near a positive level r , while the first overlap exceeds r by an amount that may tend to zero. We obtain a uniform pressure loss for that regime. Fix a minimizing measure μ and use its scalar data from Section 3. Recall $\Gamma(s) = \mathbb{E}u(s, X_s)^2$, and write $\chi_s = \chi(s, X_s)$. We work in the case where $\Gamma(s) = s$ on a fixed interval; if the identity fails there, the pair pressure gap supplies the forbidden interval used in Section 8.

Fix $t_0 > 0$ with $3t_0 < q_*$, where q_* is the constant in Proposition 6.1. Recall $\rho = n^{-\kappa}$ with $\kappa = 1/10000$, and set

$$h = \rho^8, \quad \varepsilon = \rho^{20}. \quad (7.1)$$

Constants below may depend on β, t_0 and the fixed minimizing measure. They do not depend on n or on the overlap parameters.

The comparison will alter the fields by a contrast on the two sides of a branching time, together with a short increment on the other branch. When the surrounding interval has small Parisi mass, a tent integration by parts shows that the contrast contributes little beyond its Gaussian variance. The mixed term with the other branch then provides the strict pressure loss.

Lemma 7.1 (Narrow-annulus locking). *Suppose that*

$$\Gamma(s) = s \quad \text{for } t_0/2 < s < 3t_0. \quad (7.2)$$

For three independent Gibbs configurations v, x, y , conditionally on the disorder, define

$$q = R(x, y), \quad r = \frac{R(v, x) + R(v, y)}{2}, \quad \delta = \frac{R(v, x) - R(v, y)}{2}.$$

Then, for all sufficiently large n ,

$$\mathbb{E}\pi^{\otimes 3} \left\{ \begin{array}{l} t_0/2 < r < 3t_0, \quad \mu([r-h, r+h]) \leq \rho^4, \\ q \geq r + \rho, \quad \varepsilon \leq |\delta| \leq 2\varepsilon \end{array} \right\} \leq \exp(-n^{4/5}). \quad (7.3)$$

Proof. Fix a feasible overlap matrix, in the coordinate order v, x, y ,

$$D = \begin{pmatrix} 1 & r + \delta & r - \delta \\ r + \delta & 1 & q \\ r - \delta & q & 1 \end{pmatrix}, \quad (7.4)$$

satisfying the event in (7.3). We may first impose the additional restriction

$$|q - \Gamma(q)| \leq \rho^{13}. \quad (7.5)$$

For an excluded pair overlap, (4.4) already gives a pressure deficit of at least $\rho^{26}/2$. At the end of the proof we will apply Lemma 4.2 to these deficits and sum over the at most $n + 1$ pair overlaps. The independent third draw does not change their total probability.

For the minimizing scalar data, the source of the strict loss is already visible. The identity $\Gamma'(r) = 1$ normalizes $\beta^2 \mathbb{E}\chi_r^2$ to one, whereas the conditional susceptibility evolution decreases the mixed moment:

$$\beta^2 \mathbb{E}[\chi_q \chi_r] \leq 1 - c(q - r).$$

Step 8 will verify a uniform $c > 0$. We construct a covariance perturbation whose quadratic pressure change contains this deficit. Its much larger Gaussian variance contribution will cancel against the interpolation penalty.

The proof of the three-replica deficit has three stages. In Steps 1–4 we construct an admissible perturbed hierarchy and reduce the comparison to one quadratic moment. Steps 5–6 estimate that moment using the tent on the first branch and a ramp on the second. In Steps 7–9 we pass from finite hierarchies to the minimizing scalar data, establish the negative quadratic coefficient, and sum the resulting probability bounds.

Step 1. Finite scalar data and the base hierarchy. For all large n , the overlap restrictions imply

$$0 < r - h < r < r + h < q \leq 1.$$

Take increasingly fine partitions of $[0, 1]$ containing $r - h, r, r + h, q$. On each half-open segment, approximate α by its actual value at the left endpoint, and set its value at 1 to 1. These approximations are nondecreasing step distribution functions. They converge in L^1 and preserve the actual endpoint values at the four indicated points. In particular, each satisfies

$$\alpha(r + h) - \alpha(r - h) \leq \rho^4. \quad (7.6)$$

During the finite calculations, we omit the approximation index: $\alpha, \Phi, u, \chi, X, A$ mean the step data, and $\mathcal{P}_\beta(\alpha)$ is the scalar functional of the corresponding measure. These step measures need not minimize the functional. We will use stationarity only after passing to the minimizing limit.

We index only the integrated covariance increments: $Q_0 = 0$, $Q_\ell - Q_{\ell-1} \succeq 0$, with nondecreasing masses $m_\ell \in [0, 1]$ and field covariances $\beta^2(Q_\ell - Q_{\ell-1})$. For this indexing write

$$Q_{\text{end}} = Q_N, \quad \int m \, d|Q|^2 = \sum_{\ell=1}^N m_\ell (|Q_\ell|^2 - |Q_{\ell-1}|^2).$$

Prepending a zero-covariance quenched level gives precisely the convention of Proposition 2.3. We always use zero terminal multiplier, so the terminal function is the sum of the three scalar $\log(2 \cosh)$ functions.

Build the base covariance path Q in three stages:

1. From scalar time 0 to r , give all three fields the same increment of variance $\beta^2 \, ds$, integrated at mass $\alpha(s)/3$.
2. From r to q , give v its own scalar branch at mass $\alpha(s)$ and give x, y a common scalar branch at mass $\alpha(s)/2$. The branches use independent Gaussian increments. Merge their finite schedules in nondecreasing mass order, preserving each branch's local order.
3. From q to 1, give the three coordinates independent scalar increments simultaneously on each segment, all integrated at mass $\alpha(s)$.

This is the base hierarchy of Section 4.2, now with the specified finite approximation. In the middle stage, scalar time is local to each branch: we merge by mass, preserving each branch's order. The end masses of the first stage are at most the starting middle masses, and the end middle masses are at most the starting third-stage masses. Thus the complete schedule is nondecreasing, and every increment is positive semidefinite.

Write Ψ_0 for its recursive value. The block-size cancellation proved in Section 4.2 gives

$$\Psi_0 - \frac{\beta^2}{4} \int m \, d|Q|^2 = 3\Phi(0, 0) - \frac{3\beta^2}{2} \int_0^1 s\alpha(s) \, ds = 3\mathcal{P}_\beta(\alpha). \quad (7.7)$$

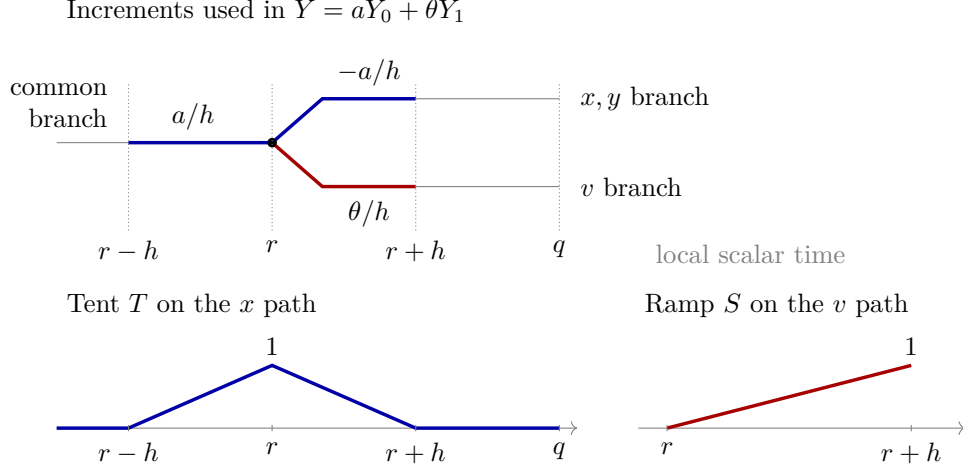


Figure 1: The three increments in the perturbation. The upper labels are coefficients of the raw field increments in Y ; $a = 1 - \theta$. Horizontal position records local scalar time, not the mass-sorted order of the hierarchy. Below, the raw covariance of Y_0 with X_s^x , divided by β^2 , is the tent $T(s)$, which vanishes before the endpoint q . Conditional on the common field X_r , the raw covariance of Y_1 with X_s^v on $[r, r+h]$, divided by β^2 , is the ramp $S(s)$. The tent isolates the small change in Parisi mass near r ; the other branch supplies the mixed susceptibility term. Lengths are schematic.

Its endpoint is D with $\delta = 0$. The same calculation identifies the base tilted law: after the common path through r , the x, y branch and the v branch each have the scalar transition law. Denote their scalar paths by X^x and X^v ; they coincide through r and then run to q on their separate clocks. A frozen branch's additive value cancels from every active normalized kernel, so merging the schedules does not change their conditional independence.

To record the conditional independence that will be used below, let $\mathcal{X}_0$ be the common prefix through r . If $K_{r,z}$ denotes the scalar path law starting at (r, z) , then

$$\mathcal{L}(X_{[r,q]}^x, X_{[r,q]}^v \mid \mathcal{X}_0) = K_{r,X_r} \otimes K_{r,X_r}.$$

In particular, knowing the entire x branch reveals no additional future noise of the v branch.

Step 2. A perturbation preserving positive semidefiniteness. Set

$$\theta = \rho^2, \quad a = 1 - \theta,$$

and form the following linear combinations of field increments:

$$Y_0 = \frac{(X_r^x - X_{r-h}^x) - (X_{r+h}^x - X_r^x)}{h}, \quad Y_1 = \frac{X_{r+h}^v - X_r^v}{h}, \quad Y = aY_0 + \theta Y_1. \quad (7.8)$$

Under the raw law these are Gaussian linear expressions. Under the base tilted law the same expressions use the scalar paths just identified. The contrast Y_0 includes the common increment before the split and the x increment after the split; Y_1 uses the post-split increment on the v branch. Figure 1 shows these three increments and their covariance kernels.

At each level of the first two stages, add δ times that level's contribution to Y to the x field, and subtract it from the y field. Leave v unchanged. Each modified increment is a deterministic linear image of its original Gaussian increment. Consequently its covariance is positive semidefinite,

and increments at different levels remain independent. This gives an admissible path $\tilde{Q}$ with the original masses. At the end of the second stage the added fields are $0, \delta Y, -\delta Y$.

For any prefix, write K_c for the accumulated raw variance of Y , divided by β^2 , and write g_v, g_x for its raw covariances with the base v, x fields in the same units. Its covariance with the base y field equals g_x . The coefficients on the three active intervals are $a/h, -a/h, \theta/h$, so

$$K = \frac{2a^2 + \theta^2}{h}, \quad g_v^{\text{end}} = a + \theta = 1, \quad g_x^{\text{end}} = a - a = 0. \quad (7.9)$$

Let e_v, e_x, e_y be the coordinate vectors and set $L_2 = (e_x - e_y)(e_x - e_y)^\top$. At every prefix,

$$\tilde{Q} = Q + \delta L_1 + \delta^2 K_c L_2, \quad L_1 = \begin{pmatrix} 0 & g_v & -g_v \\ g_v & 2g_x & 0 \\ -g_v & 0 & -2g_x \end{pmatrix}. \quad (7.10)$$

For the Frobenius product $U : V = \sum_{i,j} U_{ij} V_{ij}$, both Q and L_2 are orthogonal to L_1 . Also $|L_1|^2 = 4g_v^2 + 8g_x^2$ and $|L_2|^2 = 4$. Expanding gives

$$|\tilde{Q}|^2 - |Q|^2 = \delta^2(4g_v^2 + 8g_x^2 + 2K_c Q : L_2) + 4\delta^4 K_c^2. \quad (7.11)$$

The endpoint is

$$\tilde{Q}_{\text{end}} = D + \delta^2 K L_2. \quad (7.12)$$

Its diagonal need not be one. We therefore use precisely the arbitrary-endpoint comparison (2.11), proved in Proposition 2.3. It assigns the endpoint discrepancy the cost

$$\frac{\beta^2}{4} |D - \tilde{Q}_{\text{end}}|^2 = \beta^2 \delta^4 K^2. \quad (7.13)$$

Step 3. The penalty gain. We next compute the quadratic terms in the interpolation penalty. The base x, y fields coincide before the third stage, so $Q : L_2 = 0$ there. In the third stage, $Q : L_2 = 2(s - q)$ and $K_c = K$ is fixed. The term $2K_c Q : L_2$ in (7.11) therefore contributes $\beta^2 \delta^2 K A(q)$ to the penalty.

For the other quadratic terms put

$$I = \int m (d(g_v^2) + 2d(g_x^2)).$$

The three active pieces have separate roles. Before the split, on $[r - h, r]$, both covariances grow from 0 to a at mass $\alpha(s)/3$; this contributes $\alpha(r)a^2 + O(\rho^4)$. On the active v steps, g_v grows from a to $a + \theta$ and g_x is fixed; this contributes $\alpha(r)(2a\theta + \theta^2) + O(\rho^4)$. On the active shared x, y steps, g_x decreases from a to zero and g_v is fixed. The factor 2 in I cancels the factor $1/2$ in their masses, giving $-\alpha(r)a^2 + O(\rho^4)$. The error in replacing each mass by $\alpha(r)$ is controlled by (7.6) times the total variation of the relevant covariance square. These total variations are uniformly bounded, independently of the partition and its interleaving. Thus

$$I = \alpha(r)(2a\theta + \theta^2) + O(\rho^4). \quad (7.14)$$

Finally, K_c is a nondecreasing cumulative variance and all masses are nonnegative. The fourth-order contribution $\int m d(K_c^2)$ is therefore nonnegative. Combining the terms proves

$$\frac{\beta^2}{4} \int m d(|\tilde{Q}|^2 - |Q|^2) \geq \beta^2 \delta^2 (K A(q) + \alpha(r)(2a\theta + \theta^2) - C\rho^4). \quad (7.15)$$

Step 4. The recursive-value cost. Let $\tilde{\Psi}$ denote the modified one-site value. Integrating the unchanged third stage backwards leaves the boundary difference

$$U = \Phi(q, X_q^x + \delta Y) + \Phi(q, X_q^x - \delta Y) - 2\Phi(q, X_q^x). \quad (7.16)$$

Initially view this as a function of the full raw Gaussian prefix through the second stage. Let $\mathbb{E}_0$ denote expectation under its base tilted law. We claim that

$$\tilde{\Psi} - \Psi_0 \leq \log \mathbb{E}_0 e^U. \quad (7.17)$$

Apply the normalized-kernel comparison (4.11) with boundary change U . Here the state at each level is the entire raw Gaussian prefix: Y uses several increments, so the current field alone need not determine the modified future value. If K_i is the normalized base kernel and Δ_i is the change in recursive value, the ratio of the defining integrals still gives

$$\Delta_i = \frac{1}{m_i} \log K_i e^{m_i \Delta_{i+1}} \quad (m_i > 0), \quad \Delta_i = K_i \Delta_{i+1} \quad (m_i = 0).$$

The kernel induction used for (4.11) therefore applies without change and proves (7.17). This argument uses no inverse transformation of the fields, so singular covariance increments cause no difficulty. Every kernel is from the base hierarchy, whose tilted prefix law was identified in Step 1.

Write $\chi_q = \chi(q, X_q^x)$. By Lemma 3.1,

$$0 \leq U \leq \delta^2 Y^2, \quad |U - \delta^2 \chi_q Y^2| \leq C_\beta |\delta Y|^3. \quad (7.18)$$

The diffusion representation under $\mathbb{E}_0$ writes $Y = G + B_*$, where G is a centered Gaussian linear combination of Brownian increments with variance $\beta^2 K$, and $|B_*| \leq C_\beta$. Indeed, each drift in (7.8) is bounded by β^2 , integrated for time h , and then divided by h . No independence between G and B_* is required. Since $\delta^2/h = O(\rho^{32})$, Gaussian exponential moments and $|Y| \leq |G| + C_\beta$ imply, for all large n ,

$$\mathbb{E}_0 |Y|^3 \leq C_\beta h^{-3/2}, \quad \mathbb{E}_0 [Y^4 e^{\delta^2 Y^2}] \leq C_\beta h^{-2}.$$

Use $e^u \leq 1 + u + \frac{1}{2}u^2 e^u$ for $u \geq 0$, followed by $\log(1 + z) \leq z$, and then (7.18). We obtain

$$\log \mathbb{E}_0 e^U \leq \delta^2 \mathbb{E}_0 [\chi_q Y^2] + C_\beta (|\delta|^3 h^{-3/2} + \delta^4 h^{-2}). \quad (7.19)$$

We have now reduced the comparison to estimating $\mathbb{E}_0 [\chi_q Y^2]$. Its Gaussian variance term will cancel the $KA(q)$ part of the penalty up to the pair deviation. The strict gain must come from the mixed moment of Y_0 and Y_1 .

Step 5. The tent on the first branch. We will establish the quadratic-moment bound

$$\mathbb{E}_0 [\chi_q Y^2] \leq \beta^2 K \mathbb{E}_0 \chi_q + 2a\theta\beta^4 \alpha(r) \mathbb{E}_0 [\chi_q \chi_r] + C_\beta (\rho^4 + \theta^2), \quad \chi_r = \chi(r, X_r^x). \quad (7.20)$$

Begin with the scalar x path through q . Write its partition as $0 = s_0 < \dots < s_N = q$ and its scalar masses as $m_j^{\text{sc}} = \alpha(s_j)$. Its normalized tilted density D_x relative to the raw Gaussian increments has logarithm

$$\ell := \log D_x = \sum_{j=0}^{N-1} m_j^{\text{sc}} (\Phi(s_{j+1}, X_{s_{j+1}}^x) - \Phi(s_j, X_{s_j}^x)). \quad (7.21)$$

The scalar masses appear here because the base-law factorization in Step 1 has already canceled the divided matrix masses.

Let T be the continuous piecewise linear tent supported on $[r-h, r+h]$, zero at both endpoints and equal to one at r . For functions of the raw cumulative coordinates, denote by ∂ the derivative obtained by replacing X_s^x with $X_s^x + tT(s)$ and differentiating at $t = 0$. Under the raw law,

$$\text{Cov}_{\text{raw}}(Y_0, X_s^x) = \beta^2 T(s), \quad \text{Var}_{\text{raw}}(Y_0) = \frac{2\beta^2}{h}.$$

Finite-dimensional Gaussian integration by parts gives

$$\mathbb{E}_{\text{raw}}[Y_0 f] = \beta^2 \mathbb{E}_{\text{raw}}[\partial f], \quad \mathbb{E}_{\text{raw}}[Y_0^2 f] = \frac{2\beta^2}{h} \mathbb{E}_{\text{raw}} f + \beta^4 \mathbb{E}_{\text{raw}}[\partial^2 f]. \quad (7.22)$$

For a direct verification, multiply the Gaussian density by $\exp\{tY_0 - t^2(2\beta^2/h)/2\}$. This shifts every cumulative coordinate by $t\beta^2 T(s)$; one and two differentiations give the two formulas. These derivatives act on raw coordinates, not on the solution map of the tilted stochastic differential equation.

Telescope (7.21). Since $T(0) = T(q) = 0$, only internal coefficients $m_{j-1}^{\text{sc}} - m_j^{\text{sc}}$ remain:

$$\partial \ell = \sum_{j=1}^{N-1} (m_{j-1}^{\text{sc}} - m_j^{\text{sc}}) T(s_j) u(s_j, X_{s_j}^x), \quad \partial^2 \ell = \sum_{j=1}^{N-1} (m_{j-1}^{\text{sc}} - m_j^{\text{sc}}) T(s_j)^2 \chi(s_j, X_{s_j}^x).$$

Only points strictly between $r-h$ and $r+h$ contribute. The sum of the absolute coefficients at those points is at most $\alpha(r+h) - \alpha(r-h)$. Thus

$$|\partial \ell| \leq \rho^4, \quad |\partial^2 \ell| \leq \rho^4. \quad (7.23)$$

This estimate includes an atom at r . Atoms at $r-h$ or $r+h$ contribute zero because the tent vanishes there. No continuity of α at these points is assumed.

The endpoint q lies outside the tent, so $\partial \chi_q = 0$. Apply the second identity of (7.22) to $f = D_x \chi_q$:

$$\begin{aligned} \mathbb{E}_0[\chi_q Y_0^2] &= \frac{2\beta^2}{h} \mathbb{E}_0 \chi_q + \beta^4 \mathbb{E}_0 [\chi_q ((\partial \ell)^2 + \partial^2 \ell)] \\ &= \frac{2\beta^2}{h} \mathbb{E}_0 \chi_q + O_\beta(\rho^4). \end{aligned} \quad (7.24)$$

At the tent's peak, put $u_r = u(r, X_r^x)$; then $\partial u_r = \chi_r$. The first identity applied to $f = D_x \chi_q u_r$ gives

$$\begin{aligned} \mathbb{E}_0[\chi_q Y_0 u_r] &= \beta^2 \mathbb{E}_0[\chi_q \chi_r] + \beta^2 \mathbb{E}_0[\chi_q u_r \partial \ell] \\ &= \beta^2 \mathbb{E}_0[\chi_q \chi_r] + O_\beta(\rho^4). \end{aligned} \quad (7.25)$$

Both integrations by parts are legitimate for every fixed finite partition. Scalar values grow at most linearly, their needed derivatives are bounded, and the density grows at most exponential-linearly in the finite Gaussian vector. Gaussian integrability therefore justifies differentiation directly, or after truncation and passage to the limit. The error estimates above are uniform over the partitions.

Step 6. The ramp on the other branch and the mixed moment. Condition now on the entire path X^x through q . The product law from Step 1 leaves the future v branch as a scalar diffusion starting at the fixed value X_r . On $[r, r+h]$, its conditional tilted density has the form (7.21), with starting time r . Use the ramp $S(s) = (s-r)/h$ and denote the corresponding raw-coordinate

derivative by ∂_v . The raw covariance of Y_1 with the cumulative coordinate at s is $\beta^2 S(s)$, and its raw variance is β^2/h .

Let ℓ_v be this branch's conditional log density. Telescoping its derivative produces the terminal term $m_{\text{last}} u(r+h, X_{r+h}^v)$ and internal terms with total absolute coefficient at most $m_{\text{last}} - m_{\text{first}} \leq 1$. The initial value is fixed and the ramp is zero there. Using $|u| \leq 1$ and $\chi \leq 1$ gives

$$|\partial_v \ell_v| \leq 2, \quad |\partial_v^2 \ell_v| \leq 2.$$

The conditional second Gaussian integration-by-parts identity therefore yields

$$\mathbb{E}_0[Y_1^2 | X^x] = \frac{\beta^2}{h} + \beta^4 \mathbb{E}_0[(\partial_v \ell_v)^2 + \partial_v^2 \ell_v | X^x] = \frac{\beta^2}{h} + O_\beta(1). \quad (7.26)$$

The error is uniform in the starting field X_r . Multiplying by the X^x -measurable variable χ_q and averaging gives

$$\mathbb{E}_0[\chi_q Y_1^2] = \frac{\beta^2}{h} \mathbb{E}_0 \chi_q + O_\beta(1).$$

For the mixed moment we need the conditional first moment of Y_1 . The Brownian increment has conditional mean zero. The arbitrary-start martingale identity (3.5) gives $\mathbb{E}_0[u(z, X_z^v) | X^x] = u_r$, since the independent branch starts at X_r . Hence its drift representation gives exactly

$$\mathbb{E}_0[Y_1 | X^x] = \frac{\beta^2}{h} \int_r^{r+h} \alpha(z) \mathbb{E}_0[u(z, X_z^v) | X^x] dz = \beta^2 \bar{\alpha} u_r, \quad \bar{\alpha} = \frac{1}{h} \int_r^{r+h} \alpha(z) dz. \quad (7.27)$$

Conditioning on the whole x path introduces no additional conditioning on the v branch's future noise. Moreover, (7.6) implies $|\bar{\alpha} - \alpha(r)| \leq \rho^4$. Since Y_0, χ_q, u_r are X^x -measurable, combine (7.25) with (7.27) to obtain

$$\mathbb{E}_0[\chi_q Y_0 Y_1] = \beta^2 \bar{\alpha} \mathbb{E}_0[\chi_q Y_0 u_r] = \beta^4 \alpha(r) \mathbb{E}_0[\chi_q \chi_r] + O_\beta(\rho^4). \quad (7.28)$$

Finally expand $Y^2 = a^2 Y_0^2 + 2a\theta Y_0 Y_1 + \theta^2 Y_1^2$. Equations (7.9), (7.24), (7.26), and (7.28) prove (7.20). The distinction between the two errors matters: the tent contributes $O_\beta(\rho^4)$, while the ramp's $O_\beta(1)$ error is multiplied by θ^2 .

Step 7. Passing the pressure comparison to the minimizer. We can now apply (2.11) to the modified finite hierarchy. Combine the base value, mismatch, penalty, recursive comparison, and moment estimates in (7.7), (7.13), (7.15), (7.17), (7.19), and (7.20). Since $K = O(h^{-1})$, the endpoint cost is part of the $\delta^4 h^{-2}$ error. Discard the additional nonpositive term $-\beta^2 \delta^2 \alpha(r) \theta^2$ to get

$$\begin{aligned} F_{3,n}(D) &\leq 3\mathcal{P}_\beta(\alpha) + \delta^2 \{ \beta^2 K (\mathbb{E}_0 \chi_q - A(q)) + 2a\theta \beta^2 \alpha(r) (\beta^2 \mathbb{E}_0[\chi_q \chi_r] - 1) + C_\beta(\rho^4 + \theta^2) \} \\ &\quad + C_\beta(|\delta|^3 h^{-3/2} + \delta^4 h^{-2}). \end{aligned} \quad (7.29)$$

The constants do not depend on the partition.

Keep n, r, q, δ fixed and send the mesh size to zero. Lemma 3.1 gives uniform convergence of the scalar derivatives and L^1 stability of the values. Couple the diffusions with the same Brownian motion. Their drifts are uniformly bounded and Lipschitz in space, so L^1 convergence of α together with uniform convergence of u gives path convergence by Gronwall's inequality. The bounded expectations in (7.29) consequently converge. So do $A(q)$ and the scalar functional, the latter to P_β . By construction, every approximation uses the actual value $\alpha(r)$.

This limit is taken at fixed h , so it requires no approximation rate uniform as $h \downarrow 0$. Returning to the minimizing data, and writing $\mathbb{E}$ for its scalar-process expectation, we have

$$F_{3,n}(D) \leq 3P_\beta + \delta^2 \{ \beta^2 K(\mathbb{E}\chi_q - A(q)) + 2a\theta\beta^2\alpha(r)(\beta^2\mathbb{E}[\chi_q\chi_r] - 1) + C_\beta(\rho^4 + \theta^2) \} + C_\beta(|\delta|^3 h^{-3/2} + \delta^4 h^{-2}). \quad (7.30)$$

In the product $\chi_q\chi_r$, both susceptibilities belong to the same scalar path. It remains to show that their evolution makes the quadratic coefficient negative by more than all displayed errors.

Step 8. The strict quadratic margin and the scale choices. The assumed identity (7.2) and the scalar evolution imply

$$\beta^2 \mathbb{E}\chi_r^2 = \Gamma'(r) = 1.$$

Multiply the conditional susceptibility identity (3.7) by χ_r and average. It gives

$$\beta^2 \mathbb{E}[\chi_q\chi_r] = 1 - \beta^4 \int_r^q \alpha(z) \mathbb{E}[\chi_z^2\chi_r] dz \leq 1 - c(q - r) \leq 1 - c\rho \quad (7.31)$$

with a uniform $c > 0$ over the allowed r, q . To justify uniformity, including $q = 1$, use the drift bound to obtain $\sup_{s \leq 1} |X_s| \leq \beta \sup_{s \leq 1} |B_s| + \beta^2$. Choose a fixed M so that the event that the right side is at most M has positive probability p_M . Continuity and strict positivity of χ imply

$$c_M := \min_{0 \leq s \leq 1, |z| \leq M} \chi(s, z) > 0.$$

Thus $\mathbb{E}[\chi_z^2\chi_r] \geq p_M c_M^3$ for all the times under consideration. Also $\alpha(z) \geq \alpha(t_0/2) > 0$ when $z \geq r$. We may take $c = \beta^4 \alpha(t_0/2) p_M c_M^3$ in (7.31). This argument uses positivity on a compact set, not a lower bound for χ on the whole real line.

The all-time stationarity identity (3.15) and the pair restriction (7.5) give

$$\mathbb{E}\chi_q - A(q) = \alpha(q)(q - \Gamma(q)) \leq \rho^{13}.$$

Since $K = O(\rho^{-8})$, the first term in braces in (7.30) is at most $C_\beta \rho^5$. For all large n , $a \geq 1/2$; using $\theta = \rho^2$, $\alpha(r) \geq \alpha(t_0/2) > 0$, and (7.31), the second term is at most $-c_1 \rho^3$. The remaining brace term is $O_\beta(\rho^4)$. The last two errors, after division by δ^2 , satisfy

$$C_\beta |\delta| h^{-3/2} \leq C_\beta \rho^8, \quad C_\beta \delta^2 h^{-2} \leq C_\beta \rho^{24}.$$

Every positive error is therefore smaller than the negative ρ^3 contribution for all large n . Uniformly over the feasible matrices in question, this proves

$$F_{3,n}(D) \leq 3P_\beta - c_2 \delta^2 \rho^3 \leq 3P_\beta - c_2 \rho^{43}. \quad (7.32)$$

Step 9. From pressure deficits to the claimed probability. The choice $\kappa = 1/10000$ satisfies $43\kappa < 1/24$, so the deficit in (7.32) dominates the finite-size error required by Lemma 4.2. For each feasible triple matrix that lemma gives two exponential bounds, with exponents at least constant multiples of $n\rho^{43}$ and $n\rho^{86}$. The latter equals $n^{1-86\kappa} = n^{0.9914}$. Sum these bounds over the at most $(n+1)^3$ triple overlap matrices.

For the pairs removed in (7.5), (4.4) and the same lemma give exponents at least constant multiples of $n\rho^{26}$ and $n\rho^{52} = n^{0.9948}$. There are at most $n+1$ pair overlaps to sum over. The fixed constants and polynomial counts in both sums are absorbed by $\exp(-n^{4/5})$ for all sufficiently large n . This proves (7.3). $\square$

8 Reference banks, crossings and slow mixing

We have established all static inputs. It remains to use them along a path: coverage selects references, fixed-scale locking controls incompatible references, and narrow-annulus locking preserves constraints during backwards searches. We first reduce Theorem 1.1 to a sign passage, then construct the required references, and finally bound the probability of all resulting threshold constraints. As before, $L = n^\kappa$, $\rho = n^{-\kappa}$, and constants may depend on the fixed inverse temperature but not on n .

8.1 Stationary paths and a sign-passage criterion

Given J , start the discrete heat-bath chain $(x_k)_{k \geq 0}$ at $x_0 \sim \pi^J$. Each step chooses a uniform site and resamples its spin from the conditional Gibbs law. Stationarity gives marginal π^J at every deterministic index. If N is an independent rate- n Poisson process, $x_{N(t)}$ has conditional transition kernel P_t^J from the realized x_0 . We work up to the deterministic index

$$m = \lceil e^{2L} \rceil. \quad (8.1)$$

Markov's inequality gives

$$\mathbb{P}\{N(e^L) > m\} \leq \frac{ne^L}{m} \leq ne^{-L} = o(1). \quad (8.2)$$

For every fixed configuration v , the overlap $R(v, x_k)$ changes by at most $2/n$ at a single step.

A *bank* is an ordered list of independent π^J samples. Given J , all raw bank entries are mutually independent and independent of the complete path. Their sizes are deterministic functions of n . A chosen reference may be reversed in sign at selection; that orientation is thereafter fixed.

Suppose a first bank is used to select a reference v_1 satisfying $R(v_1, x_0) \geq q_*$, with the choice and orientation depending only on J, x_0 and this bank. Write $\mathcal{S}$ for successful selection, and define its sign-passage event by

$$\mathcal{A} = \mathcal{S} \cap \{R(v_1, x_k) \leq 0 \text{ for some } 0 \leq k \leq m\}.$$

Also write

$$\mathcal{M} = \left\{ \|P_{e^L}^J(x_0, \cdot) - \pi^J\|_{\text{TV}} \leq \frac{1}{4} \right\}.$$

Spin-flip symmetry implies, for every oriented v ,

$$\pi^J\{z : R(v, z) \leq 0\} \geq \frac{1}{2}.$$

Condition on J, x_0 and successful first-bank selection. Since that selection uses no future updates, the path and clock still have their original conditional laws. On $\mathcal{M}$, total variation therefore puts conditional probability at least $1/4$ on the opposite-sign half-space at time e^L . Removing clock overruns gives

$$\mathbb{P}(\mathcal{A}) \geq \frac{1}{4} \mathbb{P}(\mathcal{M} \cap \mathcal{S}) - \mathbb{P}\{N(e^L) > m\}. \quad (8.3)$$

Thus the remaining task is concrete: make the first reference available with probability tending to one, and make its sign passage have probability tending to zero.

Choose a fixed $b \in (0, 1)$ no larger than the small-overlap cutoff in Lemma 4.1 with large-overlap cutoff q_* . Next choose $b' \in (0, b)$ no larger than that lemma's small-overlap cutoff with large-overlap cutoff b . Fix $t_0 > 0$ so small that

$$4t_0 < \min\{q_*, b', b^2/2\}. \quad (8.4)$$

We use the notation

$$\eta_n = n^{-1/100} = \rho^{100} \quad (8.5)$$

for the scale in fixed-scale locking.

8.2 The case with a fixed forbidden overlap interval

There are two possible scalar regimes. In the first, $\Gamma(z) = z$ fails at some point of $(t_0/2, 3t_0)$. Continuity then supplies fixed constants

$$t_0/2 < a_0 < a_1 < 3t_0, \quad d_0 > 0,$$

such that $|q - \Gamma(q)| \geq d_0$ for all $q \in [a_0, a_1]$. The pair gap (4.4), Lemma 4.2, and the polynomial number of feasible pair overlaps imply, for some $c > 0$ and all sufficiently large n ,

$$\mathbb{E}[(\pi^J)^{\otimes 2}\{(v, x) : R(v, x) \in [a_0, a_1]\}] \leq e^{-cn}. \quad (8.6)$$

Apply Proposition 6.1 with $D_1 = 3$, let D_3 be its constant, and take a single bank with $K = \lceil \exp((D_3 + 1)L) \rceil$ entries. Select its first entry with $|R(v, x_0)| \geq q_*$, oriented to give positive overlap. On the disorder event of coverage for the whole configuration space, the Gibbs mass of starting states whose qualifying neighborhood has mass below $e^{-D_3 L}$ is at most e^{-3L} . For every other starting state, the independent bank misses that neighborhood with probability at most e^{-e^L} . Hence $\mathbb{P}(\mathcal{S}) \rightarrow 1$.

At a deterministic $k \leq m$, the path point and a fixed signed bank entry are independent Gibbs draws conditional on J . Sum (8.6) over entries, signs and indices:

$$\mathbb{P}\{R(\sigma v, x_k) \in [a_0, a_1] \text{ for some bank entry } v, \sigma \in \{-1, 1\}, k \leq m\} \leq 2K(m+1)e^{-cn} = o(1).$$

For large n , a sequence beginning at overlap at least $q_* > a_1$ and reaching nonpositive overlap, with jumps at most $2/n < a_1 - a_0$, must visit $[a_0, a_1]$. Therefore $\mathbb{P}(\mathcal{A}) \rightarrow 0$. Together with (8.3), this proves the theorem in this case.

This completes the forbidden-interval case. In the second regime we have

$$\Gamma(z) = z \quad (t_0/2 < z < 3t_0), \quad (8.7)$$

and put

$$h = \rho^8, \quad \varepsilon = \rho^{20}. \quad (8.8)$$

8.3 Thresholds and simultaneous bank properties

Define the fixed integers and the number of levels by

$$H_0 = \lceil 2/b^2 \rceil, \quad B = H_0 + 2, \quad p = B \left\lfloor \frac{t_0}{5B\rho} \right\rfloor. \quad (8.9)$$

For large n , $p \geq B$. Divide the p levels into consecutive blocks of B levels each. At the first level of every block after the first, we *reset* the selection cell to the whole space. Older retained references are kept unless they meet the conflict rule specified below. Independently of the disorder, banks and path, draw independent thresholds r_i , uniform on

$$I_i = [t_0 + 4(i-1)\rho, t_0 + (4(i-1) + 1)\rho]. \quad (8.10)$$

These intervals lie in $[t_0, 2t_0]$, and

$$r_j - r_i \geq 3\rho \quad (i < j). \quad (8.11)$$

Fubini's theorem gives

$$\mathbb{E}[\mu([r_i - 2h, r_i + 2h])] \leq \frac{4h}{\rho}.$$

Consequently Markov's inequality and a union bound show that the event

$$\mathcal{T} = \{\mu([r_i - 2h, r_i + 2h]) \leq \rho^4 \text{ for every } 1 \leq i \leq p\} \quad (8.12)$$

satisfies

$$\mathbb{P}(\mathcal{T}^c) \leq \frac{4ph}{\rho^5} = O(\rho^2). \quad (8.13)$$

The Fubini calculation includes atoms; no regularity of the distribution of the minimizing measure is imposed here.

There will be a bank $\mathcal{B}_j$ at every level. Its size must compensate for the number of earlier reference tuples that can define the selection cell. Using all preceding levels would give a recursion of growing depth p ; Proposition 6.1 does not control its constants uniformly in n . The reset confines this recursion to B fixed steps. Bank sizes can then depend only on the position within a block, keeping the total number of samples at $\exp(O(L))$. Older retained references will still enter the crossing constraints below.

We require coverage for every possible earlier tuple in the current block. For entries $u_i \in \mathcal{B}_i$ from those earlier levels, define

$$S = \{z : |R(u_i, z)| \leq 3t_0 \text{ for every index in this tuple}\}. \quad (8.14)$$

The empty tuple gives the whole space. We require

$$x_k \in S, \quad 0 \leq k \leq m \implies \text{some } w \in \mathcal{B}_j \cap S \text{ satisfies } |R(w, x_k)| \geq q_*. \quad (8.15)$$

Absolute values make this property insensitive to orientations.

Choose the bank sizes recursively over the fixed positions $s = 1, \dots, B$. Once the constants at smaller positions are fixed, set

$$C_s = \sum_{u < s} (D_{3,u} + 2), \quad D_{1,s} = C_s + 5.$$

Let $D_{3,s}$ be a constant supplied by Proposition 6.1 for $D_{1,s}$, and use

$$K_s = \lceil \exp((D_{3,s} + 1)L) \rceil$$

entries at position s . For large n , the number of earlier tuples in the block is at most $\exp(C_s L)$. Only B fixed applications of coverage are needed, so their disorder events have an intersection $\mathcal{D}$ with $\mathbb{P}(\mathcal{D}) \rightarrow 1$.

Fix a level at position s , a realization $J \in \mathcal{D}$, and its earlier bank lists. Each cell S has a bad subset

$$B_S = \{x \in S : \pi^J\{z \in S : |R(z, x)| \geq q_*\} < e^{-D_{3,s}L}\}$$

of Gibbs mass less than $e^{-D_{1,s}L}$. The path is independent of those lists given J , and each deterministic-index marginal is π^J . A union bound over tuples and deterministic indices therefore bounds the probability that any x_k lies in its cell's bad subset by

$$(m+1)e^{(C_s - D_{1,s})L} = O(e^{-3L}).$$

After excluding these bad targets, condition also on the complete path. The bank at the present level is still fresh. Its probability of missing a qualifying neighborhood inside the cell is at most

$$(1 - e^{-D_{3,s}L})^{K_s} \leq e^{-e^L}.$$

Union over the same tuples and indices, and then over the $p = O(L)$ levels, shows that (8.15) holds simultaneously with probability $1 - o(1)$. The recursion has only B steps, so the total number of entries across all p banks is $\exp(O(L))$.

We also impose simultaneous avoidance of the exceptional triples in Lemma 4.1, for the two cutoff pairs (q_*, b) and (b, b') , and in Lemma 7.1. We do so for every ordered triple of signed entries from distinct banks, and for every ordered triple consisting of two signed entries from distinct banks and one path point x_k , $0 \leq k \leq m$. For each deterministic choice, the conditional law given J is $(\pi^J)^{\otimes 3}$: bank independence, stationarity at that deterministic index, and spin-flip symmetry suffice. The sampled configurations themselves need not be different.

There are $\exp(O(L))$ such choices, including all orders and signs. The exceptional probability for each is bounded by a constant times $e^{-n^{9/10}} + e^{-n^{4/5}}$. A union bound thus makes all these locking assertions simultaneous with probability $1 - o(1)$. Let $\mathcal{G}$ be the intersection of this event, the simultaneous bank property, and $\mathcal{T}$. We have proved

$$\mathbb{P}(\mathcal{G}^c) = o(1). \quad (8.16)$$

We have made all these estimates simultaneously before choosing references or crossing times. Later application at a backwards crossing therefore uses this deterministic-index event, not an assertion of stationarity at a random time.

8.4 Backwards searches and deletion of conflicts

Select v_1 from $\mathcal{B}_1$ as the first entry whose absolute overlap with x_0 is at least q_* , and orient it positively. If it exists, let τ_1 be the first $k \leq m$ with $R(v_1, x_k) \leq r_1$. On success initialize the ordered retained list as $A_1 = \{1\}$.

At a subsequent level j , assuming all earlier levels have succeeded, form the cell (8.14) using the selected references in earlier positions of the current block. Choose as v_j the first entry of $\mathcal{B}_j$ lying in this cell and having absolute overlap at least q_* with $x_{\tau_{j-1}}$. Orient it so that $R(v_j, x_{\tau_{j-1}}) \geq q_*$. From A_{j-1} delete the conflicts, namely those indices i with

$$|R(v_i, v_j)| > b. \quad (8.17)$$

The remaining old indices are called compatible. Search backwards from τ_{j-1} , stopping at the first point encountered with overlap at most r_j with v_j . Equivalently, if the set is nonempty, define

$$\tau_j = \max\{0 \leq k \leq \tau_{j-1} : R(v_j, x_k) \leq r_j\}. \quad (8.18)$$

Add j to the compatible list to obtain A_j .

To use independence of each fresh threshold, define v_j on failure realizations as well. A missing candidate or missing crossing ends the procedure. A candidate already selected before its crossing fails is kept as that level's reference. At a level with no candidate, and at all levels after an earlier failure, use the first entry of the corresponding bank without sign reversal. These defaults serve the final conditional-probability calculation. Selection and orientation of v_j use only thresholds earlier than r_j ; its own threshold is used solely for the subsequent crossing. The first reference, when selected successfully, uses only J, x_0 and $\mathcal{B}_1$ as required by (8.3).

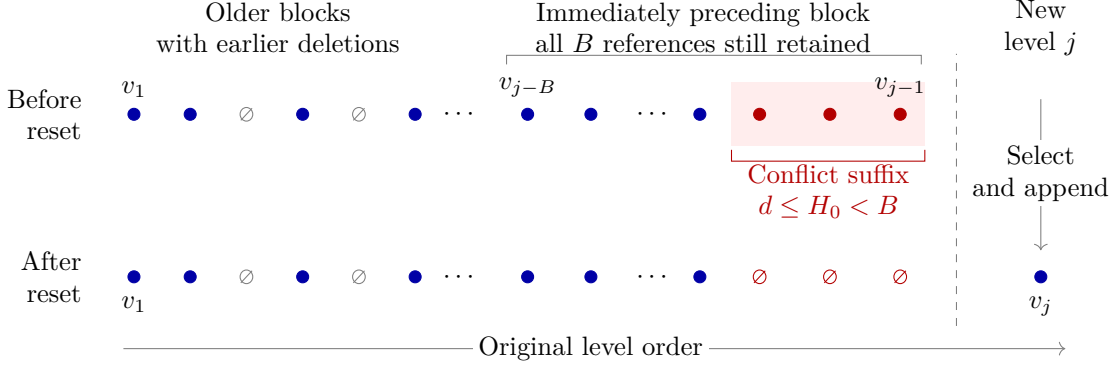


Figure 2: Why a reset preserves the first reference. The dots are schematic slots in the original level order; omitted entries are indicated by ellipsis, and crossed gray slots represent deletions at earlier resets. Immediately before a reset at level j , the preceding block $\{j - B, \dots, j - 1\}$ is complete. Conflicts form a suffix of the retained list of length $d \leq H_0 < B$, illustrated in red. The suffix therefore misses both the first reference of that complete block and all older retained references. Thus v_1 survives at the first reset, when it belongs to the preceding block, and at every later reset, as drawn. The new reference v_j is then appended. Older blocks may contain holes and need not be complete. The diagram illustrates a nonempty conflict suffix; the proof also permits $d = 0$.

We now work deterministically on $\mathcal{G} \cap \mathcal{A}$. For all sufficiently large n , we will show that every level succeeds, that reference 1 survives, and that

$$|R(v_i, x_{\tau_j}) - r_i| \leq 6j\varepsilon \quad (i \in A_j), \quad (8.19)$$

$$|R(v_i, v_\ell) - r_i| \leq 6j\varepsilon \quad (i, \ell \in A_j, i < \ell). \quad (8.20)$$

The induction also keeps the current block complete: after level j , all references from that block's first level through j remain on the list. At the next reset, the preceding block is therefore intact. Older blocks may already have gaps, but the complete preceding block will prevent a new deletion from reaching them.

Deletions occur only at the first level of a new block, and at most H_0 indices, forming a suffix of the old list, are deleted at each such level.

At level 1, coverage of the whole space supplies the reference. Sign passage forces the first crossing, where the one-step overlap bound puts the value in $[r_1 - 2/n, r_1]$. This starts the induction. In the following induction we use the scale relations

$$\frac{1}{n} \ll \eta_n = \rho^{100} \ll \varepsilon = \rho^{20}, \quad p\varepsilon = O(\rho^{19}) \ll h = \rho^8 \ll \rho. \quad (8.21)$$

Suppose the assertions hold through level $j - 1$. All earlier references in the current block are still retained, since there have been no deletions within that block after its first level. Their overlaps with $x_{\tau_{j-1}}$ have absolute value at most $2t_0 + 6p\varepsilon < 3t_0$. Thus this target belongs to the required cell, and the new selection $w = v_j$ exists by (8.15).

For each compatible old index i , the two small overlaps in the triple $(v_i, w, x_{\tau_{j-1}})$ have absolute value at most b , whereas $R(w, x_{\tau_{j-1}}) \geq q_*$. Fixed-scale locking gives

$$|R(v_i, w) - R(v_i, x_{\tau_{j-1}})| \leq 2\eta_n. \quad (8.22)$$

In particular,

$$|R(v_i, w) - r_i| \leq 6(j - 1)\varepsilon + 2\eta_n \quad (i \text{ compatible}). \quad (8.23)$$

Conflicts occupy a bounded final segment. A conflicting old index i cannot precede a compatible old index ℓ . Indeed, by (8.20) and (8.23), the magnitudes of $R(v_\ell, v_i)$ and $R(v_\ell, w)$ are within $6(j-1)\varepsilon$ and $6(j-1)\varepsilon + 2\eta_n$ of r_i and r_ℓ , respectively. Both magnitudes are less than b' , by (8.4). Their difference in magnitude is at least

$$3\rho - 12p\varepsilon - 2\eta_n > 2\eta_n.$$

On the other hand $|R(v_i, w)| > b$. Reverse one of this pair if necessary to make their mutual overlap positive. The signed difference of the two small overlaps has absolute value at least their difference in magnitude. This contradicts fixed-scale locking with cutoff pair (b, b') , which holds for all signs on $\mathcal{G}$. The conflicts consequently form a suffix of the old retained list.

The number of conflicts is bounded by H_0 . Indeed, the induction bounds all absolute overlaps between retained old references by $2t_0 + 6p\varepsilon < 3t_0$. If $d \geq 1$ of them conflict with w , normalize them to unit vectors and reverse each sign to give inner product greater than b with $w/\sqrt{n}$. Denote these vectors by $u_1, \dots, u_d$. Then

$$d^2 b^2 < \left\langle \sum_{a=1}^d u_a, \frac{w}{\sqrt{n}} \right\rangle^2 \leq \left\| \sum_{a=1}^d u_a \right\|^2 \leq d + 3t_0 d(d-1).$$

Since $3t_0 < b^2/2$, this implies

$$d < \frac{1 - 3t_0}{b^2 - 3t_0} < \frac{2}{b^2} \leq H_0.$$

In particular $d \leq H_0$.

If j is not the first level of its block, the most recent old index $j-1$ is compatible: the cell condition gives $|R(v_{j-1}, w)| \leq 3t_0 < b$. A suffix of conflicts is therefore empty. At the first level of a new block, the additional invariant says that all B entries selected in the preceding block are still present. Removing at most $H_0 < B$ final indices cannot reach any older block, or the first index of the preceding block. In particular index 1 survives, including at the first reset. Adding the new index starts the current block as a retained singleton; at a nonreset it extends the complete current block because there is no deletion. This establishes the additional invariant and the deletion claims before proving that the new crossing exists.

The backwards search preserves old constraints. Follow the search for τ_j , including the crossing step when there is one. The new reference satisfies throughout this search

$$R(w, x_k) \geq r_j - 2/n. \tag{8.24}$$

If there is no crossing, this follows with the stronger strict bound $R(w, x_k) > r_j$ at every index down to zero. If there is a crossing, all indices strictly after it have overlap greater than r_j , and the jump bound gives (8.24) at the crossing itself.

For each compatible old reference v_i , (8.23) places the fixed value $R(v_i, w)$ near r_i . The larger threshold r_j separates the new overlap from that old level throughout the search. We will use narrow-annulus locking to prove

$$|R(v_i, x_k) - R(v_i, w)| < 2\varepsilon \tag{8.25}$$

on this entire searched portion. Equation (8.22) proves the claim at the target. At a first violation while moving backwards, the one-step bound puts the left side in $[2\varepsilon, 2\varepsilon + 2/n]$. For this triple use the parameters of Lemma 7.1:

$$r = \frac{R(v_i, x_k) + R(v_i, w)}{2}, \quad \delta = \frac{R(v_i, x_k) - R(v_i, w)}{2}, \quad q = R(w, x_k).$$

Then

$$\varepsilon \leq |\delta| \leq \varepsilon + 1/n \leq 2\varepsilon, \quad |r - r_i| \leq 6(j-1)\varepsilon + 2\eta_n + \varepsilon + 1/n < h.$$

It follows that $t_0/2 < r < 3t_0$ and $[r-h, r+h] \subseteq [r_i-2h, r_i+2h]$. Thus $\mathcal{T}$ gives the required bound $\mu([r-h, r+h]) \leq \rho^4$. Also, by (8.11), (8.24), and (8.21),

$$q - r \geq 3\rho - 6(j-1)\varepsilon - 2\eta_n - \varepsilon - 3/n \geq \rho.$$

All conditions of the forbidden narrow-annulus event hold, contradicting its simultaneous avoidance on $\mathcal{G}$. This proves (8.25).

The crossing must exist. Otherwise the search reaches index zero, and (8.23)–(8.25) apply there to the surviving compatible reference 1. They would give

$$R(v_1, x_0) < r_1 + 6(j-1)\varepsilon + 2\eta_n + 2\varepsilon < q_*,$$

contradicting the first selection. Hence the crossing exists. Its new reference overlap belongs to $[r_j - 2/n, r_j]$. For each retained old reference, the error at the new crossing is at most $6(j-1)\varepsilon + 2\eta_n + 2\varepsilon \leq 6j\varepsilon$. The old-new reference errors satisfy the same claimed bound by (8.23), and old-old reference errors are unchanged. This proves (8.19)–(8.20) and completes the induction.

There are at most p/B reset occasions, so the final list has size at least

$$|A_p| \geq p - H_0 p/B \geq p/B. \quad (8.26)$$

Consequently $\mathcal{G} \cap \mathcal{A}$ implies that, for some index k in $\{0, \dots, m\}$ and some subset of p/B indices, the inequalities

$$|R(v_i, x_k) - r_i| \leq 6p\varepsilon \quad (8.27)$$

hold simultaneously. We have reduced sign passage to this existence event. The probability bound that follows will not condition on $\mathcal{G}$ or on the random retained list.

8.5 Predictable thresholds and completion of the proof

Let $\mathcal{F}_0$ be generated by J , all raw bank lists, and the path $(x_0, \dots, x_m)$, and set

$$\mathcal{F}_i = \sigma(\mathcal{F}_0, r_1, \dots, r_i), \quad 1 \leq i \leq p.$$

Every r_i is uniform on I_i and independent of $\mathcal{F}_{i-1}$. The failure conventions ensure that each oriented v_i is $\mathcal{F}_{i-1}$ -measurable. Here is the complete induction. At level 1 its selection does not use any threshold. Inductively, whether a prior level failed, the previously selected references, and the preceding crossing time are all determined by earlier thresholds. Selection at level i uses only that information and the bank and path data in $\mathcal{F}_0$. The subsequent crossing may use r_i , but never changes that selected reference. A default caused by an earlier failure likewise depends only on earlier thresholds. This proves the claim, including on failure events.

For each deterministic $k \leq m$, let

$$E_i(k) = \{|R(v_i, x_k) - r_i| \leq 6p\varepsilon\}, \quad \gamma_n = \frac{12p\varepsilon}{\rho}.$$

The interval defining $E_i(k)$ has length at most $12p\varepsilon$, while I_i has length ρ . Thus, for all sufficiently large n ,

$$\mathbb{P}(E_i(k) \mid \mathcal{F}_{i-1}) \leq \gamma_n < 1. \quad (8.28)$$

For a deterministic subset $\{i_1 < \dots < i_s\}$, the earlier events $E_{i_1}(k), \dots, E_{i_{s-1}}(k)$ are $\mathcal{F}_{i_s-1}$ -measurable. Conditioning first at the largest selected index and using (8.28) yields

$$\mathbb{P}\left(\bigcap_{a=1}^s E_{i_a}(k)\right) \leq \gamma_n \mathbb{P}\left(\bigcap_{a=1}^{s-1} E_{i_a}(k)\right) \leq \gamma_n^s.$$

The elimination order is essential: the latest selected threshold is integrated out first. Later centers may depend on earlier thresholds, but the displayed conditioning remains valid. Neither the random retained subset nor the good event has been conditioned on.

Taking $s = p/B$, then taking a union over all deterministic subsets and all deterministic indices k , proves

$$\mathbb{P}\{(8.27) \text{ holds for some } k \text{ and some subset of size } p/B\} \leq (m+1)2^p \left(\frac{12p\varepsilon}{\rho}\right)^{p/B} = o(1). \quad (8.29)$$

Indeed $p = (t_0/5)L + O(1)$, $\rho = L^{-1}$, and $12p\varepsilon/\rho = O(\rho^{18})$. The logarithm of the right-hand side is therefore bounded above by

$$2L + O(p) - \frac{18p}{B} \log(1/\rho) = -\Omega(L \log L).$$

The deterministic induction and (8.16) now imply

$$\mathbb{P}(\mathcal{A}) \leq \mathbb{P}(\mathcal{G}^c) + \mathbb{P}(\mathcal{G} \cap \mathcal{A}) = o(1).$$

The first selection succeeds on the simultaneous bank event at level 1, so $\mathbb{P}(\mathcal{S}^c) = o(1)$ as well. Finally, (8.3) and (8.2) give

$$\mathbb{P}(\mathcal{M}) \leq \mathbb{P}(\mathcal{S}^c) + 4\mathbb{P}(\mathcal{A}) + 4\mathbb{P}\{N(e^L) > m\} = o(1).$$

This proves Theorem 1.1 also under (8.7), and the two cases exhaust the possibilities.

Finally take $T_n = \lfloor e^L \rfloor$ discrete update attempts. This deterministic index lies below m for large n . If its transition law is within $1/4$ of equilibrium, the opposite-sign half-space has conditional probability at least $1/4$ at that index. Thus the same sign-passage inequality holds with no clock-overflow term. All bank, locking and threshold estimates already concern the discrete path, so they prove the second assertion of Theorem 1.1 at exactly $\lfloor e^L \rfloor$ attempts.

A Coverage inside sets of polynomial Gibbs mass

The fixed-depth crossing argument in Appendix B uses polynomially many reference samples. The neighborhood bound $e^{-D_3 n^\kappa}$ in Proposition 6.1 is smaller than every fixed inverse power of n , so it does not supply those references. We therefore apply the same energy and rotation mechanism with a perturbation variance of order $\log n/n$. This yields polynomial overlap mass inside sets of polynomial Gibbs mass, uniformly over every subset of the cube. We give the argument from the energy and rotation estimates in Section 5, so that the fixed-depth route can be read independently of the growing-depth construction.

Throughout this appendix, fix $\beta > 1$ and use the off-diagonal Hamiltonian of Equation (1.1). Write $H^J = H_n^J$, $Z^J = Z_{n,\beta}^J$, and $\pi^J = \pi_{n,\beta}^J$. For a nonempty $S \subset \{-1, 1\}^n$, define its conditional Gibbs measure by

$$\pi_S^J(x) = \frac{\pi^J(x) \mathbf{1}_{\{x \in S\}}}{\pi^J(S)}.$$

Every nonempty set has positive Gibbs mass. All Gaussian disorder vectors below have independent standard normal coordinates indexed by $i < j$.

Proposition A.1 (Polynomial overlap coverage). *There exists $q_* = q_*(\beta) \in (0, 1)$ such that, for every fixed $D_1 > 0$, there exists a fixed $D_2 > 0$ for which*

$$\mathbb{P}_J(\text{for every } S \subset \{-1, 1\}^n \text{ with } \pi^J(S) \geq n^{-D_1}, \quad (\pi_S^J)^{\otimes 2}\{|R| \geq q_*\} \geq n^{-D_2}) \longrightarrow 1. \quad (\text{A.1})$$

The threshold q_* does not depend on D_1 .

The proof compares two estimates for the change in $\log Z$ under a small Gaussian rotation. If a set has too few overlapping pairs, a fresh Gaussian perturbation increases its restricted partition sum. The energy upper tail limits the loss caused by decreasing the coefficient of the original disorder. Together these give an increase that the rotation estimate makes unlikely. We first prove the required gain for an arbitrary deterministic probability measure, so that it can later be applied after conditioning on J .

A.1 Gain under independent Gaussian disorder

Lemma A.2 (Gaussian gain from few overlapping pairs). *Fix $M > 0$, $q \in (0, 1)$, and $D \geq \beta^2 M/2 + 2$. For all sufficiently large n , set $t = M \log n/n$. If ν is any deterministic probability measure on $\{-1, 1\}^n$ satisfying*

$$\nu^{\otimes 2}\{|R| \geq q\} \leq 4n^{-D},$$

and G is standard Gaussian, define $W = \nu(e^{\beta\sqrt{t}H^G})$. There is a constant C_β , independent of M, q, D, n, ν , such that

$$\mathbb{P}_G\left(\log W \geq \frac{\beta^2 t(n-1)}{4} - \beta^2 ntq - C_\beta \sqrt{nt} - \log 2\right) \geq \frac{1}{2}. \quad (\text{A.2})$$

Proof. For $x \in \{-1, 1\}^n$ let $h_x = (n^{-1/2}x_i x_j)_{i < j}$, so that $H^G(x) = h_x \cdot G$. In particular,

$$|h_x|^2 = \frac{n-1}{2}, \quad \mathbb{E}_G[H^G(x)H^G(y)] = \frac{nR(x, y)^2 - 1}{2}.$$

Set

$$m_0 = \frac{\beta^2 t(n-1)}{4}, \quad A = \frac{\beta^2 ntq^2}{2}.$$

The Gaussian exponential moment formula gives $\mathbb{E}_G W = e^{m_0}$. For the second moment, split pairs according to whether $|R| < q$:

$$\begin{aligned} \frac{\mathbb{E}_G W^2}{(\mathbb{E}_G W)^2} &= \nu^{\otimes 2}\left(e^{\beta^2 t(nR^2-1)/2}\right) \\ &\leq e^A + 4n^{-D}e^{\beta^2 nt/2} \leq e^A + 4n^{-2} \leq 2e^A. \end{aligned} \quad (\text{A.3})$$

The exponent assumption on D gives the second inequality, and the last one holds for $n \geq 2$. Paley–Zygmund therefore implies

$$\mathbb{P}_G(\log W \geq m_0 - \log 2) \geq \frac{1}{8}e^{-A}. \quad (\text{A.4})$$

This probability may be small. Gaussian concentration will turn it into a lower bound of probability at least one half, while controlling the loss in the value of $\log W$.

Write $Y(G) = \log W$. Differentiation expresses its gradient as $\beta\sqrt{t}$ times a convex combination of the vectors h_x . Consequently Y is Lipschitz with constant at most

$$L_W = \beta\sqrt{\frac{t(n-1)}{2}}.$$

The Gaussian moment bound in Equation (2.3) of Lemma 2.1, followed by Chernoff's inequality, gives the following one-sided tails for this deterministic function of G :

$$\mathbb{P}_G(Y - \mathbb{E}_G Y \geq s) \leq e^{-s^2/(2L_W^2)}, \quad \mathbb{P}_G(Y - \mathbb{E}_G Y \leq -s) \leq e^{-s^2/(2L_W^2)} \quad (s > 0).$$

If $m_0 - \log 2 > \mathbb{E}_G Y$, comparison of the upper-tail bound with Equation (A.4) gives

$$\mathbb{E}_G Y \geq m_0 - \log 2 - L_W \sqrt{2(A + \log 8)}.$$

If $m_0 - \log 2 \leq \mathbb{E}_G Y$, the same inequality holds immediately. Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, we thus obtain

$$\mathbb{E}_G Y \geq m_0 - \log 2 - \frac{\beta^2 ntq}{\sqrt{2}} - \beta\sqrt{nt \log 8}. \quad (\text{A.5})$$

The lower-tail bound also gives $Y \geq \mathbb{E}_G Y - L_W \sqrt{2 \log 2}$ with probability at least $1/2$. Combining this with Equation (A.5), and bounding $\beta^2 ntq/\sqrt{2}$ by $\beta^2 ntq$, proves the stated estimate. One admissible choice is

$$C_\beta = \beta(\sqrt{\log 8} + \sqrt{\log 2}).$$

This constant is independent of the measure ν and of the parameters M, q, D . $\square$

A.2 Uniform coverage over all subsets

Proof of Proposition A.1. Let $e_1 \in (0, \beta/2)$ and $c_E > 0$ be the constants from Lemma 5.1. Thus, outside a disorder event of probability at most $e^{-c_E n}$,

$$\pi^J\{H^J > ne_1\} \leq e^{-c_E n}. \quad (\text{A.6})$$

Define

$$g = \frac{\beta^2}{4} - \frac{\beta e_1}{2} > 0.$$

Choose the overlap threshold before choosing any mass exponent:

$$0 < q_* < \min\left\{1, \frac{g}{8\beta^2}\right\}. \quad (\text{A.7})$$

It depends only on β . We choose this same threshold in Proposition 6.1: both coverage arguments use the same e_1 and permit the strict bound in Equation (A.7). Now fix $D_1 > 0$ and choose

$$M \geq \frac{8(D_1 + 1)}{g}, \quad D_2 = \frac{\beta^2 M}{2} + 2, \quad t = \frac{M \log n}{n}, \quad \theta = \arcsin \sqrt{t}. \quad (\text{A.8})$$

For all sufficiently large n , $0 < t < 1/2$. All constants have been fixed independently of n and of the disorder.

Let $\mathcal{B}_n$ be the event that Equation (A.6) holds and that some $S \subset \{-1, 1\}^n$ satisfies

$$\pi^J(S) \geq n^{-D_1}, \quad (\pi_S^J)^{\otimes 2}\{|R| \geq q_*\} < n^{-D_2}. \quad (\text{A.9})$$

We will bound the probability of this existence event by selecting one violating set. Fix a deterministic ordering of the subsets of the finite cube and take its first member satisfying Equation (A.9). For each fixed nonempty set, both Gibbs quantities in that equation are continuous functions of J . The resulting selector $S = S(J)$ is therefore measurable on $\mathcal{B}_n$. In particular, it is fixed when we condition on J .

For such a fixed J , remove the part of S above the energy threshold:

$$T = S \cap \{x : H^J(x) \leq ne_1\}, \quad \nu = \pi_T^J.$$

For all sufficiently large n , uniformly over $J \in \mathcal{B}_n$,

$$\pi^J(T) \geq \pi^J(S) - e^{-c_E n} \geq \frac{1}{2} \pi^J(S) \geq \frac{1}{2} n^{-D_1}. \quad (\text{A.10})$$

In particular, T is nonempty. Since $T \subset S$, the two conditioning denominators give

$$\nu^{\otimes 2}\{|R| \geq q_*\} \leq \left(\frac{\pi^J(S)}{\pi^J(T)}\right)^2 (\pi_S^J)^{\otimes 2}\{|R| \geq q_*\} < 4n^{-D_2}. \quad (\text{A.11})$$

The trimmed measure is measurable in J as well. Draw G independently of J . Conditional on J , the measure ν is deterministic, so Lemma A.2 applies with $q = q_*$ and $D = D_2$. Write

$$W = \nu(e^{\beta\sqrt{t}H^G}), \quad J_\theta = \sqrt{1-t}J + \sqrt{t}G.$$

No part of the choice of S, T , or ν uses this fresh disorder.

We now compare this conditional gain with the full partition sum. Linearity in the disorder, restriction to T , and $\sqrt{1-t} - 1 < 0$ imply

$$\begin{aligned} \log \frac{Z^{J_\theta}}{Z^J} &= \log \pi^J \left(e^{\beta(\sqrt{1-t}-1)H^J + \beta\sqrt{t}H^G} \right) \\ &\geq -D_1 \log n - \log 2 - \beta(1 - \sqrt{1-t})ne_1 + \log W. \end{aligned}$$

Thus Lemma A.2 gives, with conditional probability at least $1/2$,

$$\begin{aligned} \log \frac{Z^{J_\theta}}{Z^J} &\geq -D_1 \log n - 2 \log 2 - \beta(1 - \sqrt{1-t})ne_1 + \frac{\beta^2 t(n-1)}{4} - \beta^2 ntq_* - C_\beta \sqrt{nt} \\ &\geq nt \left(g - \beta^2 q_* - \frac{D_1}{M} - \frac{\beta e_1 t}{2} \right) - C_\beta \sqrt{nt} - 2 \log 2 - \frac{\beta^2 t}{4}. \end{aligned} \quad (\text{A.12})$$

The last line uses $1 - \sqrt{1-t} \leq t/2 + t^2/2$ for $0 \leq t < 1$. Our choices ensure $\beta^2 q_* < g/8$ and $D_1/M < g/8$. Since $t \rightarrow 0$ and $nt = M \log n \rightarrow \infty$, the remaining errors in Equation (A.12) are $o(nt)$. Consequently, for all sufficiently large n and every $J \in \mathcal{B}_n$,

$$\mathbb{P}_G \left(\log Z^{J_\theta} - \log Z^J \geq \frac{g}{2} nt \mid J \right) \geq \frac{1}{2}. \quad (\text{A.13})$$

The lower bound on n depends only on the fixed parameters and is uniform in J and its selected set.

It remains to bound this increment under the joint law of (J, G) . For $t \leq 1/2$, $\theta \leq \sqrt{2t}$. The unconditional rotation estimate in Lemma 5.2 therefore yields

$$\begin{aligned} \mathbb{P}_{J,G} \left(\log Z^{J_\theta} - \log Z^J \geq \frac{g}{2} nt \right) &\leq \exp \left(-\frac{g^2 nt}{8\beta^2} \right) \\ &= n^{-g^2 M/(8\beta^2)}. \end{aligned}$$

Integrating Equation (A.13) over $\mathcal{B}_n$ bounds $\mathbb{P}_J(\mathcal{B}_n)$ by twice this probability. Adding the exceptional event from Lemma 5.1, we conclude that the failure probability in Equation (A.1) is at most

$$e^{-c_E n} + 2n^{-g^2 M/(8\beta^2)} \longrightarrow 0. \quad (\text{A.14})$$

The selector controls the existence of any violating subset, without a union bound over subsets of the cube. The choice of q_* in Equation (A.7) preceded the choice of D_1 , as required. $\square$

A.3 Target coverage within arbitrary subsets

For the crossing argument, we need a bound on the total Gibbs mass of targets lacking polynomial Gibbs mass of overlapping references in the same subset. Both masses below use the original Gibbs measure. This is the form needed when targets have Gibbs marginals and reference banks contain unconditioned Gibbs samples. Uniformity of the pair statement gives the target bound by applying it to the set of points that would violate it.

Corollary A.3 (Target coverage within arbitrary subsets). *Let q_* be as in Proposition A.1. For every fixed $D_1 > 0$, there exists a fixed $D_3 > 0$ such that, with probability tending to one over J , every $S \subset \{-1, 1\}^n$ satisfies*

$$\pi^J \left(\left\{ x \in S : \pi^J \{ b \in S : |R(x, b)| \geq q_* \} < n^{-D_3} \right\} \right) < n^{-D_1}. \quad (\text{A.15})$$

Proof. Fix $D_1 > 0$, take the corresponding D_2 from Proposition A.1, and choose any fixed $D_3 > D_1 + D_2$. Work on the simultaneous event in that proposition. Given $S \subset \{-1, 1\}^n$, denote the set inside the outer measure in Equation (A.15) by B . If $m = \pi^J(B) \geq n^{-D_1}$, then B is nonempty, and $B \subset S$ implies

$$\begin{aligned} (\pi_B^J)^{\otimes 2} \{|R| \geq q_*\} &= \frac{1}{m^2} \sum_{x \in B} \pi^J(x) \pi^J \{ b \in B : |R(x, b)| \geq q_* \} \\ &< \frac{n^{-D_3}}{m} \leq n^{-D_3 + D_1} < n^{-D_2}. \end{aligned}$$

This contradicts Proposition A.1 applied to B . Hence $\pi^J(B) < n^{-D_1}$. This implication is deterministic on the same disorder event for every S , including cells selected after observing J . For $S = \emptyset$ the assertion is immediate. $\square$

B A fixed number of crossings on polynomial time scales

The polynomial coverage estimate in Proposition A.1 has a direct dynamical application. Together with the absolute-overlap locking estimate of Proposition 4.4, it rules out mixing on each fixed polynomial time scale by using a fixed number of nested crossings. We give the argument in full. Its number of levels and threshold intervals remain fixed as n grows.

Theorem B.1 (Polynomial-time obstruction from fixed-depth crossings). *For every fixed $\beta > 1$ and every fixed $a > 0$,*

$$\mathbb{P}_{J, x_0 \sim \pi^J} \left\{ d^J(x_0, n^a) > \frac{1}{4} \right\} \longrightarrow 1, \quad (\text{B.1})$$

$$\mathbb{P}_{J, x_0 \sim \pi^J} \left\{ d_{\text{disc}}^J(x_0, \lfloor n^a \rfloor) > \frac{1}{4} \right\} \longrightarrow 1. \quad (\text{B.2})$$

Consequently both $\mathbb{P}\{T^J(x_0) > n^a\}$ and $\mathbb{P}\{T_{\text{disc}}^J(x_0) > n^a\}$ tend to one under the same joint law. Each assertion is equivalently a statement that the Gibbs mass of the indicated initial states tends to one in probability over J .

As in the main theorem, the transition law in each distance starts from the realized state x_0 . The proof uses stationarity to sample possible targets along a path, and then uses total variation from that fixed initial state to force a sign passage. Independent random thresholds will make such a passage unlikely.

B.1 A stationary path and fixed overlap scales

Fix $\beta > 1$ and $a > 0$, and abbreviate π^J to π . Put

$$t_n = n^a, \quad m = \lceil 2n^{a+1} \rceil, \quad e_* = \frac{1}{100}, \quad \delta_n = n^{-e_*}. \quad (\text{B.3})$$

Conditional on J , start x_0 with law π and generate an infinite discrete heat-bath chain $(x_k)_{k \geq 0}$. At each step choose a uniform site and use fresh resampling randomness, including possible self-transitions. Let $N(t)$ be an independent rate- n Poisson process. The process $x_{N(t)}$ is the continuous-time chain with rate one at each site. Indeed, if K_i^J denotes resampling at site i , then $K^J = n^{-1} \sum_i K_i^J$ and its generator is $\sum_i (K_i^J - I) = n(K^J - I)$. Therefore

$$\mathbb{E}[(K^J)^{N(t)}] = \exp\{nt(K^J - I)\} = P_t^J.$$

Stationarity gives, for every deterministic k ,

$$\mathcal{L}(x_k \mid J) = \pi. \quad (\text{B.4})$$

Moreover, Chernoff's inequality at parameter $\log 2$, with $\mathbb{E}N(t_n) = n^{a+1}$ and $m \geq 2n^{a+1}$, gives

$$\eta_n := \mathbb{P}\{N(t_n) > m\} \leq \exp\{-(2 \log 2 - 1)n^{a+1}\} = o(1). \quad (\text{B.5})$$

Use the common threshold $q_* = q_*(\beta) > 0$ of Proposition A.1. With the same energy cutoff e_1 as in the main coverage argument, its choice satisfies

$$g = \frac{\beta^2}{4} - \frac{\beta e_1}{2} > 0, \quad 0 < q_* < \min\{1, g/(8\beta^2)\}.$$

Choose an integer $p \geq 2$ such that

$$pe_* > a + 2. \quad (\text{B.6})$$

This integer is fixed throughout the limit. First choose $t_p > 0$ with $2t_p < q_*/2$. For $j = p, p-1, \dots, 2$, take a cutoff $b_j \in (0, q_*/2)$ allowed by Proposition 4.4 for large-overlap threshold $h = t_j/2$, and then choose $t_{j-1} > 0$ with

$$t_{j-1} < t_j/10, \quad 4t_{j-1} < b_j. \quad (\text{B.7})$$

Thus $0 < t_1 < \dots < t_p$, and all scale gaps are fixed positive constants. Independently of the disorder and of all other randomness introduced below, draw

$$r_j \sim \text{Unif}[t_j, 2t_j], \quad 1 \leq j \leq p, \quad (\text{B.8})$$

independently for different j .

B.2 Polynomial reference banks cover every possible target

We next sample finite lists of Gibbs configurations. These lists provide references for the later crossing construction. Their sizes are chosen so that coverage holds for every earlier reference choice and every deterministic index of the path segment $x_0, \dots, x_m$.

Conditional on J , all entries in all banks are independent with law π , and all banks are independent of the entire chain. Choose their sizes recursively. Having chosen $D_{3,i}$ for $i < j$, set

$$B_j = \sum_{i < j} (D_{3,i} + 1), \quad B_1 = 0,$$

choose $D_{1,j} > B_j + a + 3$, and use Corollary A.3 with exponent $D_{1,j}$ to obtain $D_{3,j} > 0$. Bank j consists of

$$M_j = \lceil n^{D_{3,j}+1} \rceil \text{ entries } V_{j,1}, \dots, V_{j,M_j}. \quad (\text{B.9})$$

The constants are chosen before any bank entries or thresholds are sampled. In particular, the number of tuples with one entry from each of the first $j-1$ banks satisfies

$$Q_j := \prod_{i < j} M_i \leq 2^{j-1} n^{B_j}. \quad (\text{B.10})$$

For the empty product, $Q_1 = 1$.

For such a tuple $\mathbf{v} = (v_1, \dots, v_{j-1})$, define its cell by

$$S_j(\mathbf{v}) = \{z \in \{-1, 1\}^n : |R(v_i, z)| \leq 3t_i \text{ for every } i < j\}. \quad (\text{B.11})$$

In particular $S_1 = \{-1, 1\}^n$. The cell restricts the small overlaps with the earlier references, while coverage will supply a large overlap with the current target inside that cell.

Lemma B.2 (Simultaneous bank coverage). *With probability $1 - o(1)$, for every $1 \leq j \leq p$, every tuple $\mathbf{v}$ from banks $1, \dots, j-1$, and every $0 \leq k \leq m$,*

$$x_k \in S_j(\mathbf{v}) \implies \text{there is } \ell \leq M_j \text{ such that } V_{j,\ell} \in S_j(\mathbf{v}), \quad |R(V_{j,\ell}, x_k)| \geq q_*.$$

This event depends only on J , the chain and the bank lists, and is independent of the thresholds.

Proof. Intersect the p disorder events from Corollary A.3. Since p is fixed, the intersection has probability $1 - o(1)$. On it, for every $S \subseteq \{-1, 1\}^n$ the set

$$\mathcal{B}_j(S) = \{x \in S : \pi\{z \in S : |R(z, x)| \geq q_*\} < n^{-D_{3,j}}\}$$

has π -mass less than $n^{-D_{1,j}}$.

Fix a disorder in this intersection and condition on the complete first $j-1$ banks. Each cell in (B.11) is then fixed. The path remains independent of these banks conditional on J , so (B.4) applies to every deterministic k . A union bound over all tuples and all indices gives

$$\mathbb{P}\{x_k \in \mathcal{B}_j(S_j(\mathbf{v})) \text{ for some } \mathbf{v}, k \mid J, \text{ banks } 1, \dots, j-1\} \leq Q_j(m+1)n^{-D_{1,j}}. \quad (\text{B.12})$$

This bound uses marginal stationarity, without requiring independence between different times.

Now condition also on the entire chain. For a target x_k in its cell but outside its bad set, acceptable references have π -mass at least $n^{-D_{3,j}}$. The entries of bank j are still independent Gibbs draws under this conditioning. The chance that this bank misses that set is at most

$$(1 - n^{-D_{3,j}})^{M_j} \leq \exp(-M_j n^{-D_{3,j}}) \leq e^{-n}.$$

Another union bound gives $Q_j(m+1)e^{-n}$. Combining the two bounds and summing over the fixed number of banks, the total failure probability is at most

$$o(1) + \sum_{j=1}^p Q_j(m+1)(n^{-D_{1,j}} + e^{-n}) = o(1). \quad (\text{B.13})$$

Indeed $Q_j(m+1) = O(n^{B_j+a+1})$ and $D_{1,j} > B_j + a + 3$. Neither part of the event uses any threshold. $\square$

Lemma B.3 (Simultaneous bank locking). *With probability $1 - o(1)$, for every $i < j \leq p$, every entry v of bank i , every entry w of bank j , and every $0 \leq k \leq m$,*

$$\begin{aligned} |R(x_k, w)| &\geq t_j/2, & |R(v, x_k)|, |R(v, w)| &\leq b_j \\ \implies & ||R(v, x_k)| - |R(v, w)|| &\leq 2\delta_n. \end{aligned} \quad (\text{B.14})$$

Proof. For fixed bank indices and deterministic k , the triple (v, x_k, w) has law $\pi^{\otimes 3}$ conditional on J . This follows from the independence of the two distinct banks and their independence from the stationary chain. Proposition 4.4, with $h = t_j/2$ and cutoff b_j , bounds the averaged failure probability by $e^{-n^{9/10}}$. There are only finitely many choices of j for the large- n threshold in that proposition. A union bound therefore gives the failure estimate

$$\sum_{i < j} (m+1) M_i M_j e^{-n^{9/10}} = o(1). \quad (\text{B.15})$$

Every bank exponent and p are fixed, so the prefactor is polynomial. $\square$

Let $\mathcal{G}_n$ be the intersection of the events in the two lemmas. Then

$$\mathbb{P}(\mathcal{G}_n^c) = o(1). \quad (\text{B.16})$$

It is determined by $J, x_0, \dots, x_m$ and the bank lists. We have now obtained coverage and locking simultaneously for every choice the crossing construction can make; the thresholds remain independent of this event.

B.3 A measurable construction of nested crossings

For a reference $v \in \{-1, 1\}^n$, write $D_v(k) = |R(v, x_k)|$. A single update changes at most one spin, so

$$|D_v(k+1) - D_v(k)| \leq 2/n \quad \text{for every } v, k. \quad (\text{B.17})$$

We select the first acceptable entry in each bank, in its given order. To keep every selected reference defined even on exceptional events, we specify both successful selections and defaults.

At level 1, choose v_1 as the first entry of bank 1 satisfying $D_{v_1}(0) \geq q_*$. If there is no such entry, set $v_1 = V_{1,1}$, $\tau_1 = 0$, and declare the construction inactive. If selection succeeds, set

$$\tau_1 = \min\{0 \leq k \leq m : D_{v_1}(k) \leq r_1\} \quad (\text{B.18})$$

when the set is nonempty, and declare the construction active. If that set is empty, set $\tau_1 = 0$ and declare it inactive.

Suppose $j \geq 2$ and the earlier levels are defined. If the construction is active and $x_{\tau_{j-1}} \in S_j(v_1, \dots, v_{j-1})$, select v_j as the first entry of bank j in that cell with

$$|R(v_j, x_{\tau_{j-1}})| \geq q_*.$$

After a successful selection, search backwards from τ_{j-1} :

$$\tau_j = \max\{0 \leq k \leq \tau_{j-1} : D_{v_j}(k) \leq r_j\} \quad (\text{B.19})$$

if this set is nonempty, keeping the construction active. If it is empty, retain the selected v_j , set $\tau_j = 0$, and declare the construction inactive. If the construction was already inactive, the target was outside its cell, or the reference selection failed, set $v_j = V_{j,1}$ and $\tau_j = 0$, and declare it inactive. An inactive construction stays inactive at all later levels. At every successful selection, the depth at the target is at least $q_* > 2t_j \geq r_j$; hence any successful backwards crossing has $\tau_j < \tau_{j-1}$.

The information used in these choices is central to the proof. Define

$$\mathcal{W} = \sigma(J, x_0, \dots, x_m, \text{ all bank lists}), \quad \mathcal{F}_0 = \mathcal{W}, \quad \mathcal{F}_j = \mathcal{W} \vee \sigma(r_1, \dots, r_j).$$

Induction through the explicit rules above gives

$$v_j \text{ is } \mathcal{F}_{j-1}\text{-measurable,} \quad \tau_j \text{ and the active flag after level } j \text{ are } \mathcal{F}_j\text{-measurable.} \quad (\text{B.20})$$

In particular, neither the selection of v_j nor any default for it uses r_j or a later threshold.

Let C_1 denote successful selection of the first reference. On C_1 define $s_1 = \text{sgn } R(v_1, x_0) \in \{-1, 1\}$ and let

$$\mathcal{S}_n = C_1 \cap \{\text{there exists } 0 \leq k \leq m : s_1 R(v_1, x_k) \leq 0\}. \quad (\text{B.21})$$

Take this event to be false outside C_1 . Starting from signed overlap at least q_* , a sign passage must visit absolute overlap at most $2/n$, because signed overlaps also change by at most $2/n$. For sufficiently large n , it therefore forces the first crossing in (B.18).

Lemma B.4 (Deterministic nesting). *For all sufficiently large n , on $\mathcal{G}_n \cap \mathcal{S}_n$ every reference selection and every crossing succeeds. For every $1 \leq j \leq p$,*

$$|D_{v_j}(\tau_j) - r_j| \leq 2/n, \quad |D_{v_i}(\tau_j) - r_i| \leq 6j\delta_n \quad (1 \leq i \leq j). \quad (\text{B.22})$$

Proof. At level 1, simultaneous coverage ensures successful selection and $\mathcal{S}_n$ ensures a crossing. The initial depth exceeds r_1 . Minimality of τ_1 and (B.17) therefore put its depth in $[r_1 - 2/n, r_1]$, proving the assertions at that level.

Assume the assertions through level $j - 1$, where $j \geq 2$. For every $i < j$,

$$D_{v_i}(\tau_{j-1}) \leq 2t_i + 6(j-1)\delta_n < 3t_i$$

for large n . Thus the target is in its cell, and simultaneous coverage supplies an entry v_j in that cell with new depth at least q_* . Both $D_{v_i}(\tau_{j-1})$ and $|R(v_i, v_j)|$ are at most $3t_i$, which is less than b_j because $3t_i \leq 3t_{j-1} < b_j$. The new depth at the target is at least $q_* > t_j/2$. Simultaneous locking gives

$$||R(v_i, v_j)| - r_i| \leq |D_{v_i}(\tau_{j-1}) - r_i| + 2\delta_n \leq (6(j-1) + 2)\delta_n. \quad (\text{B.23})$$

We next verify the locking hypotheses throughout the backwards search. At any visited index k where the new depth is at least $t_j/2$ and the old depths are at most b_j , locking implies

$$D_{v_i}(k) \leq |R(v_i, v_j)| + 2\delta_n \leq 3t_i + 2\delta_n.$$

The gap $b_j - 3t_{j-1}$ is a fixed positive number. Hence (B.17) ensures that every old depth remains below b_j at the next backwards step, for all sufficiently large n . Before a crossing, the new depth exceeds $r_j \geq t_j$; at the first crossing step it is at least $r_j - 2/n \geq t_j/2$. Starting from the target,

these observations prove by induction over the backwards steps that locking continues to apply up to and including the first crossing. If no crossing exists, the induction reaches index 0.

Throughout that portion of the search, locking and (B.23) give, for $i < j$,

$$|D_{v_i}(k) - r_i| \leq (6(j-1) + 4)\delta_n. \quad (\text{B.24})$$

A crossing must occur: otherwise this inequality at $k = 0$ and $i = 1$ would imply

$$q_* \leq D_{v_1}(0) \leq 2t_1 + (6(j-1) + 4)\delta_n < q_*.$$

At the backwards crossing, the next index towards the target has depth strictly greater than r_j . Consequently $D_{v_j}(\tau_j) \in [r_j - 2/n, r_j]$ by the one-step bound. Combining this with (B.24) proves (B.22). Since p , q_* , and all t_i, b_j are fixed, one sufficiently large value of n makes every buffer used in the induction valid simultaneously. $\square$

B.4 Independent thresholds make the final shells unlikely

The deterministic construction has reduced a sign passage to a single time at which all selected reference depths lie in narrow intervals about their thresholds. Define this event, using the references from the construction including its defaults, by

$$\mathcal{A}_n = \{\text{there exists } 0 \leq k \leq m : |D_{v_j}(k) - r_j| \leq 6p\delta_n \text{ for every } 1 \leq j \leq p\}. \quad (\text{B.25})$$

Lemma B.4, applied at τ_p , shows that

$$\mathcal{G}_n \cap \mathcal{S}_n \subseteq \mathcal{A}_n. \quad (\text{B.26})$$

Fix a deterministic k and set $E_{j,k} = \{|D_{v_j}(k) - r_j| \leq 6p\delta_n\}$. By (B.20), its center $D_{v_j}(k)$ is $\mathcal{F}_{j-1}$ -measurable, while r_j is independent of $\mathcal{F}_{j-1}$ and uniform on an interval of length t_j . Thus

$$\mathbb{P}(E_{j,k} \mid \mathcal{F}_{j-1}) \leq \min\{1, 12p\delta_n/t_j\}. \quad (\text{B.27})$$

The earlier events $E_{i,k}$, $i < j$, are $\mathcal{F}_{j-1}$ -measurable. Applying conditional expectation successively for $j = p, p-1, \dots, 1$ therefore gives

$$\mathbb{P}\left(\bigcap_{j=1}^p E_{j,k} \mid \mathcal{W}\right) \leq \prod_{j=1}^p \min\{1, 12p\delta_n/t_j\}.$$

The centers may depend on earlier thresholds; the displayed filtration is exactly what allows this product estimate. A union bound over the deterministic indices $k = 0, \dots, m$ now yields

$$\mathbb{P}(\mathcal{A}_n) \leq (m+1) \prod_{j=1}^p \frac{12p\delta_n}{t_j} \leq C_{\beta,a} n^{a+1-pe_*} = o(1). \quad (\text{B.28})$$

The exponent is less than -1 by (B.6).

B.5 A fast realized initial state forces a sign passage

Proof of Theorem B.1. First consider continuous time and write

$$q_n = \mathbb{P}_{J, x_0 \sim \pi^J} \{d^J(x_0, t_n) \leq 1/4\}.$$

The first reference is chosen using only J, x_0 and bank 1. Conditional on these variables, the future chain retains its transition law from the realized x_0 . On C_1 , define

$$A = A(J, x_0, v_1) = \{z \in \{-1, 1\}^n : s_1 R(v_1, z) \leq 0\}.$$

Spin reversal preserves π and interchanges positive and negative overlaps, so $\pi(A) \geq 1/2$. For any realized J, x_0 with $d^J(x_0, t_n) \leq 1/4$, and any bank 1 for which C_1 holds, the definition of total variation gives

$$\mathbb{P}\{x_{N(t_n)} \in A \mid J, x_0, \text{bank 1}\} \geq 1/4.$$

If $N(t_n) \leq m$, this endpoint event implies $\mathcal{S}_n$. The Poisson process is independent of all the conditioning variables, so the error in (B.5) is uniform under this conditioning. Consequently

$$\begin{aligned} \mathbb{P}(\mathcal{S}_n) &\geq (1/4 - \eta_n) \mathbb{P}\{d^J(x_0, t_n) \leq 1/4, C_1\} \\ &\geq q_n/4 - \eta_n - \frac{1}{4} \mathbb{P}(C_1^c). \end{aligned} \tag{B.29}$$

The first-bank case of Lemma B.2 gives $\mathbb{P}(C_1^c) = o(1)$. Adding the independent thresholds does not change any probability in this comparison.

Combining (B.16), (B.26), (B.28), and (B.29) gives the explicit estimate

$$\frac{q_n}{4} \leq \eta_n + \frac{1}{4} \mathbb{P}(C_1^c) + \mathbb{P}(\mathcal{G}_n^c) + C_{\beta, a} n^{a+1-pe_*} = o(1). \tag{B.30}$$

This proves (B.1).

For discrete time set $k_n = \lfloor n^a \rfloor \leq m$ and

$$q_n^{\text{disc}} = \mathbb{P}_{J, x_0 \sim \pi^J} \{d_{\text{disc}}^J(x_0, k_n) \leq 1/4\}.$$

Use the same stationary discrete path, banks, scales and thresholds. Whenever the realized initial state is in the event defining q_n^{disc} and C_1 holds, total variation now gives

$$\mathbb{P}\{x_{k_n} \in A \mid J, x_0, \text{bank 1}\} \geq 1/4.$$

This endpoint is among $x_0, \dots, x_m$, so it forces $\mathcal{S}_n$ with no clock-tail error. The same argument therefore gives

$$\frac{q_n^{\text{disc}}}{4} \leq \frac{1}{4} \mathbb{P}(C_1^c) + \mathbb{P}(\mathcal{G}_n^c) + C_{\beta, a} n^{a+1-pe_*} = o(1), \tag{B.31}$$

proving (B.2) directly.

For each realized finite chain, total variation is nonincreasing in time, and in continuous time it is also continuous. Hence the first conclusion implies $T^J(x_0) > n^a$. In discrete time it implies $T_{\text{disc}}^J(x_0) > \lfloor n^a \rfloor$; this integer inequality gives $T_{\text{disc}}^J(x_0) > n^a$. Finally, the Gibbs mass of each exceptional set lies in $[0, 1]$ and has expectation tending to zero. Markov's inequality makes that mass tend to zero in probability over J ; conversely, convergence in probability of a bounded mass implies convergence of its expectation. This proves the asserted equivalent formulations. $\square$

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